National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Rectal Cancer

Version 2.2026 — April 7, 2026

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

NCCN Guidelines for Patients® available at www.nccn.org/patients

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Rectal Cancer

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See NCCN Categories of Evidence and Consensus.

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See NCCN Categories of Preference.

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Updates in Version 2.2026 of the NCCN Guidelines for Rectal Cancer:

REC-3A

- Text box modified: For stage II and III, if somatic PI3K pathway alterations, start aspirin (unless contraindicated) 100–162 mg PO daily for 3 years after recovery from surgery (concurrent with chemotherapy if given). ~~Therefore,~~ Molecular profiling*, to include somatic PI3K pathway alterations,* should be done on all stage II and III CRC tumors. (Also for REC-5, REC-6, REC-14)

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Updates in Version 1.2026 of the NCCN Guidelines for Rectal Cancer:

REC-1
- Workup, footnote c added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
- Branches for pedunculated polyp vs sessile polyp removed
  - Footnote i revised: ~~Observation may be considered, with the understanding that there is significantly greater incidence of adverse outcomes (residual disease, recurrent disease, mortality, or hematogenous metastasis, but not lymph node metastasis) than pedunculated malignant polyps.~~ *Historical data have suggested that endoscopically resected sessile malignant polyps with submucosal invasion harbored higher risk of residual disease and/or lymph node metastases than similar lesions that were of pedunculated morphology. More recently, evidence suggests that margin status and the presence or absence of unfavorable histologic features may be the most important predictors of residual disease/ lymph node metastases regardless of morphologic type.* See Principles of Pathologic Review (REC-B) - Endoscopically Removed Malignant Polyps.

REC-2
- Workups separated onto separate pages
  - Workup, Rectal cancer without suspected or proven distant metastases
    - Bullet 3 revised: ~~Consider PIK3CA~~ Testing *for somatic PI3K pathway alterations* for stage II-III
    - Footnote k revised: The rectum lies below ~~a virtual line~~ *the peritoneal reflection* from the sacral promontory to the upper edge of the symphysis as determined by MRI. (Also for REC-3, REC-3A, REC-A 1 of 5)
    - Footnote l modified: ~~FDA added a Black Boxed warning to the capecitabine label, recommending to test patients for genetic variants of DPYD prior to initiating capecitabine unless immediate treatment is necessary.~~ For information on DPYD testing and fluoropyrimidine-associated toxicity, see the NCCN Guidelines for Colon Cancer. (Also for REC-2A)
    - Footnote m added: Somatic PI3K pathway alterations include mutations in PI3KCA exon 9 and 20; other PIK3CA, PI3K3R1, and PTEN mutations; and deep deletions of PTEN. Aspirin should not be initiated until after recovery from surgery. For patients receiving adjuvant therapy and aspirin, aspirin can be given concurrently with adjuvant chemotherapy. (Also for REC-3A, REC-5, REC-6, REC-14)

REC-2A
- Workup, Rectal cancer with suspected or proven distant metastases
  - Bullet 6 and sub-bullets revised:
    - ~~Molecular testing~~ *Biomarker testing ordered as rapidly as possible following diagnosis*, including:
      - RAS and BRAF *V600E* mutations; HER2 *(ERBB2) overexpression/amplifications*~~;~~ MMR or MSI status (if not previously done)
      - Testing should be conducted as part of *multigene panel testing (MGPT)* ~~broad molecular profiling~~, which would identify rare and actionable mutations and fusions such as POLE/POLD1, RET, and NTRK *1/2/3*.
    - Biomarker terminology changes have been applied throughout
- Footnote n revised: Tissue- or blood-based *analysis using MGPT* ~~next-generation sequencing (NGS) panels~~ have the ability to pick up rare and actionable mutations and fusions.

REC-3
- Clinical stage, branches revised:
  - T1 *(favorable risk)*, N0
    - Branches for nonsurgical candidate vs surgical candidate removed
    - Primary treatment revised: Endoscopic submucosal dissection (ESD) or Transanal local excision~~, if appropriate~~ *or Transabdominal resection*

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Updates in Version 1.2026 of the NCCN Guidelines for Rectal Cancer:

REC-3 continued
- T1 *(high risk)*,T2, N0
  - Primary treatment option modified: Transabdominal resection *(preferred)*
- Footnote removed: T1–2, N0 should be based on assessment of pelvis MRI (preferred) or endorectal ultrasound.
- Footnote removed: Bolus fluorouracil/leucovorin/RT is an option for patients not able to tolerate capecitabine or infusional fluorouracil. (Also for REC-4, REC-5, REC-6, REC-8, REC-9, REC-11A, REC-14, REC-16, REC-D 1 of 2)
- Footnote r modified: ...the 5-year follow-up of the RAPIDO trial now indicates a statistically higher locoregional recurrence rate ~~(10%) in the experimental arm of~~ *with* short-course RT. ~~→ chemotherapy → surgery versus control arm (6%) of chemoRT → surgery → adjuvant chemotherapy.~~ (Also for REC-4, REC-6)
- Footnote s modified: In those patients who achieve a complete clinical response ~~with no evidence of residual disease on DRE, rectal MRI, and direct endoscopic evaluation~~, a "watch and wait," nonoperative ~~(chemotherapy and/or RT)~~ management approach may be considered in... (Also for REC-4, REC-6, REC-14)

REC-3A
- Text box added: For stage II and III, if somatic PI3K pathway alterations, start aspirin (unless contraindicated) 100–162 mg PO daily for 3 years after recovery from surgery (concurrent with chemotherapy if given). Therefore, molecular profiling should be done on all stage II and III CRC tumors. (Also for REC-5, REC-6, REC-14)
- Footnote removed: For stage II-III disease, if PIK3CA mutation, add aspirin 100-162 mg PO daily for 3 years following surgery. (Also for REC-5, REC-6, REC-14)

REC-5
- pT3, N0, M0, adjuvant treatment modified: ~~Consider~~ FOLFOX or CAPEOX alone or Observation
  - Footnote w added: The use of postoperative RT should be highly selective (eg, for incomplete mesorectal grade) due to severe toxicity concerns.

REC-8
- Footnote cc revised: Resection is preferred over locally ablative procedures *for liver metastases* ~~(eg, image-guided thermal ablation or stereotactic body RT [SBRT])~~. *For small liver lesions (≤3 cm), thermal ablation is equivalent to resection.* ~~However, these~~ *Other* local techniques can be considered for liver ~~or lung~~ oligometastases *when thermal ablation is not indicated* (REC-C and REC-E). ~~For small lesions (≤3 cm), thermal ablation is equivalent to resection.~~ (Also for REC-9, REC-12, REC-13, REC-16, REC-17)
- Footnote dd added: For lung metastases, surgery, thermal ablation, and stereotactic RT (SBRT) can be considered (REC-C and REC-E). (Also for REC-9, REC-12, REC-13, REC-16, REC-17)

REC-9
- Footnote removed and added to REC-F 1 of 17: An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines. (Also for REC-13)
- Footnote ee modified: ...The use of bevacizumab may interfere with wound healing. *Similarly, the combination of bevacizumab and SBRT has been associated with an increased risk of pulmonary hemorrhage, particularly for ultracentral tumors.*

REC-11
- Footnote mm modified: Determination of tumor gene status for RAS and BRAF mutations and HER2 *(ERBB2) overexpression/*amplifications~~~~ (individually or as part of tissue- or blood-based *MGPT* ~~next-generation sequencing [NGS] panel~~). See Principles of Pathologic Review (REC-B)- KRAS, NRAS, and BRAF Mutation Testing and MSI or MMR Testing. ~~NGS panels~~ *MGPT* has the ability to pick up rare and actionable mutations and fusions. (Also for REC-11A)

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Updates in Version 1.2026 of the NCCN Guidelines for Rectal Cancer:

REC-13
- pMMR/MSS Unresectable Metachronous Metastases
  - Initial Treatment, regimen modified to allow for additional options for chemotherapy: Encorafenib + (Cetuximab or Panitumumab) ± *FOLFIRI or FOLFOX or CAPEOX* (BRAF V600E mutation)
    - ◊ Footnote ss added: The addition of FOLFOX or CAPEOX can be considered for patients without residual neuropathy and good prior tolerance of chemotherapy.
  - Lapatinib removed as a treatment option (Also for REC-F)

REC-14
- dMMR/MSI-H or POLE/POLD1; Neoadjuvant/Definitive Immunotherapy
  - Checkpoint inhibitor immunotherapy options modified to include all checkpoint inhibitors on REC-F 5 of 17 as options
  - Additional branch added after Re-evaluate disease status every 2–3 mo: If consistent response, continue checkpoint inhibitor immunotherapy
  - Footnote removed: If no previous treatment with a checkpoint inhibitor. (Also for REC-16)

REC-16
- dMMR/MSI-H or POLE/POLD1; Neoadjuvant Treatment
  - Branch modified: ~~Consider holding radiation if complete response to neoadjuvant therapy or~~ *If no complete response, consider:...*
  - Footnote removed: Checkpoint inhibitor therapy options include: nivolumab ± ipilimumab, pembrolizumab, dostarlimab-gxly, cemiplimab-rwlc, retifanlimab-dlwr, toripalimabtpzi, or tislelizumab-jsgr. Combination PD-L1/CTLA-4 blockade has been associated with improved survival in unresectable or metastatic CRC, but also higher toxicity. (Also for REC-17)

REC-17
- dMMR/MSI-H or POLE/POLD1; Resectable Metachronous Metastases
  - 'Preferred' removed for FOLFOX and CAPEOX
  - Capecitabine or Fluorouracil/Leucovorin removed as treatment options
  - Initial Treatment, No previous immunotherapy modified: Checkpoint inhibitor immunotherapy *(preferred) (REC-F 5 of 17)*

REC-18
- New page added to algorithm for dMMR/MSI-H or POLE/POLD1 unresectable metastases with content from previous page in REC-F

REC-A 1 of 5
- Principles of Imaging; Initial Workup/Staging
  - Bullet 1 modified: Pelvis MRI with or without contrast or endorectal ultrasound (*only for T1 tumors* or if MRI is contraindicated)...
  - Bullet removed: Pelvis MRI or CT can be used for workup of synchronous metastatic disease.
  - Bullet 2, sub-bullet removed: Evaluate local extent of tumor or infiltration into surrounding structures.
  - Bullet 2, sub-bullet 4 modified: ...IV iodinated contrast material ~~is~~ *may* also *be* contraindicated, and...

REC-A 2 of 5
- Restaging
  - Bullet 2 modified: FDG-PET/CT is not *usually* indicated*; however, it may be necessary in case of equivocal additional imaging findings.*
- Follow-up/Surveillance
  - Bullet 2, sub-bullet 3 modified: FDG-PET/CT examination is *usually* not recommended.

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Updates in Version 1.2026 of the NCCN Guidelines for Rectal Cancer:

REC-A 2 of 5 continued
- Follow-up/Surveillance
  - Bullet 3, sub-bullet 3 added: If liver-directed therapy is contemplated, magnetic resonance with hepatobiliary agent should be considered to better delineate number and location of lesions.
  - Bullet 4 modified: FDG-PET/CT is *usually* not indicated with the exception of...
  - Bullet 5 and sub-bullets added: Therapy-dependent follow-up interval recommendations
    - ◊ For chemo/RT: Initial assessment 12 weeks after start of treatment, reassessment 16–20 weeks after start of treatment.
    - ◊ For total neoadjuvant therapy: Response assessment at 24 weeks after start of treatment, reassessment 34–38 weeks after the start of treatment.

REC-A 3 of 5
- Pelvic MRI Requirements: Criteria updated throughout the tables. (Also for Pelvic MRI Reporting before/after neoadjuvant therapy on REC-A 4 of 5)

REC-A 5 of 5
- References have been updated.

REC-B 1 of 10 through REC-B 10 of 10
- Principles of Pathologic and Molecular Review
  - This section has been extensively revised. Terminology and references have been updated throughout.

REC-C 4 of 9
- Principles of Surgery and Locoregional Therapies; Criteria for Resectability of Metastases and Locoregional Therapies Within Surgery
  - Text added: Local therapies for liver and lung oligometastases should be selected through multidisciplinary review based on technical feasibility, patient comorbidities, and disease distribution.
  - Liver, bullet 13 added: For liver metastases, in general, resection may be prioritized over local thermal ablation. However, for select patients with small lesions (≤3 cm), thermal ablation should be prioritized, when feasible, due to equivalent outcomes and reduced toxicity compared to surgery.34 SBRT can be considered in select cases where other local therapy options are not possible or in patients who prefer a noninvasive approach.
  - Lung, bullet 10 added: For lung metastases, surgery, thermal ablation, and SBRT can be considered based on multidisciplinary discussion and patient preference.

REC-C 8 of 9
- Reference added: van der Lei S, Puijk RS, Dijkstra M, et al. Thermal ablation versus surgical resection of small-size colorectal liver metastases (COLLISION): an international, randomised, controlled, phase 3 non-inferiority trial. Lancet Oncol 2025;26:187-199.

REC-E 1 of 3
- Principles of Radiation Therapy
  - General Principles
    - ◊ Bullet 2 added: Endorectal brachytherapy can be considered as a means of dose escalation to achieve organ preservation in select patients at institutions with experience with this treatment technique.
  - Treatment Information
    - ◊ Bullet 3 modified: ...consider use of hyperfractionated pelvic re-irradiation *or proton beam therapy* if retreatment is planned.

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Updates in Version 1.2026 of the NCCN Guidelines for Rectal Cancer:

REC-E 2 of 3
- RT Dosing
  - Sub-bullet 1 modified: 45–54 Gy in 25–30 fractions to the pelvis. *Alternatively, a radiation plan using a simultaneous integrated boost can be used to deliver 45 Gy to the pelvis and 50 Gy to the tumor bed in 25 fractions or 45 Gy to the pelvis and 54 Gy to the tumor bed in 27 fractions.*
  - Short-course RT, diamond added: Small bowel max point dose should be limited to D0.03 cc 26.5 Gy, and V20 Gy should be 50 cc for individual small bowel loops. Depending upon anatomy and clinical circumstances, it may be reasonable to accept a D0.03cc up to 28.75 Gy.

REC-E 3 of 3
- References have been updated.

REC-F
- Principles of Systemic Therapy for Advanced or Metastatic Disease
  - This section has been extensively revised. Treatment recommendations have been broken down by biomarker status and put into new tables.
    - Landing page added
    - Bullet list of principles added on REC-F 1 of 17
    - Dosing and references updated

REC-F 4 of 17
- KRAS WT/NRAS WT/BRAF V600E Mutation
  - Initial therapy, Preferred
    - Regimen added as a category 2A recommendation: FOLFIRI/Encorafenib + (Cetuximab or Panitumumab)
  - Initial therapy, Other Recommended
    - Regimen added as a category 2A recommendation: CAPEOX/Encorafenib + (Cetuximab or Panitumumab)

REC-F 5 of 17
- Checkpoint Inhibitor Therapy for Any Line of Therapy
  - The following agents have been added as category 2A recommendations:
    - PD-1
      - Penpulimab-kcqx
    - PD-L1
      - Atezolizumab
      - Avelumab
      - Cosibelimab-ipdl
      - Durvalumab
  - Asterisk modified: Patients should be followed closely for 10 weeks to assess for response. *Some of these agents have not been specifically studied in CRC but are included in this Guideline based on the presumed class effect from data demonstrated in other cancer types. See Discussion section for more information.*

REC-G 1 of 2
- Principles of Survivorship; Counseling Regarding Healthy Lifestyle and Wellness
  - Bullet 3 modified: Adopt a physically active lifestyle ~~(at least 30 minutes of moderateintensity activity on most days of the week)~~. *Discuss and facilitate options to achieve a goal of at least 150 minutes of moderate to vigorous physical activity per week.*
  - Bullet 7: Link added to NCCN Guidelines for Smoking Cessation

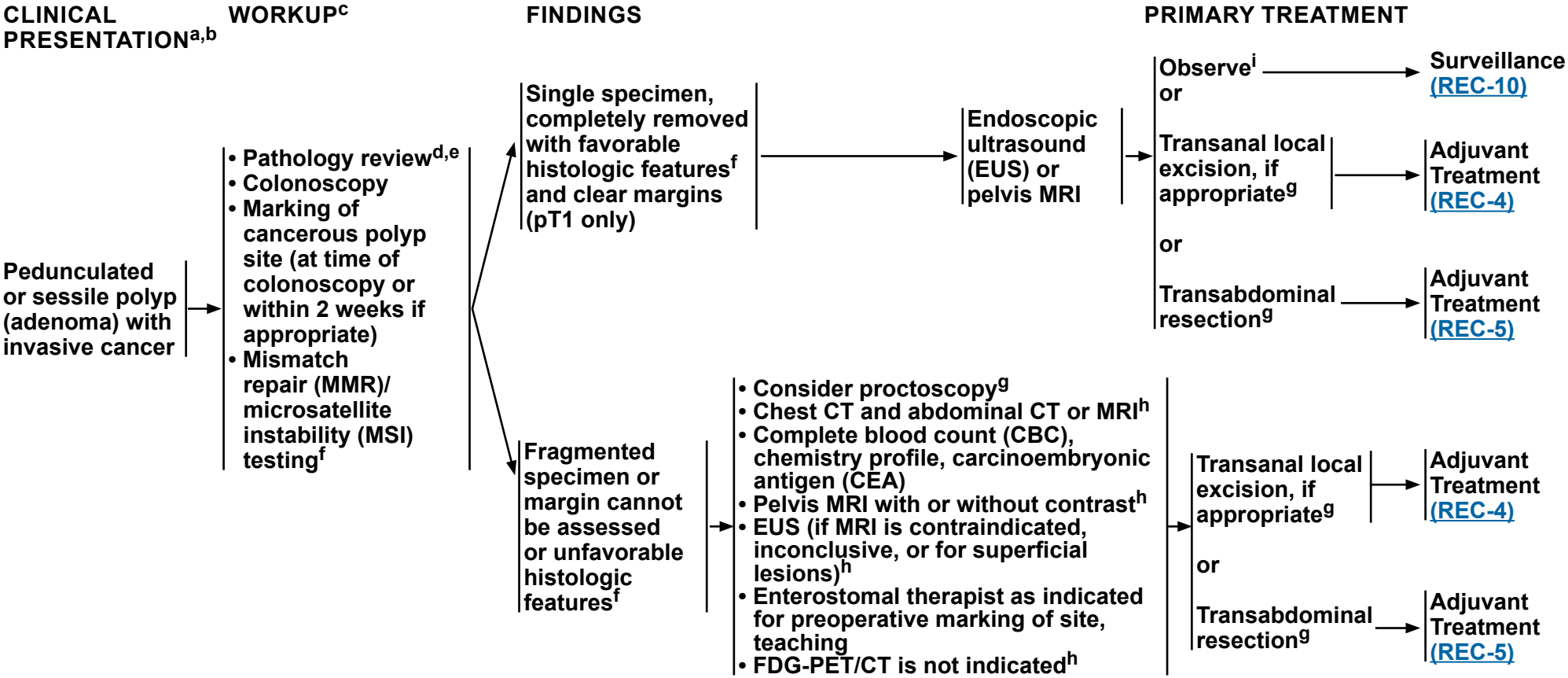
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[a] All patients with rectal cancer should be counseled for family history. Patients with suspected Lynch syndrome (LS), familial adenomatous polyposis (FAP), and attenuated FAP, see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.
[b] For melanoma histology, see the NCCN Guidelines for Melanoma: Cutaneous.
[c] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
[d] Confirm the presence of invasive cancer (pT1). pTis has no biological potential to metastasize.
[e] It has not been established if molecular markers (other than MSI-high [MSI-H]/ MMR deficient [dMMR]) are useful in treatment determination (predictive markers) and prognosis. Compton CC, et al. Arch Pathol Lab Med 2000;124:979-994.
[f] Principles of Pathologic Review (REC-B).
[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[i] Historical data have suggested that endoscopically resected sessile malignant polyps with submucosal invasion harbored higher risk of residual disease and/ or lymph node metastases than similar lesions that were of pedunculated morphology. More recently, evidence suggests that margin status and the presence or absence of unfavorable histologic features may be the most important predictors of residual disease/lymph node metastases regardless of morphologic type. See Principles of Pathologic Review (REC-B) - Endoscopically Removed Malignant Polyps.

Note: All recommendations are category 2A unless otherwise indicated.

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CLINICAL PRESENTATION[a,b]

WORKUP[l]

Rectal cancer without suspected or proven distant metastases[j,k] →

- **Biopsy**
- **MMR/MSI testing[f]**
- **Testing for somatic PI3K pathway alterations[m] for stage II–III**
- **Pathology review**
- **Colonoscopy**
- **Consider proctoscopy[g]**
- **Chest CT and abdominal CT or MRI[h]**
- **CBC, chemistry profile, CEA**
- **Pelvis MRI with or without contrast[h]**
- **Endorectal ultrasound (if MRI is contraindicated or inconclusive, or for superficial lesions)[h]**
- **Enterostomal therapist as indicated for preoperative marking of site, teaching**
- **FDG-PET/CT is not indicated[h]**
- **Multidisciplinary team evaluation, including formal surgical evaluation**
- **Fertility risk discussion/counseling in appropriate patients**

→ REC-3

Rectal cancer with suspected or proven distant metastases → REC-2A

[a] All patients with rectal cancer should be counseled for family history. Patients with suspected LS, FAP, and attenuated FAP, see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.
[b] For melanoma histology, see the NCCN Guidelines for Melanoma: Cutaneous.
[f] Principles of Pathologic Review (REC-B).
[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[j] For tools to aid optimal assessment and care of older adults with cancer, see the NCCN Guidelines for Older Adult Oncology.
[k] The rectum lies below the peritoneal reflection from the sacral promontory to the upper edge of the symphysis as determined by MRI.
[l] For information on *DPYD* testing and fluoropyrimidine-associated toxicity, see the NCCN Guidelines for Colon Cancer.
[m] Somatic PI3K pathway alterations include mutations in *PIK3CA* exon 9 and 20; other *PIK3CA, PIK3R1*, and *PTEN* mutations; and deep deletions of *PTEN*. Aspirin should not be initiated until after recovery from surgery. For patients receiving adjuvant therapy and aspirin, aspirin can be given concurrently with adjuvant chemotherapy.

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CLINICAL PRESENTATION[a,b]

WORKUP[l]

Rectal cancer with suspected or proven distant metastases[j] →

- **Colonoscopy**
- **Consider proctoscopy**
- **Chest CT and abdominal CT or MRI[h]**
- **Pelvis MRI with or without contrast[h]**
- **CBC, chemistry profile, CEA**
- **Biomarker testing ordered as rapidly as possible following diagnosis, including[f,n]:**
  - ***KRAS, NRAS,* and *BRAF* V600E mutations; HER2 (*ERBB2*) overexpression/amplification; MMR or MSI status (if not previously done)**
  - **Testing should be conducted as part of multigene panel testing (MGPT), which would identify rare and actionable mutations and gene fusions such as *POLE/POLD1*, *RET*, and *NTRK 1/2/3*.**
- **Biopsy, if clinically indicated**
- **Consider FDG-PET/CT (skull base to mid-thigh) if potentially surgically curable M1 disease in selected cases[h]**
  - **Consider MRI of liver for patients who have potentially resectable disease**
- **If potentially resectable, then multidisciplinary team evaluation, including a surgeon experienced in the resection of hepatobiliary or lung metastases**

→ **Proficient MMR (pMMR)/ microsatellite stable (MSS)** → **REC-7**

→ **MMR deficient (dMMR)/MSI-high (MSI-H) or *POLE/POLD1* mutation with ultra-hypermutated phenotype (eg, TMB >50 mut/Mb)** → **REC-15**

[a] All patients with rectal cancer should be counseled for family history. Patients with suspected LS, FAP, and attenuated FAP, see the NCCN Guidelines for Genetic/ Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.
[b] For melanoma histology, see the NCCN Guidelines for Melanoma: Cutaneous.
[f] Principles of Pathologic Review (REC-B).
[h] Principles of Imaging (REC-A).
[j] For tools to aid optimal assessment and care of older adults with cancer, see the NCCN Guidelines for Older Adult Oncology.
[l] For information on *DPYD* testing and fluoropyrimidine-associated toxicity, see the NCCN Guidelines for Colon Cancer.
[n] Tissue- or blood-based analysis using MGPT have the ability to pick up rare and actionable mutations and fusions.

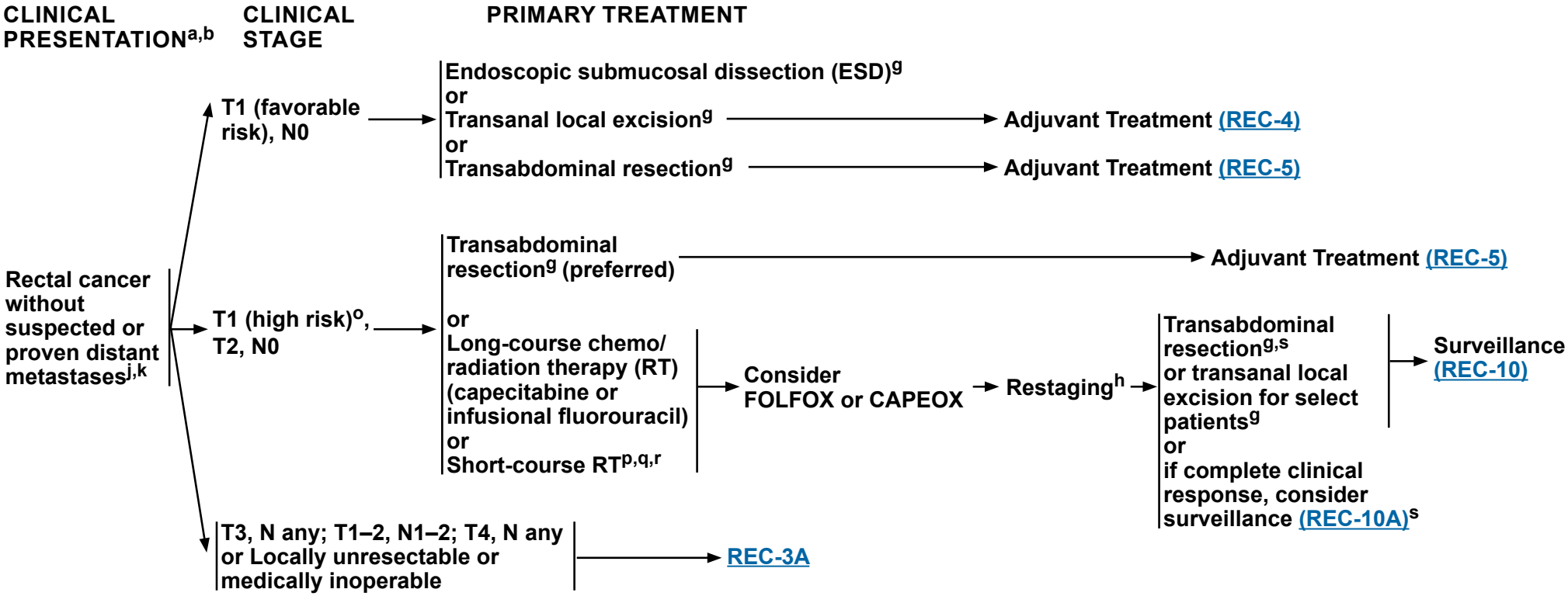
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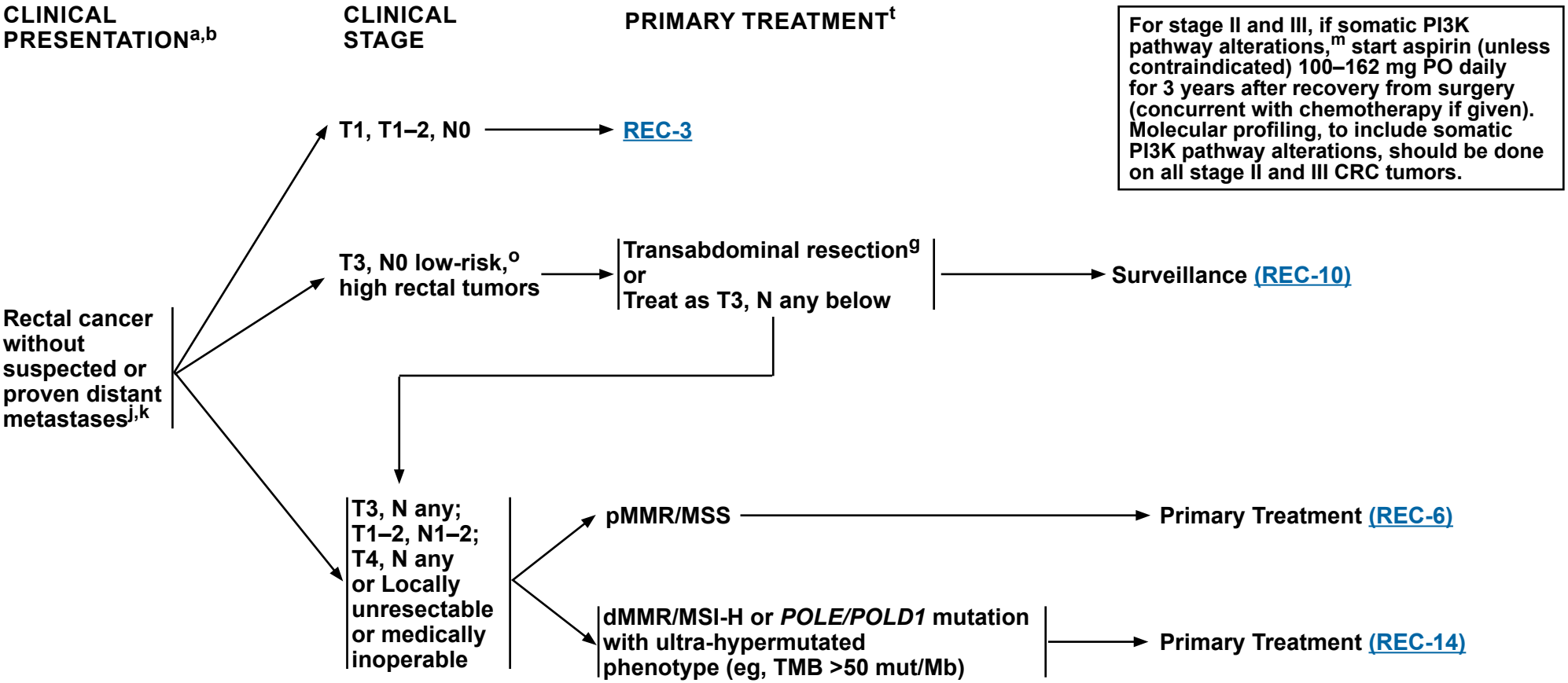


[a] All patients with rectal cancer should be counseled for family history. Patients with suspected LS, FAP, and attenuated FAP, see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.
[b] For melanoma histology, see the NCCN Guidelines for Melanoma: Cutaneous.
[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[j] For tools to aid optimal assessment and care of older adults with cancer, see the NCCN Guidelines for Older Adult Oncology.
[k] The rectum lies below the peritoneal reflection from the sacral promontory to the upper edge of the symphysis as determined by MRI.
[o] High-risk features include positive margins, lymphovascular invasion, poorly differentiated tumors, or sm3 invasion (submucosal invasion to the lower third of the submucosal level).
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[r] While short-course RT can be considered for preoperative radiation, for high-risk rectal cancer (clinical tumor stage cT4a or cT4b, extramural vascular invasion [EMVI], clinical nodal stage c-N2, involved mesorectal fascia [MRF], or enlarged lateral lymph nodes considered to be metastatic), the 5-year follow-up of the RAPIDO trial now indicates a statistically higher locoregional recurrence rate with short-course RT.
[s] In those patients who achieve a complete clinical response, a “watch and wait,” nonoperative management approach may be considered in centers with experienced multidisciplinary teams. The degree to which risk of local and/or distant recurrence may be increased relative to standard surgical resection has not yet been adequately characterized. Decisions for nonoperative management (NOM) should involve a careful discussion with the patient of their risk tolerance. See Principles of Nonoperative Management (REC-H).

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[a] All patients with rectal cancer should be counseled for family history. Patients with suspected LS, FAP, and attenuated FAP, see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.

[b] For melanoma histology, see the NCCN Guidelines for Melanoma: Cutaneous.

[g] Principles of Surgery and Locoregional Therapies (REC-C).

[j] For tools to aid optimal assessment and care of older adults with cancer, see the NCCN Guidelines for Older Adult Oncology.

[k] The rectum lies below the peritoneal reflection from the sacral promontory to the upper edge of the symphysis as determined by MRI.

[m] Somatic PI3K pathway alterations include mutations in *PIK3CA* exon 9 and 20; other *PIK3CA, PIK3R1*, and *PTEN* mutations; and deep deletions of *PTEN*. Aspirin should not be initiated until after recovery from surgery. For patients receiving adjuvant therapy and aspirin, aspirin can be given concurrently with adjuvant chemotherapy.

[o] High-risk features include positive margins, lymphovascular invasion, poorly differentiated tumors, or sm3 invasion (submucosal invasion to the lower third of the submucosal level).

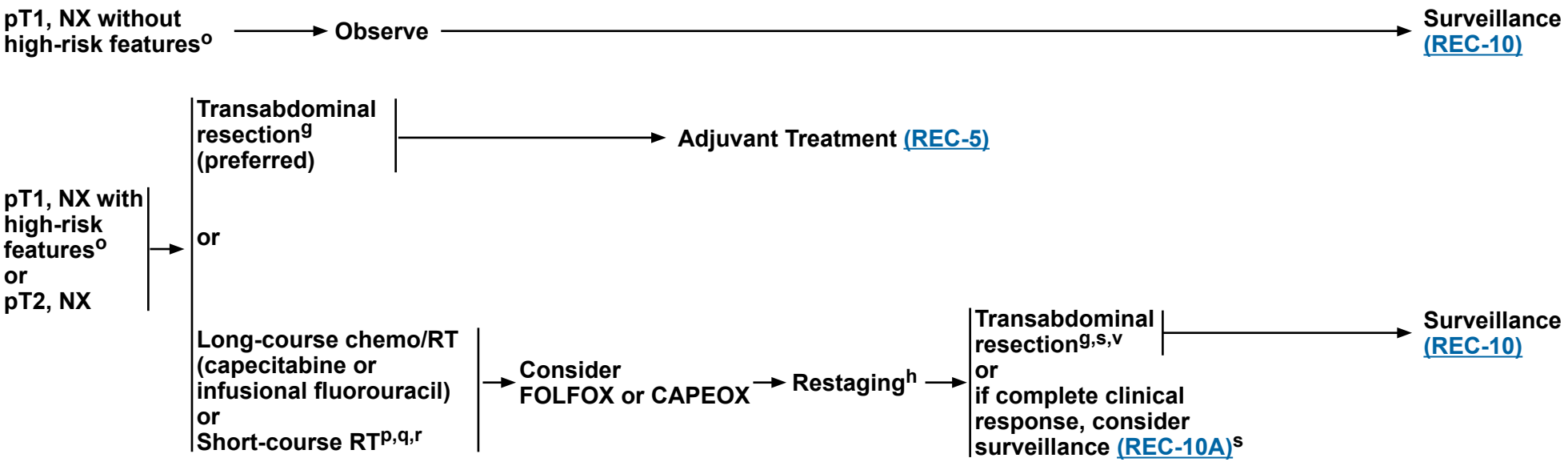
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PATHOLOGIC FINDINGS AFTER TRANSANAL LOCAL EXCISION FOR T1, N0

ADJUVANT TREATMENT[h,p,t,u] (UP TO 6 MO PERIOPERATIVE TREATMENT)



[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[o] High-risk features include positive margins, lymphovascular invasion, poorly differentiated tumors, or sm3 invasion (submucosal invasion to the lower third of the submucosal level).
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[r] While short-course RT can be considered for preoperative radiation, for high-risk rectal cancer (clinical tumor stage cT4a or cT4b, EMVI, clinical nodal stage cN2, involved MRF, or enlarged lateral lymph nodes considered to be metastatic), the 5-year follow-up of the RAPIDO trial now indicates a statistically higher locoregional recurrence rate with short-course RT.
[s] In those patients who achieve a complete clinical response, a "watch and wait," nonoperative management approach may be considered in centers with experienced multidisciplinary teams. The degree to which risk of local and/or distant recurrence may be increased relative to standard surgical resection has not yet been adequately characterized. Decisions for NOM should involve a careful discussion with the patient of their risk tolerance. See Principles of Nonoperative Management (REC-H).
[t] Principles of Perioperative Therapy (REC-D).
[u] Circulating tumor DNA (ctDNA) is a prognostic marker; however, there is currently insufficient evidence to recommend routine use of ctDNA assays outside of a clinical trial. De-escalation of care and treatment decision-making are not recommended based on ctDNA results. Participation in clinical trials is encouraged.
[v] For select patients who may be candidates for intraoperative RT (IORT), see Principles of Radiation Therapy (REC-E).

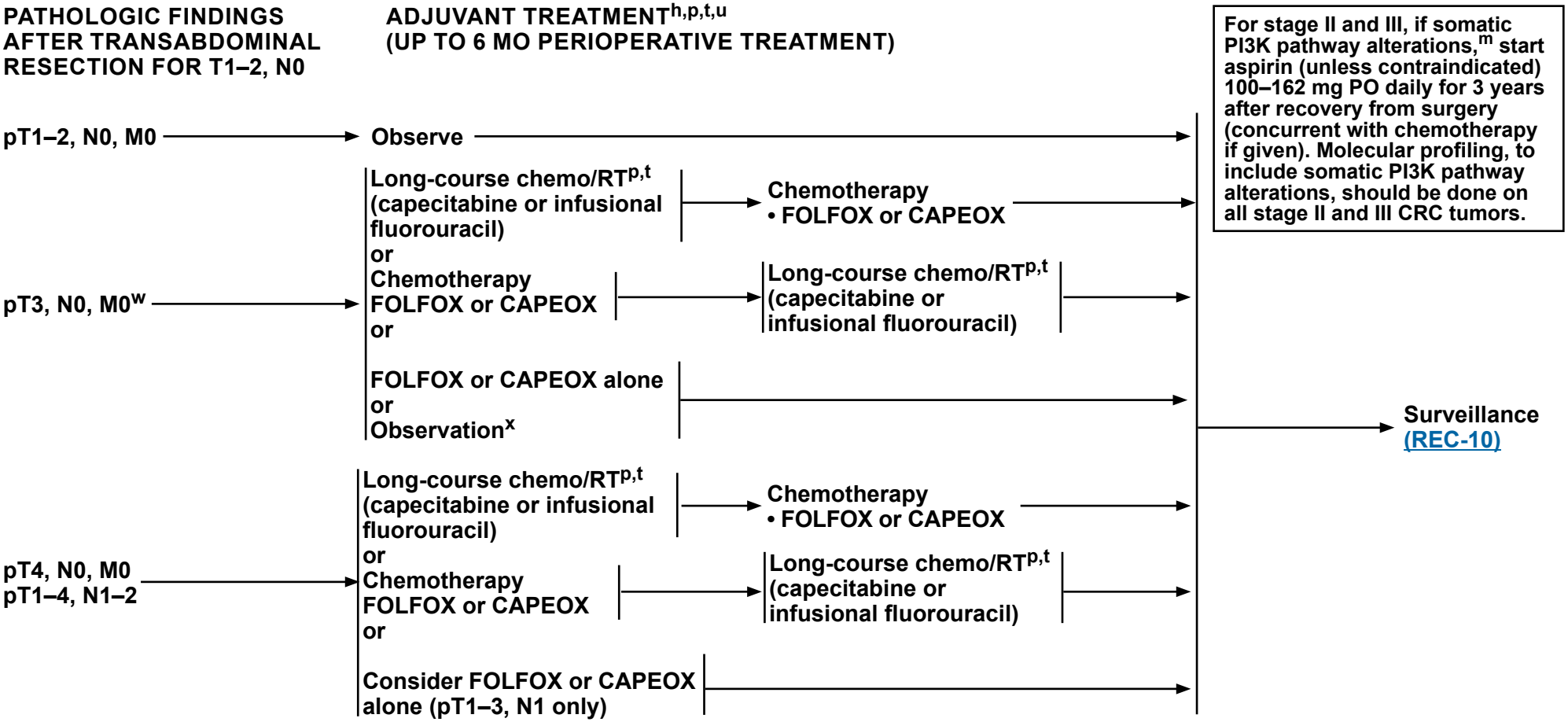
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h Principles of Imaging (REC-A).
m Somatic PI3K pathway alterations include mutations in *PIK3CA* exon 9 and 20; other *PIK3CA, PIK3R1*, and *PTEN* mutations; and deep deletions of *PTEN*. Aspirin should not be initiated until after recovery from surgery. For patients receiving adjuvant therapy and aspirin, aspirin can be given concurrently with adjuvant chemotherapy.
p Principles of Radiation Therapy (REC-E).
t Principles of Perioperative Therapy (REC-D).

u ctDNA is a prognostic marker; however, there is currently insufficient evidence to recommend routine use of ctDNA assays outside of a clinical trial. De-escalation of care and treatment decision-making are not recommended based on ctDNA results. Participation in clinical trials is encouraged.
w The use of postoperative RT should be highly selective (eg, for incomplete mesorectal grade) due to severe toxicity concerns.
x Observation following transabdominal resection can be considered in patients with pT3N0 rectal cancer if the tumor was well-differentiated or moderately well-differentiated carcinoma invading <2 mm into the mesorectum, without lymphatic or venous vessel involvement and was located in the upper rectum. Willett CG, et al. Dis Colon Rectum 1999;42:167-173.

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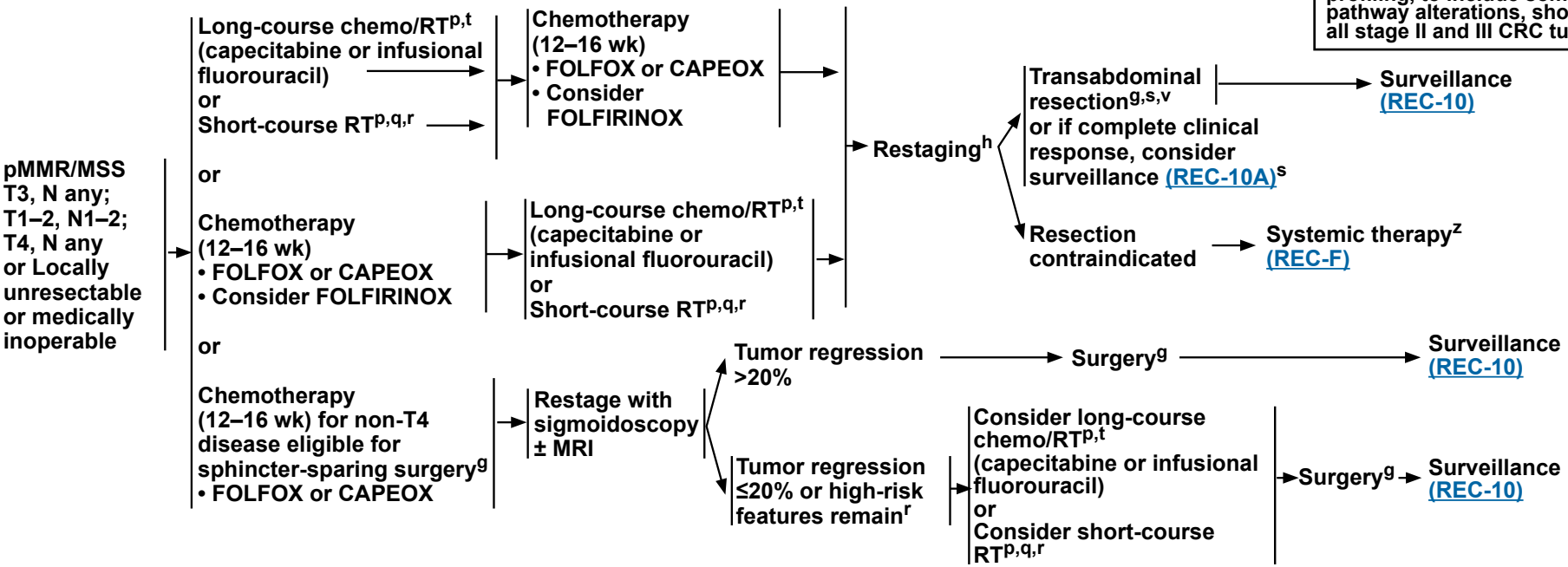
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CLINICAL STAGE

TOTAL NEOADJUVANT THERAPY[y]

PRIMARY TREATMENT

For stage II and III, if somatic PI3K pathway alterations,[m] start aspirin (unless contraindicated) 100–162 mg PO daily for 3 years after recovery from surgery (concurrent with chemotherapy if given). Molecular profiling, to include somatic PI3K pathway alterations, should be done on all stage II and III CRC tumors.



[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[m] Somatic PI3K pathway alterations include mutations in *PIK3CA* exon 9 and 20; other *PIK3CA, PIK3R1*, and *PTEN* mutations; and deep deletions of *PTEN*. Aspirin should not be initiated until after recovery from surgery. For patients receiving adjuvant therapy and aspirin, aspirin can be given concurrently with adjuvant chemotherapy.
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[r] While short-course RT can be considered for preoperative radiation, for high-risk rectal cancer (clinical tumor stage cT4a or cT4b, EMVI, clinical nodal stage cN2, involved MRF, or enlarged lateral lymph nodes considered to be metastatic), the 5-year follow-up of the RAPIDO trial now indicates a statistically higher locoregional recurrence rate with short-course RT.
[s] In those patients who achieve a complete clinical response, a "watch and wait," nonoperative management approach may be considered in centers with experienced multidisciplinary teams. The degree to which risk of local and/or distant recurrence may be increased relative to standard surgical resection has not yet been adequately characterized. Decisions for NOM should involve a careful discussion with the patient of their risk tolerance. See Principles of Nonoperative Management (REC-H).
[t] Principles of Perioperative Therapy (REC-D).
[v] For select patients who may be candidates for IORT, see Principles of Radiation Therapy (REC-E).
[y] In select cases (eg, a patient who is not a candidate for intensive therapy) neoadjuvant therapy with chemo/RT or RT alone may be considered prior to surgery.
[z] FOLFIRINOX is not recommended in this setting.

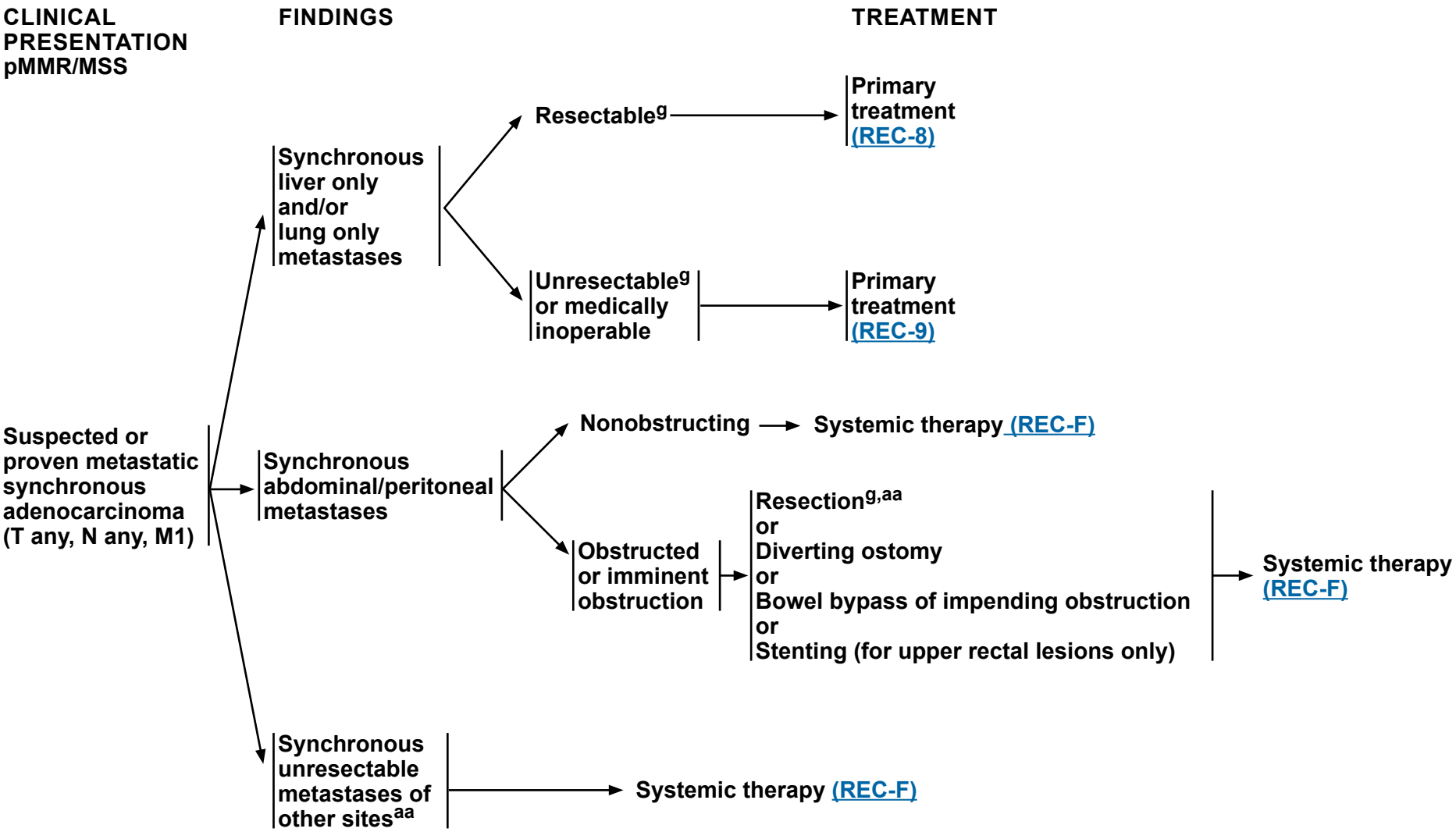
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[g] Principles of Surgery and Locoregional Therapies (REC-C).
[aa] Consider resection only if imminent risk of obstruction, significant bleeding, perforation, or other significant tumor-related symptoms.

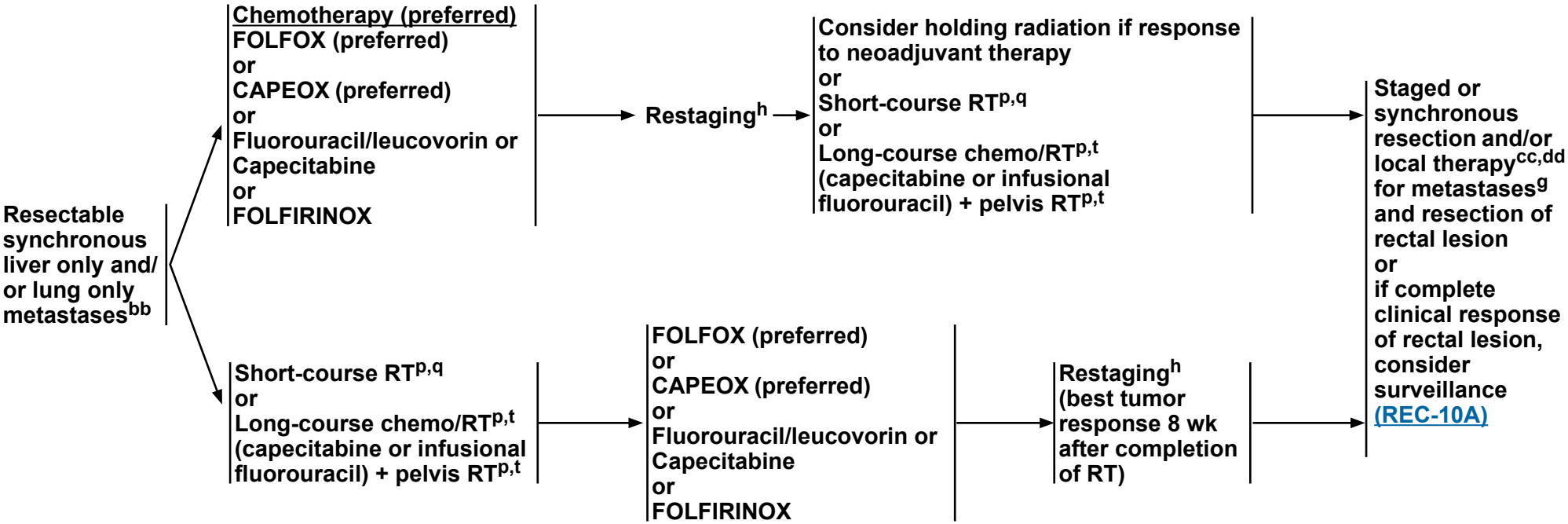
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FINDINGS
pMMR/MSS

NEOADJUVANT TREATMENT



[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[t] Principles of Perioperative Therapy (REC-D).
[bb] If obstructing lesion, consider diversion or resection (REC-7).

[cc] Resection is preferred over locally ablative procedures for liver metastases. For small liver lesions (≤3 cm), thermal ablation is equivalent to resection. Other local techniques can be considered for liver oligometastases when thermal ablation is not indicated (REC-C and REC-E).
[dd] For lung metastases, surgery, thermal ablation, and stereotactic RT (SBRT) can be considered (REC-C and REC-E).

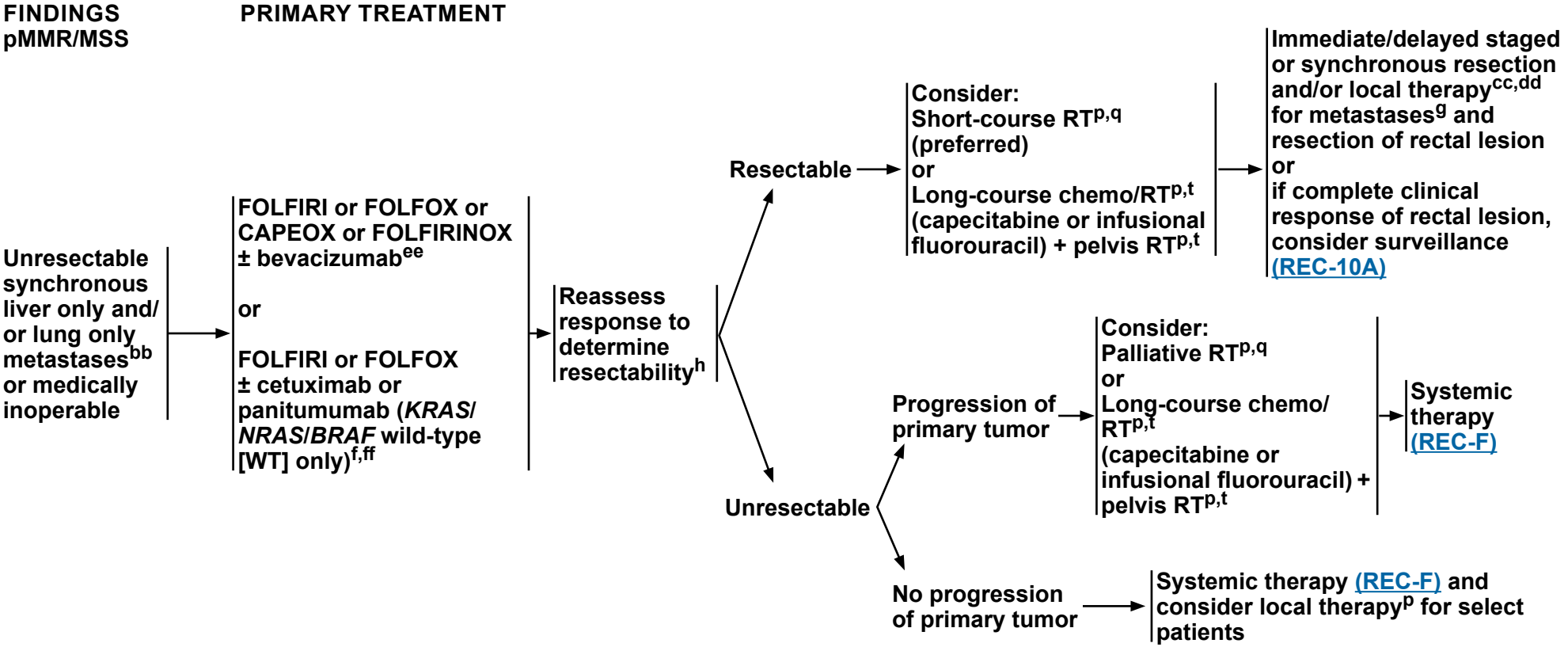
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[f] Principles of Pathologic Review (REC-B).
[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[t] Principles of Perioperative Therapy (REC-D).
[bb] If obstructing lesion, consider diversion or resection (REC-7).
[cc] Resection is preferred over locally ablative procedures for liver metastases. For small liver lesions (≤3 cm), thermal ablation is equivalent to resection. Other local techniques can be considered for liver oligometastases when thermal ablation is not indicated (REC-C and REC-E).
[dd] For lung metastases, surgery, thermal ablation, and SBRT can be considered (REC-C and REC-E).
[ee] There should be at least a 6-week interval between the last dose of bevacizumab and elective surgery, and re-initiation of bevacizumab should be delayed at least 6 to 8 weeks postoperatively. There is an increased risk of stroke and other arterial events, especially in those aged ≥65 years. The use of bevacizumab may interfere with wound healing. Similarly, the combination of bevacizumab and SBRT has been associated with an increased risk of pulmonary hemorrhage, particularly for ultracentral tumors.
[ff] Patients with *BRAF* mutations other than V600E may be considered for anti-EGFR therapy.

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SURVEILLANCE FOLLOWING OPERATIVE MANAGEMENT[h]

Low-risk polyps removed by polypectomy

- Physical exam and proctoscopy every 3–6 mo for the first 2 y
- Colonoscopy[a] at 1 y after polypectomy

Transanal local excision only

- Proctoscopy (with EUS or MRI with contrast) every 3–6 mo for the first 2 y, then every 6 mo for a total of 5 y
- Colonoscopy[a] at 1 y after surgery
  - If advanced adenoma, repeat in 1 y
  - If no advanced adenoma,[hh] repeat in 3 y, then every 5 y[ii]

Stage I with full surgical staging

- Colonoscopy[a] at 1 y after surgery
  - If advanced adenoma, repeat in 1 y
  - If no advanced adenoma,[hh] repeat in 3 y, then every 5 y[ii]

Stage II–IV[gg]

- History and physical examination every 3–6 mo for 2 y, then every 6 mo for a total of 5 y
- CEA[jj] every 3–6 mo for 2 y, then every 6 mo for a total of 5 y
- Chest/abdomen/pelvis (C/A/P) CT
  - Stage II, III: every 6–12 mo (category 2B for frequency <12 mo) for a total of 5 y
  - Stage IV: every 3–6 mo (category 2B for frequency <6 mo) x 2 y, then every 6–12 mo for a total of 5 y
- Colonoscopy[a,kk] in 1 y after surgery except if no complete preoperative colonoscopy, then colonoscopy in 3–6 mo
  - If advanced adenoma, repeat in 1 y
  - If no advanced adenoma,[hh] repeat in 3 y, then every 5 y[ii]
- FDG-PET/CT is not recommended
- Principles of Survivorship (REC-G)

→ Serial CEA elevation or documented recurrence → Workup and Treatment (REC-11)

For surveillance following ESD, see REC-C 6 of 9

[a] All patients with rectal cancer should be counseled for family history. Patients with suspected LS, FAP, and attenuated FAP, see the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.
[h] Principles of Imaging (REC-A).
[gg] ctDNA is not recommended for surveillance.
[hh] Villous polyp, polyp >1 cm, or high-grade dysplasia.
[ii] Kahi CJ, et al. Gastroenterology 2016;150:758-768.
[jj] If patient is a potential candidate for resection of isolated metastasis.
[kk] In patients with stage IV disease managed nonoperatively with complete clinical response, initiate colonoscopy surveillance from first documentation of complete response.

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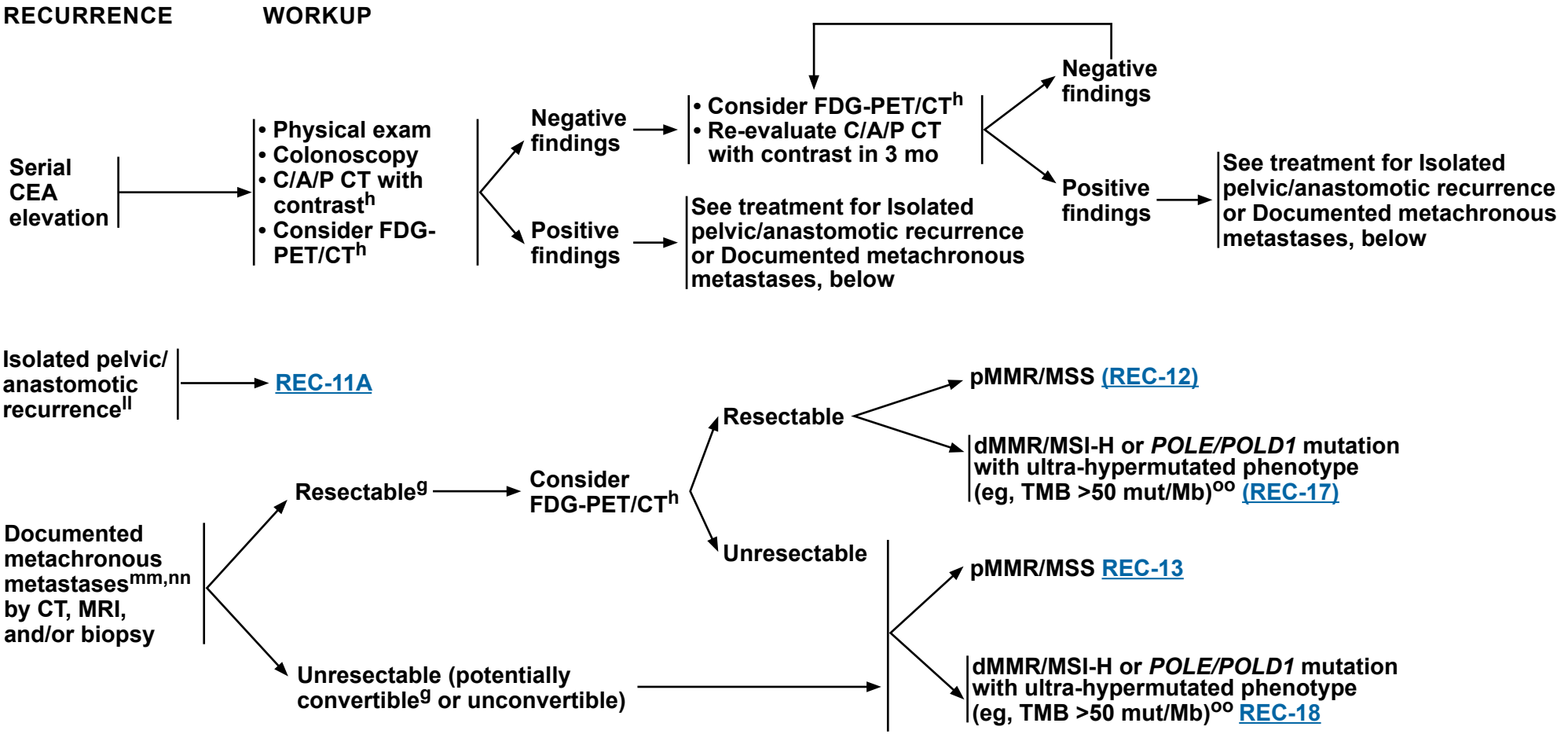
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SURVEILLANCE FOLLOWING NONOPERATIVE MANAGEMENT

- **History and physical examination every 3–6 months for 2 years and then every 6 months for a total of 5 years**
- **CEA every 3–6 months for 2 years, then every 6 months for a total of 5 years**
- **Digital rectal examination (DRE) and proctoscopy or flexible sigmoidoscopy every 3–4 months for 2 years, then every 6 months for a total of 5 years**
- **MRI rectum every 6 months for up to 3 years**
- **CT chest/abdomen every 6–12 months for a total of 5 years, CT pelvis to be included once no longer doing MRI**
- **Colonoscopy at 1 year following completion of therapy**
  - **If advanced adenoma, repeat in 1 year**
  - **If no advanced adenoma, repeat in 3 years, then every 5 years**
- **Principles of Nonoperative Management (REC-H)**

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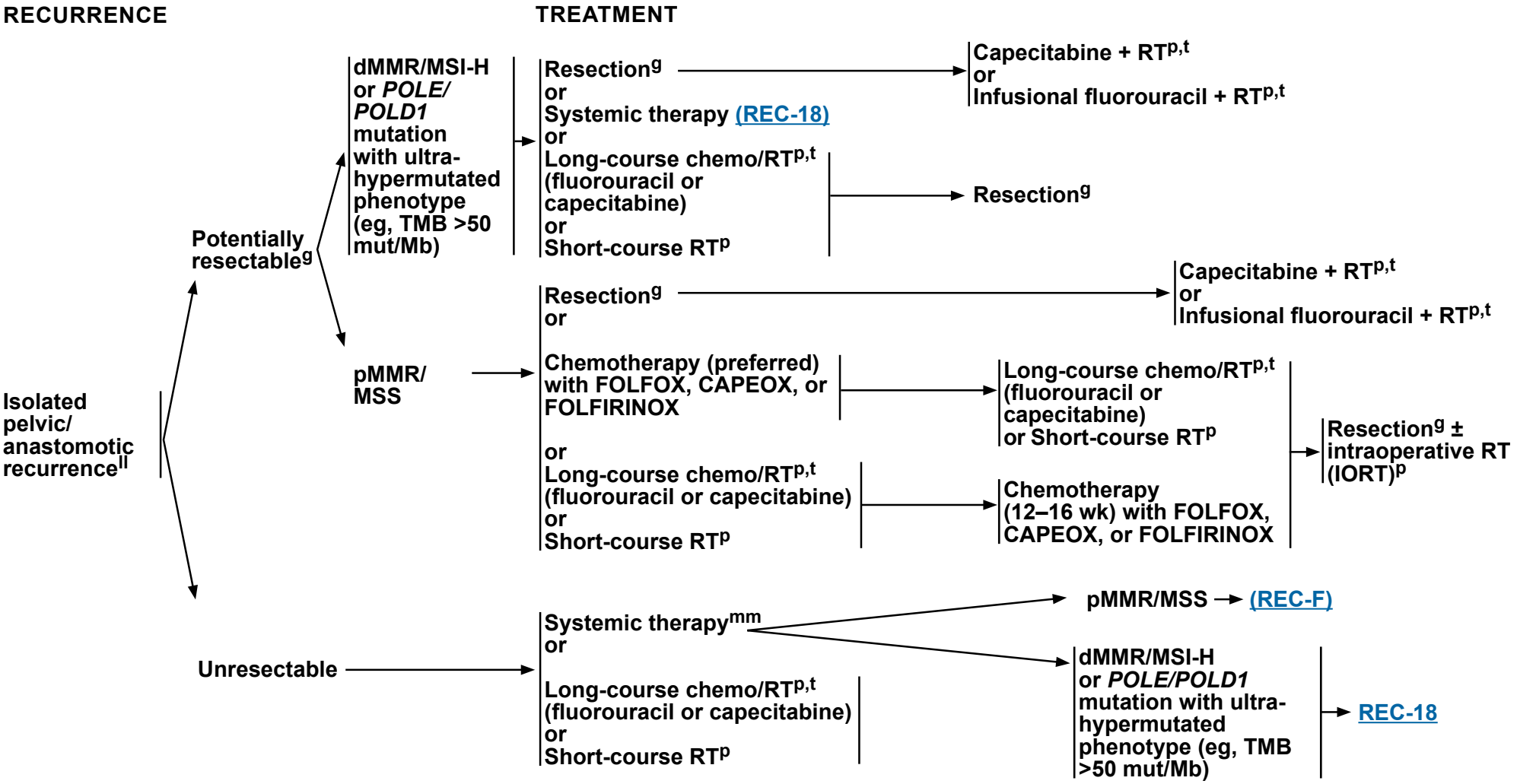


[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[ll] If previous RT given (short course or chemoradiation), see Principles of Radiation Therapy (REC-E) for further guidance.
[mm] Determination of tumor gene status for *KRAS*, *NRAS,* and *BRAF* mutations and HER2 *(ERBB2)* overexpression/amplification (individually or as part of tissue- or blood-based MGPT). See Principles of Pathologic Review (REC-B)- *KRAS*, *NRAS*, and *BRAF* Mutation Testing and MSI or MMR Testing. MGPT has the ability to pick up rare and actionable mutations and fusions.
[nn] Patients should be evaluated by a multidisciplinary team including surgical consultation for patients with potentially resectable disease.
[oo] Patients who are not candidates for immunotherapy should be treated as recommended for pMMR/MSS disease. See NCCN Guidelines for Management of Immune Checkpoint Inhibitor-Related Toxicities.

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[g] Principles of Surgery and Locoregional Therapies (REC-C).
[p] Principles of Radiation Therapy (REC-E).
[t] Principles of Perioperative Therapy (REC-D).
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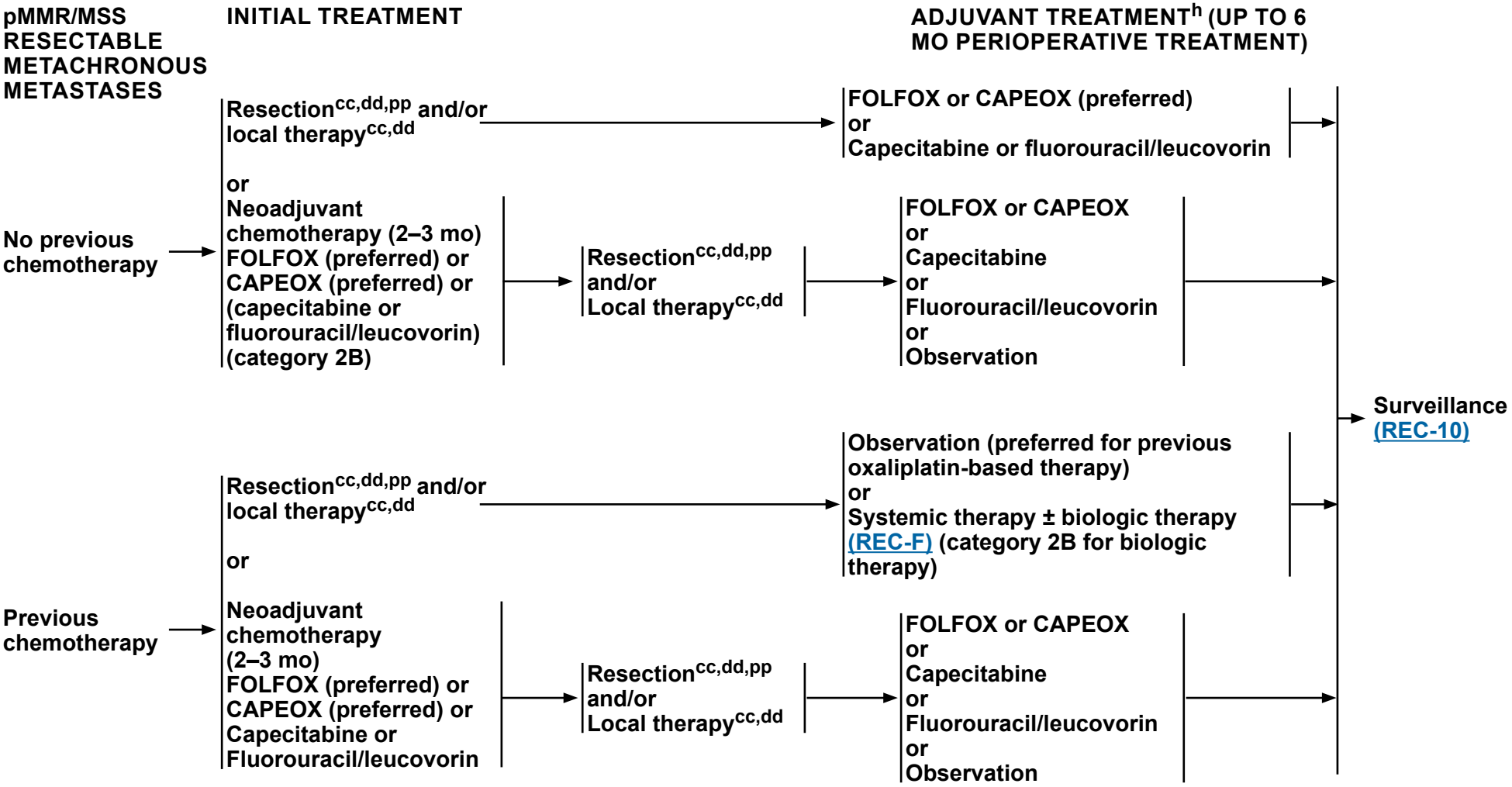
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[h] Principles of Imaging (REC-A).
[cc] Resection is preferred over locally ablative procedures for liver metastases. For small liver lesions (≤3 cm), thermal ablation is equivalent to resection. Other local techniques can be considered for liver oligometastases when thermal ablation is not indicated (REC-C and REC-E).
[dd] For lung metastases, surgery, thermal ablation, and SBRT can be considered (REC-C and REC-E).
[pp] Hepatic artery infusion ± systemic chemotherapy (vascular endothelial growth factor [VEGF] inhibitor contraindicated) (category 2B) is also an option at institutions with experience in both the surgical and medical oncologic aspects of this procedure.

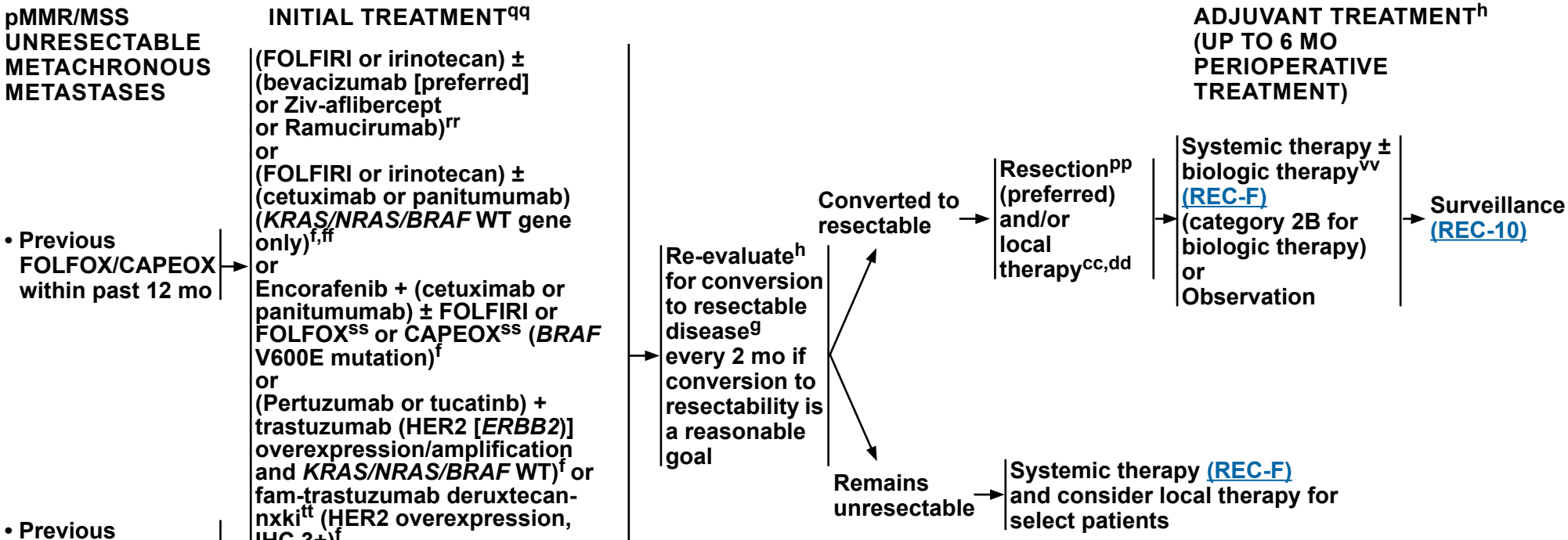
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[dd] For lung metastases, surgery, thermal ablation, and SBRT can be considered (REC-C and REC-E).
[ff] Patients with *BRAF* mutations other than V600E may be considered for anti-EGFR therapy.
[pp] Hepatic artery infusion ± systemic chemotherapy (VEGF inhibitor contraindicated) (category 2B) is also an option at institutions with experience in both the surgical and medical oncologic aspects of this procedure.
[qq] For infection risk, monitoring, and prophylaxis recommendations for targeted therapies, see INF-A in the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections.
[rr] Bevacizumab is the preferred anti-angiogenic agent based on toxicity and/or cost.
[ss] The addition of FOLFOX or CAPEOX can be considered for patients without residual neuropathy and good prior tolerance of chemotherapy.
[tt] Some activity was seen after previous treatment with a HER2-targeted regimen. May not be indicated in patients with underlying lung issues due to lung toxicity (drug-related deaths from interstitial lung disease were reported in 3.5% of patients on the DESTINY-CRC01 trial).
[uu] If patient is unable to tolerate EGFR inhibitor due to toxicity, single-agent adagrasib or sotorasib can be considered.
[vv] Biologic therapy is only appropriate for continuation of favorable response from conversion therapy.

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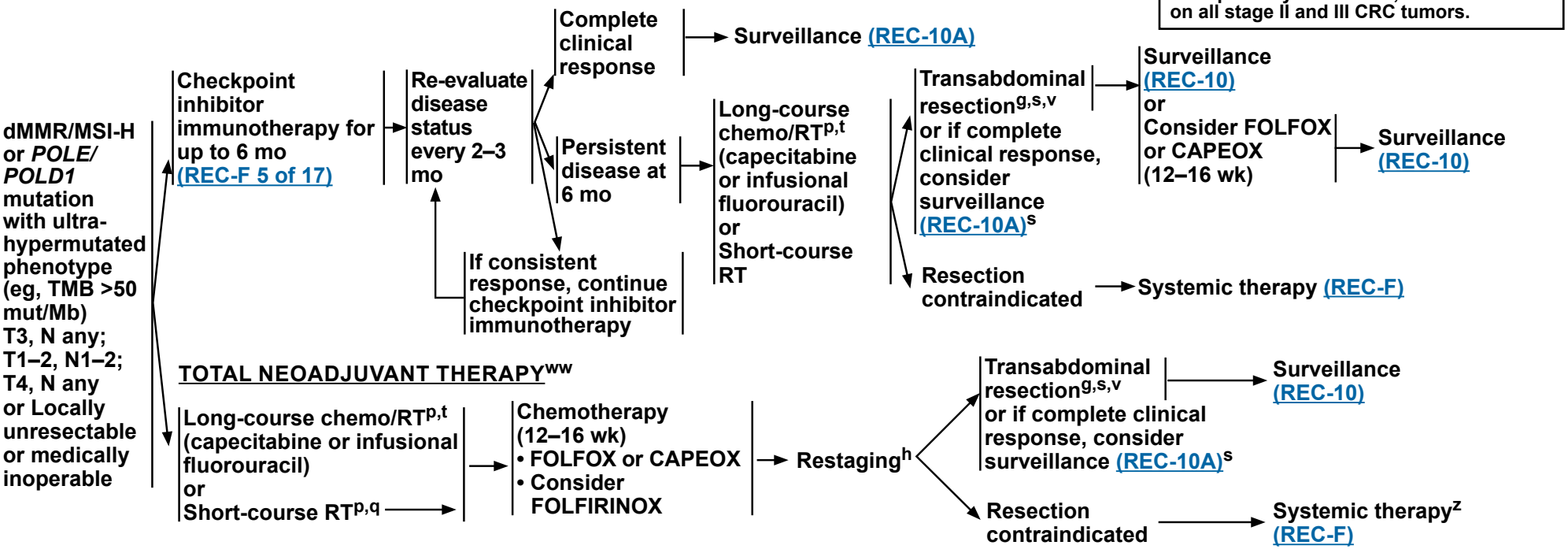
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CLINICAL STAGE

NEOADJUVANT/DEFINITIVE IMMUNOTHERAPY (PREFERRED)

For stage II and III, if somatic PI3K pathway alterations,[m] start aspirin (unless contraindicated) 100–162 mg PO daily for 3 years after recovery from surgery (concurrent with chemotherapy if given). Molecular profiling, to include somatic PI3K pathway alterations, should be done on all stage II and III CRC tumors.

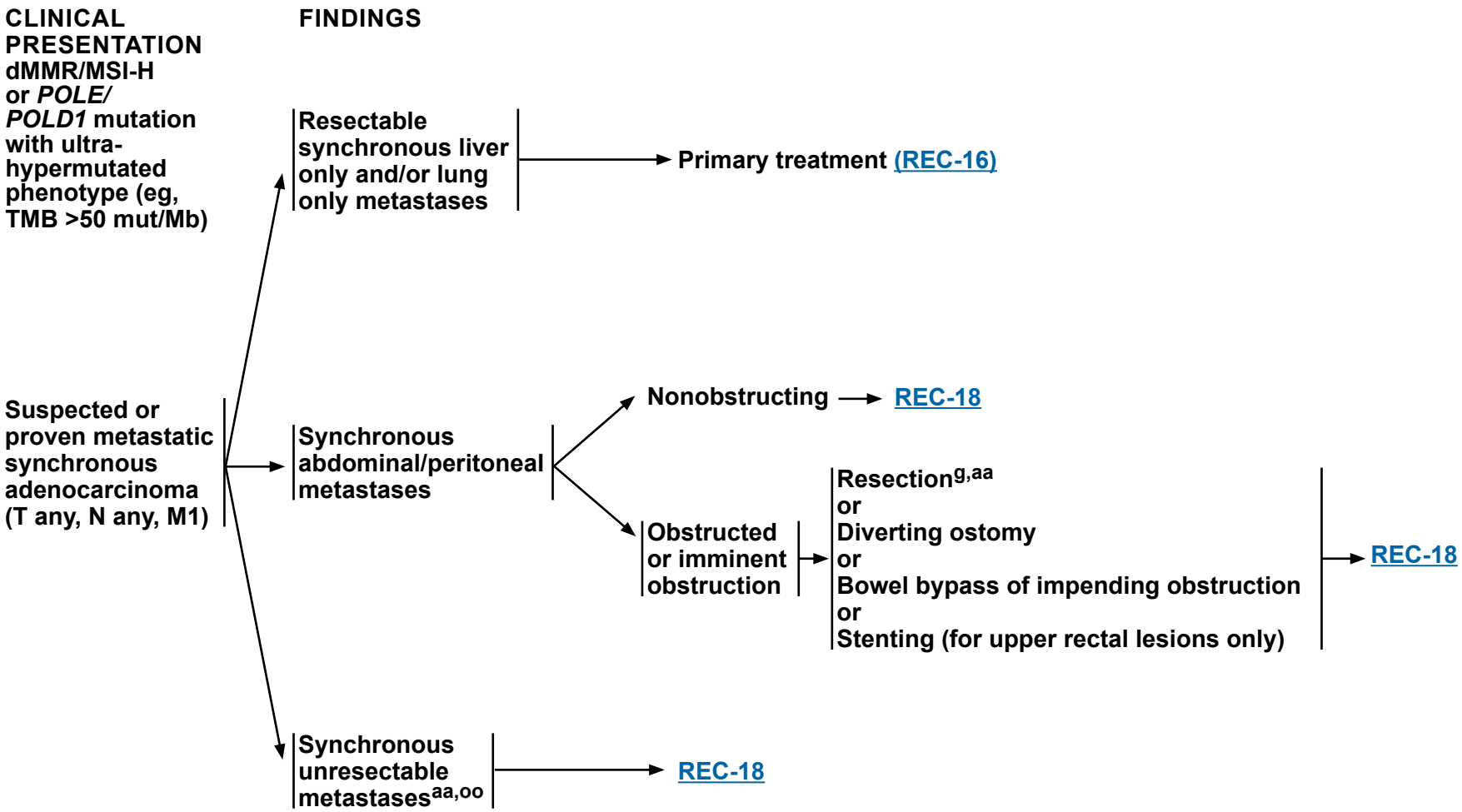


[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[m] Somatic PI3K pathway alterations include mutations in *PIK3CA* exon 9 and 20; other *PIK3CA, PIK3R1*, and *PTEN* mutations; and deep deletions of *PTEN*. Aspirin should not be initiated until after recovery from surgery. For patients receiving adjuvant therapy and aspirin, aspirin can be given concurrently with adjuvant chemotherapy.
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[s] In those patients who achieve a complete clinical response, a "watch and wait," nonoperative management approach may be considered in centers with experienced multidisciplinary teams. The degree to which risk of local and/or distant recurrence may be increased relative to standard surgical resection has not yet been adequately characterized. Decisions for NOM should involve a careful discussion with the patient of their risk tolerance. See Principles of Nonoperative Management (REC-H).
[t] Principles of Perioperative Therapy (REC-D).
[v] For select patients who may be candidates for IORT, see Principles of Radiation Therapy (REC-E).
[z] FOLFIRINOX is not recommended in this setting.
[ww] In select cases (eg, a patient who is not a candidate for intensive therapy) neoadjuvant therapy with chemo/RT or RT alone may be considered prior to surgery.

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[g] Principles of Surgery and Locoregional Therapies (REC-C).
[aa] Consider resection only if imminent risk of obstruction, significant bleeding, perforation, or other significant tumor-related symptoms.
[oo] Patients who are not candidates for immunotherapy should be treated as recommended for pMMR/MSS disease. See NCCN Guidelines for Management of Immune Checkpoint Inhibitor-Related Toxicities.

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FINDINGS

dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype (eg, TMB >50 mut/Mb)

NEOADJUVANT TREATMENT

Resectable synchronous liver only and/or lung only metastases[bb] → **Checkpoint inhibitor immunotherapy (preferred)[xx,yy]** (REC-F 5 of 17) → **Restaging[h]** → **If no complete response, consider:**
Short-course RT[p,q] (preferred)
or
Long-course chemo/RT[p,t] (capecitabine or infusional fluorouracil) →

Staged or synchronous resection and/or local therapy[cc,dd] for metastases[g] and resection of rectal lesion

or

Surveillance (REC-10A) if complete response to immunotherapy

[g] Principles of Surgery and Locoregional Therapies (REC-C).
[h] Principles of Imaging (REC-A).
[p] Principles of Radiation Therapy (REC-E).
[q] Evaluation for short-course RT should be in a multidisciplinary setting, with a discussion of the need for downstaging and the possibility of long-term toxicity.
[t] Principles of Perioperative Therapy (REC-D).
[bb] If obstructing lesion, consider diversion or resection (REC-7).

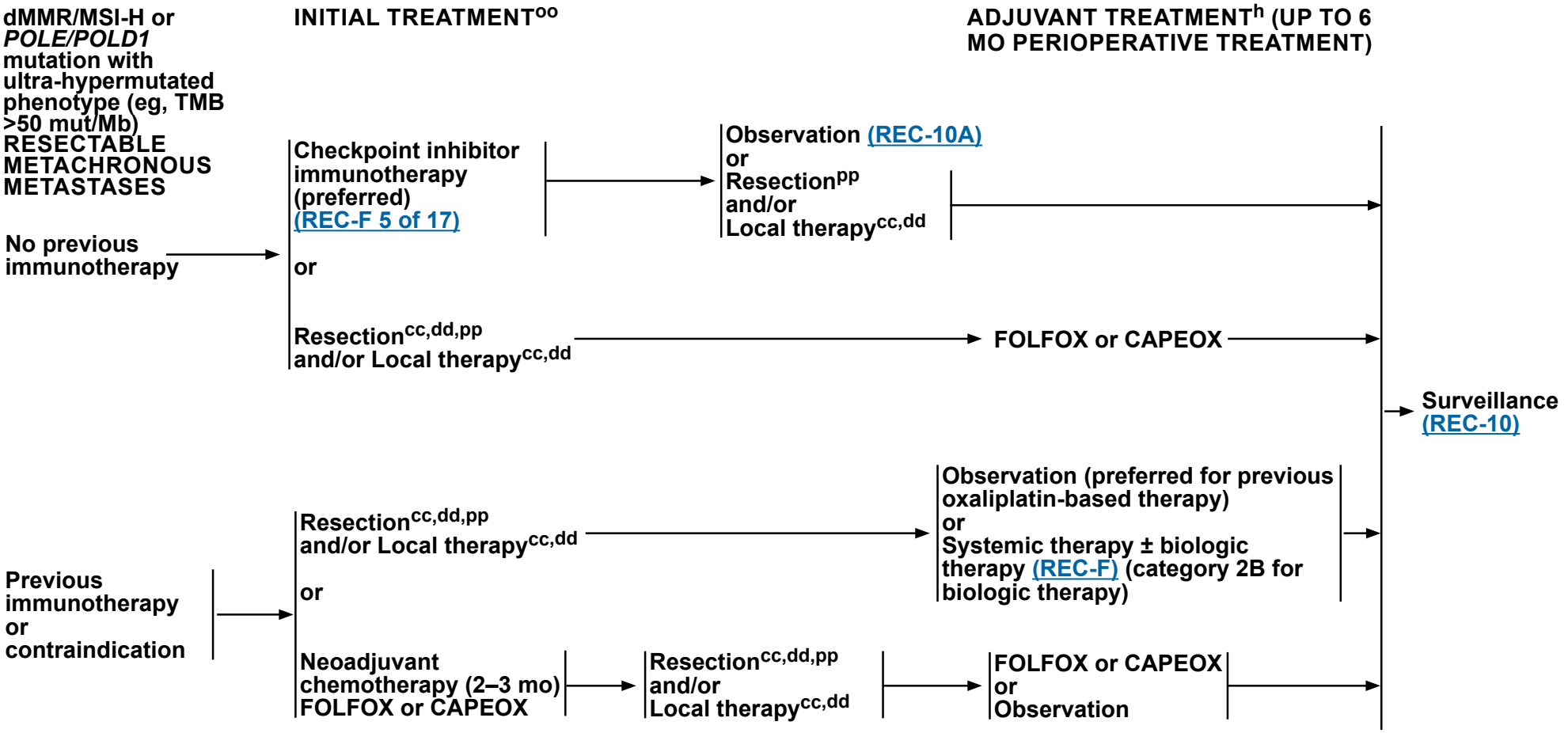
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[dd] For lung metastases, surgery, thermal ablation, and SBRT can be considered (REC-C and REC-E).
[xx] Data are limited and the risk of early progression may be higher than with chemotherapy. Andre T, et al. N Engl J Med 2020;383:2207-2218.
[yy] For patients with a contraindication to checkpoint inhibitor immunotherapy, neoadjuvant chemotherapy with FOLFOX, CAPEOX, or FOLFIRINOX is an option.

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[dd] For lung metastases, surgery, thermal ablation, and SBRT can be considered (REC-C and REC-E).

[oo] Patients who are not candidates for immunotherapy should be treated as recommended for pMMR/MSS disease. See NCCN Guidelines for Management of Immune Checkpoint Inhibitor-Related Toxicities.

[pp] Hepatic artery infusion ± systemic chemotherapy (VEGF inhibitor contraindicated) (category 2B) is also an option at institutions with experience in both the surgical and medical oncologic aspects of this procedure.

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dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype (eg, TMB >50 mut/Mb)
UNRESECTABLE METASTASES

INITIAL TREATMENT

SUBSEQUENT TREATMENT

Candidate for immunotherapy and no prior immunotherapy received → **Checkpoint inhibitor immunotherapy[zz,aaa] (REC-F 5 of 17)** → **Re-evaluate disease status every 2–3 mo** →

- **Surveillance (REC-10A)**
- **or**
- **Surgery ± RT**
- **or**
- **Continue immunotherapy (REC-F 5 of 17)**
- **or**
- **Ipilimumab + nivolumab (if checkpoint inhibitor monotherapy was previously received)**
- **or**
- **Systemic therapy (REC-F)**

Prior immunotherapy received →

- **Systemic therapy (REC-F)**
- **or**
- **Ipilimumab + nivolumab (if checkpoint inhibitor monotherapy was previously received)**

[zz] NCCN Guidelines for Management of Immune Checkpoint Inhibitor-Related Toxicities.
[aaa] If disease response, consider discontinuing checkpoint inhibitor after 2 years of treatment.

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PRINCIPLES OF IMAGING[1]

Initial Workup/Staging

- Pelvis MRI with or without contrast or endorectal ultrasound (only for T1 tumors or if MRI is contraindicated). See Pelvis MRI Requirements (REC-A 3 of 5) and Reporting (REC-A 4 of 5).
  - Assess T and N stage of the primary rectal tumor.
  - Pelvis MRI can be performed with or without IV gadolinium contrast per institutional preferences.
  - Pelvis MRI may not be required for local staging if tumor is known to be definite T1 or if patient is not a candidate for primary tumor resection (eg, widespread metastases, plan for permanent colonic diversion).
  - The rectum lies below the peritoneal reflection from the sacral promontory to the upper edge of the symphysis as determined by MRI.
- Chest CT and abdominal CT or MRI
  - Assess for distant metastatic disease to lungs, thoracic and abdominal lymph nodes, liver, peritoneal cavity, and other organs.
  - CT performed with intravenous (IV) iodinated contrast and oral contrast material unless contraindicated.
  - IV contrast is not required for the chest CT (but usually given if performed with abdominal CT).
  - If IV iodinated contrast material is contraindicated because of significant contrast allergy, then MRI examination of the abdomen with IV gadolinium-based contrast agent (GBCA) can be obtained instead. In patients with chronic renal failure (glomerular filtration rate [GFR] <30 mL/min) who are not on dialysis, IV iodinated contrast material may also be contraindicated, and IV GBCA can be administered in select cases using gadofosveset trisodium, gadoxetate disodium, gadobenate dimeglumine, or gadoteridol.
  - If iodinated and gadolinium contrast are both contraindicated due to significant allergy or chronic renal failure without dialysis, then consider MRI without IV contrast or consider FDG-PET/CT imaging.
- FDG-PET/CT is not routinely indicated.
  - FDG-PET/CT does not supplant a contrast-enhanced diagnostic CT or MRI and should only be used to evaluate an equivocal finding on a contrast-enhanced CT or MRI or in patients with strong contraindications to IV contrast administration.
- Consider FDG-PET/CT (skull base to mid-thigh):
  - If potentially surgically curable M1 disease in selected cases.
  - In patients considered for image-guided liver-directed therapies for liver metastases (ie, thermal ablation, radioembolization).[2,3]
- If liver-directed therapy or surgery is contemplated, a hepatic MRI with IV routine extracellular or hepatobiliary GBCA is preferred over CT to assess exact number and distribution of metastatic foci for local treatment planning.

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PRINCIPLES OF IMAGING[1]

Restaging

- Chest CT and abdominal CT or MRI and pelvis MRI
  - Prior to surgery for restaging
  - Prior to adjuvant treatment to assess response to primary therapy or resection
  - During re-evaluation of conversion to resectable disease
- FDG-PET/CT is not usually indicated; however, it may be necessary in case of equivocal additional imaging findings.

Follow-up/Surveillance

- Stage I disease:
  - Imaging is not routinely indicated and should only be based on symptoms and clinical concern for recurrent/metastatic disease.
- Stage II & III disease:
  - C/A/P CT every 6 to 12 months (category 2B for frequency <12 months) for a total of 5 years.
  - MRI or EUS of the rectum every 3 to 6 months for 2 years, then every 6 months for a total of 5 years (for patients with transanal local excision only).
  - FDG-PET/CT examination is usually not recommended.
- Stage IV disease:
  - C/A/P CT every 3 to 6 months (category 2B for frequency <6 months) x 2 years, then every 6 to 12 months for a total of 5 years.
  - MRI or EUS of the rectum every 3 to 6 months for 2 years, then every 6 months for a total of 5 years (for patients with transanal excision only).
  - If liver-directed therapy is contemplated, magnetic resonance with hepatobiliary agent should be considered to better delineate number and location of lesions.
- FDG-PET/CT is usually not indicated with the exception of selected patients who are considered for image-guided liver-directed therapies for hepatic metastases (ie, thermal ablation, radioembolization) or serial CEA elevation during follow-up.
- Therapy-dependent follow-up interval recommendations[4-10]:
  - For chemo/RT: Initial assessment 12 weeks after start of treatment, reassessment 16–20 weeks after start of treatment.
  - For total neoadjuvant therapy: Response assessment at 24 weeks after start of treatment, reassessment 34–38 weeks after the start of treatment.

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PRINCIPLES OF IMAGING
Pelvic MRI Requirements[1]

Patient Preparation

Rectal distension with gel	No consensus on the use of spasmolytic agents, endorectal gel, or cleansing enema. Institutional preference and protocols have been accepted in regard to these preparations.
Use of spasmolytic agents	No consensus on the use of spasmolytic agents, endorectal gel, or cleansing enema. Institutional preference and protocols have been accepted in regard to these preparations.
Use of enema or microenema	No consensus on the use of spasmolytic agents, endorectal gel, or cleansing enema. Institutional preference and protocols have been accepted in regard to these preparations.

MRI Hardware Requirement

Magnet strength	Minimum requirement 1.5 T
Coil	Coil: External phased array coil (endorectal coils expressly NOT recommended)

MRI Sequences

2D high-resolution T2-weighted	• Slice thickness 1–3 mm. 3D T2-weighted sequences are not adequate substitutes • Main sequences for T staging and detection of pathologic lymph nodes • Axial, sagittal, and coronal plane to assess extent and relationship to all surrounding structures • Axial and coronal slices should be angulated along the short (perpendicular) and long (parallel) axis of tumor for tumors in the middle and upper part of the rectum and along the anal canal for low rectal tumors
T1-weighted without contrast	T1-weighted images may be helpful for mucinous tumors on staging and restaging, as the signal can be similar to fat on T2.
Diffusion-weighted imaging (DWI)	The DWI sequence is recommended for both staging and restaging including at least a high b-value of ≥800–1000 (growing evidence of benefit).
T1-weighted with contrast	No consensus for contrast/dynamic-enhanced post-contrast sequences. No conclusive evidence for staging or restaging value at the time of consensus, not routinely recommended.

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PRINCIPLES OF IMAGING
Pelvic MRI Reporting[1]

At presentation (before neoadjuvant therapy)	• Distance from the anal verge or anorectal junction to the lower aspect of the tumor • Relationship of tumor to the peritoneal reflection (above/straddles/below) • Tumor relationship to the sphincter in low rectal tumors (none/internal/internal and external) • Tumor length • T-stage of primary mass • Tumor deposits within the mesorectum • Involvement of the mesorectal fascia (MRF) and the smallest distance (mm) between the tumor and the MRF and its location[a] • N-stage • Presence/absence of suspicious extramesorectal lymph nodes • Number of suspicious lymph nodes (N staging) • Specify locoregional [internal iliac, obturator, mesorectal (N staging)] versus distant [common and external iliac, retroperitoneal (metastatic)] chains. • Inguinal nodes may be locoregional if tumor is below the dentate line, or metastatic if tumor is above the dentate line • Presence/absence of extramural vascular invasion (EMVI) • Additional findings that can be provided in synoptic report: ▸ The circumferential location of the tumor ▸ Morphologic pattern of tumor growth (eg, annular, polypoid, mucinous, ulcerated, perforated)
After neoadjuvant therapy	• Distance from the anal verge or anorectal junction to the lower aspect of the remaining tumor • Tumor length • Presence/absence of a residual tumor (intermediate signal on T2-weighted images) • Presence/absence of fibrosis (low signal on T2-weighted images) • yT-stage and any remaining tumor deposits within the mesorectum • yN-stage and number of remaining suspicious lymph nodes • Presence of any remaining suspicious extramesorectal lymph nodes • Persistent involvement/regression from the MRF[a] • The smallest distance (mm) between the remaining tumor and the MRF and its location • EMVI should be described (if regressed/present) • Additional findings that can be provided in synoptic report: ▸ The circumferential location of the remaining tumor within the wall ▸ The morphologic pattern of tumor growth

References

[a] Circumferential resection margin (CRM) measured at the closest distance of the tumor to the MRF. Clear CRM: Greater than 1 mm from MRF and levator muscles and not invading into the intersphincteric plane. Involved CRM: within 1 mm of MRF; or, for lower third rectal tumors, within 1 mm from levator muscle; or, for anal canal lesions, invasion into or beyond the intersphincteric plane.

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PRINCIPLES OF IMAGING
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PRINCIPLES OF PATHOLOGIC AND MOLECULAR REVIEW

Endoscopically Removed Malignant Polyps

- A malignant polyp is defined as one with cancer invading through the muscularis mucosae and into the submucosa (pT1). pTis is not considered to be a "malignant polyp."
- Favorable histopathologic features: no angiolymphatic invasion, depth of submucosal invasion <1 mm, low tumor budding,[1] and negative margin of resection. A positive margin has been defined as: tumor cells present at or <1 mm from the transected margin.[2-5]
- Unfavorable histologic features, angiolymphatic invasion, depth of submucosal invasion >1 mm, high tumor budding, or a "positive margin." See above for definition of a positive margin. In several studies, tumor budding has been associated with adverse outcome and may preclude polypectomy as an adequate treatment of malignant polyps.
- There is controversy as to whether malignant colorectal polyps with a sessile configuration can be successfully treated by endoscopic removal. The literature seems to indicate that endoscopically removed sessile malignant polyps have a significantly greater incidence of adverse outcome (residual disease, recurrent disease, mortality, or hematogenous metastasis, but not lymph node metastasis) than do polypoid malignant polyps. However, when one looks closely at the data, configuration by itself is not a significant variable for adverse outcome, and endoscopically removed malignant sessile polyps with low-grade histology, <1 mm submucosal invasion, negative margin, and no lymphovascular invasion can be successfully treated with endoscopic polypectomy.[4-8]

Transanal Local Excision

- Favorable histopathologic features: <3 cm size, pT1, grade 1 or 2, no lymphatic or venous invasion, or negative margins.[9,10]
- Unfavorable histopathologic features: >3 cm in size, >pT1, with grade 3, or lymphovascular invasion, positive margin, high tumor budding, or sm3 (lower one third of the submucosa) depth of tumor invasion.[9-11]

Rectal Cancer Appropriate for Resection

- Histologic confirmation of primary malignant rectal neoplasm.
- The rectum lies below a virtual line from the sacral promontory to the upper edge of the symphysis as determined by MRI.

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PRINCIPLES OF PATHOLOGIC AND MOLECULAR REVIEW

Pathologic Stage

- The following parameters should be reported:
  - Grade of the cancer
  - Depth of invasion (pT), the pT stage, is based on viable tumor. Acellular mucin pools are not considered to be residual tumor in those patients treated with neoadjuvant therapy.
  - Number of lymph nodes evaluated and number positive (N). Acellular mucin pools are not considered to be residual tumor in those patients treated with neoadjuvant therapy.
  - Status of proximal, distal, circumferential (radial), and mesenteric margins[12,13]
  - Circumferential resection margin (CRM)[14-18]
  - Neoadjuvant treatment effect[16,17,19-21]
  - Lymphovascular invasion[16,17,22]
  - Perineural invasion (PNI)[23-25]
  - Tumor deposits[26,27]
- CRM - A positive CRM is defined as tumor ≤1 mm from the margin. This assessment includes both tumor within a lymph node as well as direct tumor extension. However, if CRM positivity is based solely on intranodal tumor, it should be stated in the pathology report. A positive CRM is a more powerful predictor of local recurrence in patients treated with neoadjuvant therapy. A positive CRM secondary to lymph node metastasis in some studies has been associated with lower recurrence rates than by direct extension.[14-18]
- Neoadjuvant treatment effect - The most recent College of American Pathologists (CAP) Guidelines on examination specimens of the rectum and the AJCC Cancer Staging Manual, Eighth Edition require commenting on treatment effect after neoadjuvant therapy. The minimum requirement is:
  - Treatment effect present.
  - No definitive response identified.
- The system used to grade tumor response as recommended by the AJCC Cancer Staging Manual, Eighth Edition and the CAP Guidelines is that as modified from Ryan R, et al. Histopathology 2005;47:141-146 and Gavioli M, et al. Dis Colon Rectum 2005;48:1851-1857.
  - 0 - Complete response: No remaining viable cancer cells.
  - 1 - Moderate response: Only small clusters or single cancer cells remaining.
  - 2 - Minimal response: Residual cancer remaining, but with predominant fibrosis.
  - 3 - Poor response: Minimal or no tumor kill; extensive residual cancer.

  According to the CAP, it is optional to grade the tumor response to treatment. However, the NCCN Rectal Cancer Guidelines Panel recommends grading tumor response. Other grading systems that are used are referenced.[16,17,19-21]

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Pathologic Stage (continued)

- PNI - The presence of PNI is associated with a significantly worse prognosis. In multivariate analysis, PNI has been shown to be an independent prognostic factor for cancer-specific, overall, and disease-free survival. For stage II rectal cancer, those with PNI have a significantly worse 5-year disease-free survival compared to those without PNI (29% vs. 82%; *P* = .0005). In stage III rectal cancer, those with PNI have a significantly worse prognosis.[22-27]
- Tumor deposits - Irregular discrete tumor deposits in pericolic or perirectal fat away from the leading edge of the tumor and showing no evidence of residual lymph node tissue, but within the lymphatic drainage of the primary carcinoma, are considered to be tumor deposits or satellite nodules and are not counted as lymph nodes replaced by tumor. Most examples are due to lymphovascular invasion or, more rarely, PNI. Because these tumor deposits are associated with reduced disease-free and overall survival, their number should be recorded in the surgical pathology report.
- Tumor budding - In recent years, tumor budding has been identified as a new prognostic factor in colon cancer. Recently, there was an international consensus conference on tumor budding reporting.[28] A tumor bud is defined as a single cell or a cluster of ≤4 cells identified on hematoxylin and eosin (H&E) at the advancing edge of the invasive carcinoma. The total number of buds are reported from a selected hot spot measuring 0.785 mm (20x ocular in most microscopes/via a conversion factor). Budding is separated into three tiers: low tier (0–4 buds), intermediate tier (5–9 buds), and high tier (≥10 buds). Two recent studies[29,30] using this scoring system have shown tumor budding to be an independent prognostic factor for stage II colon cancer. An American Society of Clinical Oncology (ASCO) guideline for stage II colon cancer designates tumor budding as an adverse (high-risk) factor.[31] Several studies have shown that high-tier tumor budding in pT1 colorectal cancers (CRCs), including malignant polyps, is associated with an increased risk of lymph node metastasis; however, methodologies for assessing tumor budding and grade were not uniform.[32-36]

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PRINCIPLES OF PATHOLOGIC AND MOLECULAR REVIEW

Lymph Node Evaluation

- The AJCC and CAP recommend examination of a minimum of 12 lymph nodes to accurately stage rectal cancer.[12,13,37] Sampling of 12 lymph nodes may not be achievable in patients who received preoperative chemotherapy. The literature lacks consensus as to the minimum number of lymph nodes needed to accurately identify stage II cancer. The minimum number of nodes has been reported as >7, >9, >13, >20, and >30.[37-45] Most of these studies have combined rectal and colon cancers and reflect those cases with surgery as the initial treatment. Two studies confined only to rectal cancer have reported 14 and >10 lymph nodes as the minimum number to accurately identify stage II rectal cancer.[41,44] The number of lymph nodes retrieved can vary with patient age, gender, tumor grade, and tumor site.[38] For stage II (pN0) colon cancer, if <12 lymph nodes are initially identified, it is recommended that the pathologist go back to the specimen and resubmit more tissue of potential lymph nodes. If 12 lymph nodes are still not identified, a comment in the report should indicate that an extensive search for lymph nodes was undertaken. The mean number of lymph nodes retrieved from rectal cancers treated with neoadjuvant therapy is significantly less than those treated by surgery alone (13 vs. 19, $P < .05$; 7 vs. 10, $P < .001$).[46,47] If 12 lymph nodes is considered the number needed to accurately stage stage II tumors, then only 20% of patients treated with neoadjuvant therapy had adequate lymph node sampling.[47] To date, the number of lymph nodes needed to accurately stage disease treated with neoadjuvant therapy is unknown. However, the clinical significance of this in the neoadjuvant setting is unknown, as postoperative therapy is indicated in all patients who receive preoperative therapy regardless of the surgical pathology results.
- Isolated tumor clusters (ITC) are defined as single cells or small groups of cells measuring <0.2 mm. The biologic significance of ITC is currently not well characterized.[48] Therefore, lymph nodes with ITC are defined as negative (N0) in the 9th edition of the AJCC staging system. However, the number of lymph nodes involved by ITC should be documented in the pathology report.
- Micrometastatic tumor deposits measure between 0.2 mm–2.0 mm and lymph nodes involved with micrometastatic deposits are considered positive lymph nodes.[48]

Evaluation of Mesorectum (total mesorectal excision, TME)

- The pathologist should evaluate the quality (completeness) of the mesorectum (only for low rectal cancer - distal 2/3)[49-51]

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PRINCIPLES OF PATHOLOGIC AND MOLECULAR REVIEW

Methods of Testing

- Testing can be performed on formalin-fixed paraffin-embedded tissue or cytology material (preferred) or blood-based circulating tumor DNA (ctDNA)/RNA.
- Repeat molecular testing should not be performed after standard cytotoxic chemotherapy, as significant molecular changes are rarely observed. Changes in the molecular profile can more commonly be seen after targeted therapies, and repeat testing may be considered to guide future targeted therapy decisions.

KRAS, NRAS, and *BRAF* Testing

- All patients with metastatic CRC should have tumor genotyped for RAS (*KRAS* and *NRAS*) and *BRAF* mutations as part of MGPT at initial diagnosis of metastatic disease. Patients with any known *KRAS* mutation (exons 2, 3, and 4) or *NRAS* mutation (exons 2, 3, and 4) should not be treated with either cetuximab or panitumumab, unless given as part of a regimen targeting a *KRAS* G12C mutation.[52-54] *BRAF* V600E mutation makes response to panitumumab or cetuximab highly unlikely.[55-57]
- *BRAF* V600E mutation testing via immunohistochemistry (IHC) is also an option.
- Testing for *KRAS*, *NRAS*, and *BRAF* mutations should be performed only in laboratories that are certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA-88) as qualified to perform high-complexity clinical laboratory (molecular pathology) testing. No specific methodology is recommended.
- When using a multigene panel it is recommended to test the metastasis for genomic additional alterations. However, testing can be performed on the primary CRCs, as literature has shown that driver *KRAS, NRAS*, and *BRAF* mutations are similar in both specimen types.[58]

Microsatellite Instability (MSI) or Mismatch Repair Testing (MMR)

- Universal MMR[a] or MSI[a] testing is recommended in all newly diagnosed patients with rectal cancer. See NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric.
- The presence of a *BRAF* V600E mutation in the setting of *MLH1* absence would preclude the diagnosis of Lynch syndrome (LS) in the vast majority of patients. However, approximately 1% of cancers with *BRAF* V600E mutations (and loss of MLH1) are LS. Caution should be exercised in excluding patients with a strong family history from germline screening in the case of *BRAF* V600E mutation.[59]
- MMR or MSI testing should be performed only in CLIA-approved laboratories.
- Testing for MSI involves looking for a change in the number of microsatellite repeats in tumor DNA compared to non-neoplastic tissue as evidence of deficient MMR mechanisms. This may be accomplished by multiplex polymerase chain reaction (PCR) followed by capillary electrophoresis or as part of a validated MGPT, the latter especially in patients with metastatic disease who require genotyping of *KRAS, NRAS,* and *BRAF*.

[a] IHC for MMR and DNA analysis for MSI are different assays and measure different biological effects caused by dMMR function.

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MSI or MMR Testing (continued)

- IHC refers to interrogating tumor tissue for protein expression. IHC performed for MMR looks at the protein expression of four MMR genes (*MLH1, MSH2, MSH6,* and *PMS2*). Patients with pathogenic germline variants in these genes have LS. However, alterations in these genes also arise in sporadically acquired CRC. Tumors that show nuclear expression of all four MMR proteins are considered pMMR. Tumors that reveal loss of expression of any MMR protein are considered dMMR. To ensure clarity, interpretation of IHC result should be reported as either “retained for all MMR proteins,” or “loss of expression of (one or more of the MMR proteins)." The words “positive” or “negative” should be avoided, although there are a few caveats such as isolated loss of MSH6 following neoadjuvant treatment. MMR proteins function as heterodimers: MLH1/PMS2 and MSH2/MSH6. Therefore, loss of expression of one protein will usually result in the loss of the heterodimer partner.
- The pattern of lost protein expression guides further testing such as *MLH1* promoter hypermethylation, or *BRAF* V600E or germline testing to assess for LS.

HER2 *(ERBB2)* Overexpression/Amplification Testing

- Diagnostic testing is via IHC, fluorescence in situ hybridization (FISH), or next-generation sequencing (NGS).
- Positive by IHC is defined as: 3+ staining in >50% of tumor cells. 3+ staining is defined as an intense membrane staining that can be circumferential, basolateral, or lateral. Those who have a HER2 score of 2+ should be reflexed to FISH testing.[60-62] HER2 *(ERBB2)* amplification detected by FISH is considered positive when the HER2 *(ERBB2)*:CEP17 ratio is ≥2 in >50% of the cells.[60-62] NGS is another methodology for testing for HER2 *(ERBB2)* amplification.[63]
- Anti-HER2 therapy with signal transduction inhibition (eg, trastuzumab/pertuzumab, trastuzumab/tucatinib) is indicated for patients with HER2 *(ERBB2)* overexpression/amplification, IHC 3+, IHC2+/ISH+, or HER2 *(ERBB2)* overexpressed/amplified (detected by NGS) cancers that are also *KRAS, NRAS,* and *BRAF*.
- Fam-trastuzumab deruxtecan-nxki is only indicated in HER2-overexpressed tumors (IHC 3+).

NTRK 1/2/3 Gene Fusions

- *NTRK 1/2/3* gene fusions are extremely rare in CRCs.[64] The overall incidence is approximately 0.35% in a cohort of 2314 CRCs, with *NTRK 1/2/3* gene fusions confined to those tumors that are pan–WT *KRAS*, *NRAS*, and *BRAF*. In one study of eight CRCs harboring *NTRK 1/2/3* gene fusions, seven were found in the small subset that were dMMR (MLH1)/MSI-H.[65] *NTRK 1/2/3* gene fusions are more frequently found among patients with dMMR.
- *TRK* inhibitors have been shown to have activity ONLY in those cases with *NTRK 1/2/3* gene fusions, and NOT with *NTRK 1/2/3* gene point mutations.
- Methodologies for detecting *NTRK 1/2/3* gene fusions are IHC,[66] FISH, DNA-based NGS, and RNA-based NGS.[65,67] In one study, DNA-based sequencing showed an overall sensitivity and specificity of 81.1% and 99.9%, respectively, for detection of *NTRK 1/2/3* gene fusions when compared to RNA-based sequencing and IHC showed an overall sensitivity of 87.9% and specificity of 81.1%. Since approximately one in five tumors identified as having an *NTRK 1/2/3* gene fusion by IHC will be a false positive, tumors that test positive by IHC should be confirmed by RNA-based NGS. That same study commented that RNA-based sequencing appears to be the optimal way to approach *NTRK 1/2/3* gene fusions, because the splicing out of introns simplifies the technical requirements of adequate coverage and because detection of RNA-level fusions provides direct evidence of functional transcription.[67] However, selection of the appropriate assay for *NTRK 1/2/3* gene fusion detection depends on tumor type and genes.

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POLE/POLD1

- Polymerase genes, *POLE* and *POLD1*, encode proteins tasked with proofreading functions to recognize and correct mispaired bases incorporated during DNA replication. Pathogenic variants (PVs) within the exonuclease domains (ED) of *POLE* and *POLD1* result in loss of this proofreading function leading to subsequent acquisition of numerous single nucleotide variants (SNVs).[68,69]
- Germline PVs within the ED of *POLE* and *POLD1* predispose patients to multiple colorectal adenomas and carcinomas, resulting in polymerase proofreading-associated polyposis (PPAP)[68,69] (see NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric).
- Somatic *POLE* mutations occur in approximately 2%–8% of patients with predominately MSS/pMMR CRC while somatic *POLD1* mutations are extremely rare.[68,70]
- NGS of CRCs in patients with either germline or somatic ED PVs demonstrate an ultra-hypermutated phenotype identified as extremely high tumor mutational burden (TMB >50 mut/Mb). TMB is calculated as the total number of somatic mutations per coding area of the tumor genome. Although calculations vary according to assay performed, TMB >10 mut/Mb is generally regarded as TMB-high (TMB-H).[70,71]
- *POLE/POLD1* mutations can be identified through single gene assays (PCR or Sanger sequencing). However, TMB calculation requires MGPT, which often includes concurrent *POLE/POLD1* sequencing. As such, performing a MGPT on CRC tumor tissue has the advantage of not only identifying *POLE/POLD1* mutations but also provides direct evidence of loss of proofreading function (TMB-H).[70-72]
- Patients with CRC harboring *POLE/POLD1* mutations have a more favorable prognosis, likely secondary to immune responses stimulated by the presence of numerous neoantigens produced as a consequence of aberrant proofreading function. Similarly, for these patients disease responds well to immune checkpoint inhibitor therapy.[72-77]

***RET* Gene Fusions**

- RET is a receptor tyrosine kinase that plays a critical role in the development and maintenance of neural and genitourinary tissues, primarily through downstream MAPK and PI3K signaling pathways.[78]
- Germline activating PVs in *RET* lead to multiple endocrine neoplasia type 2 (MEN2) (see NCCN Guidelines for Neuroendocrine and Adrenal Tumors) and loss of function PVs are associated with Hirschsprung disease and congenital abnormalities of the kidney and urinary tract.[78]
- Somatic activating alterations in *RET* include point mutations as well as gene fusions and have been identified in a variety of tumors.[78,79]
- In patients with CRC, activating *RET* gene fusions involving the C-terminal kinase domain lead to constitutive upregulation of *RET* kinase activity and subsequent promotion of cell proliferation and survival. The most common gene fusion partners reported include *KIF5B, CCDC6*, and *NCOA4*.[78-81]
- The RET-targeted inhibitor, selpercatinib, is FDA-approved for patients with solid tumors harboring activating *RET* gene fusions.[82]
- The presence of *RET* gene fusions can be interrogated through a variety of techniques, including IHC, FISH, polymerase chain reaction (PCR), and either DNA- or RNA-based NGS assays. RNA-based NGS assays are gene fusion agnostic and as such have the advantage of identifying *RET* gene fusions involving any partner gene.[79-81]
- Caution should be used in interpreting *RET* gene fusion results from ctDNA/RNA assays following targeted therapy with EGFR inhibitors.

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PRINCIPLES OF SURGERY AND LOCOREGIONAL THERAPIES

Workup

- Independent evaluation by the treating surgeon with either proctosigmoidoscopy or flexible sigmoidoscopy is recommended for all rectal tumors. Critical characteristics to be documented, in conjunction with DRE, include tumor size, distances from the anal verge and the anorectal ring, orientation within the rectal lumen (eg, anterior-posterior, laterality) and/or degree of circumferential involvement, extent of obstruction, extent of fixation to the rectal wall, degree of sphincter involvement, and sphincter tone.

Transanal Local Excision[1]

- Criteria
  - <30% circumference of bowel; <3 cm in size; clear surgical margin (>3 mm); mobile, nonfixed; within 8 cm of anal verge; T1 only; endoscopically removed polyp with cancer or indeterminate pathology; no lymphovascular invasion or PNI; well to moderately differentiated; no evidence of lymphadenopathy on pretreatment imaging
  - Surgical technique includes full-thickness excision of rectal wall with goal of ≥10 mm excision margin with an intact non-fragmented specimen.
- When the lesion can be adequately localized to the rectum, local excision of more proximal lesions may be technically feasible using advanced techniques, such as transanal endoscopic microsurgery (TEM) or transanal minimally invasive surgery (TAMIS).
- Local excision could be a viable option for patients who have a strong but incomplete response and are unable or unwilling to undergo standard TME surgery. The most suitable candidates would be those meeting the near complete response (nCR) criteria as outlined in the Principles of Nonoperative Management section (endoscopy, DRE, and MRI).
- The efficacy of local excision for patients who have achieved a clinical complete response (cCR) is not yet fully established. According to current data, cCR, as determined through MRI, DRE, and endoscopy, typically correlates with a durable response. Therefore, implementing local excision might introduce unnecessary risks without clear benefits. Given this uncertainty and potential for added risk, local excision is not routinely recommended for patients who have achieved a clear cCR.

Transabdominal Resection: Abdominoperineal resection or low anterior resection or coloanal anastomosis using TME

- Management principles
  - The treating surgeon should be experienced in rectal cancer surgery, and specifically with TME. For patients with predicted positive margins based on preoperative imaging, or lateral pelvic lymph node involvement, the surgeon should be experienced in extended resections beyond the TME plane and have a multidisciplinary team available if necessary.[2]
  - The treating surgeon should assess the distal margin before initiating treatment by DRE ± rigid or flexible endoscopy, particularly for non-palpable lesions.
  - Anticipated circumferential margins should be assessed by MRI (see Principles of Imaging, REC-A) prior to any required neoadjuvant therapy, and again considered prior to surgery. If margins are involved, assessment for feasibility of resection beyond the TME plane is required. Such an extended resection (± reconstruction) should involve careful preoperative planning and may require a multidisciplinary team.
  - For adequately staged, low-risk, upper-rectal T3, N0 tumors, surgery alone is an appropriate treatment option.
  - Remove primary tumor with adequate circumferential and distal margins.
  - Treat draining lymphatics by TME.
  - Sphincter preservation and restoration of organ integrity should be achieved without compromise of oncologic resection and consideration of anticipated patient functional outcome and quality of life.
- TME is a standard component of radical rectal cancer surgery. TME reduces the positive radial margin and local recurrence rates.
  - Extend 4 to 5 cm below distal edge of tumors for an adequate mesorectal excision. In distal rectal cancers (ie, <5 cm from anal verge), negative distal bowel wall margin of 1 to 2 cm may be acceptable.
  - Full rectal mobilization allows for a negative distal margin and adequate mesorectal excision.
  - Minimally invasive approaches (eg, laparoscopic, robotic) for resection of rectal cancer have been shown to be safe.[3,4]
  - There are no significant differences in disease-free survival and recurrence rates with minimally invasive approaches when compared to open resection.[5-7]
    - ◊ The surgeon should have experience performing minimally invasive proctectomy with TME.
    - ◊ It is not indicated for locally advanced disease with a threatened or high-risk circumferential margin based on staging. For these high-risk tumors, open surgery is preferred.
    - ◊ It is not generally indicated for acute bowel obstruction or perforation from cancer.
    - ◊ Thorough abdominal exploration is required.
- **Lymph node dissection**[8,9]
  - Clinically suspicious nodes beyond the field of resection should be biopsied and/or removed, if possible. Extensive resection of M1 lymph nodes is not indicated.
  - Extended lymph node resection is not indicated in the absence of clinically suspected nodes.

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PRINCIPLES OF SURGERY AND LOCOREGIONAL THERAPIES

Locoregional Therapies

- **Image-Guided Tumor Ablation**[10]
  - Thermal ablation creates tumor cell death through deposition of tumoricidal heat (radiofrequency or microwave) or cold (cryoablation) in the tumor and surrounding margins.
  - Non-thermal ablation such as irreversible electroporation creates tumor cell death through electrical pulses that create irreversible membrane pores and cellular lysis/destruction.

- **Liver Tumor Ablation**[10-12]
  - Thermal ablation can be considered alone, or in conjunction with surgery, in appropriately selected patients with small metastases that can be treated with margins. All original sites of disease need to be amenable to thermal ablation or resection.
  - Image-guided thermal ablation may be considered in selected surgical candidates or medically non-surgical candidates with small tumors that can be completely ablated with margins.
  - Image-guided thermal ablation can be considered in selected patients with recurrence after hepatectomy or ablation as long as all visible disease can be ablated with margins.[10-12]
  - Image-guided non-thermal ablation (irreversible electroporation) can be considered in patients whose tumors cannot be safely resected or ablated with margins due to proximity to central bile ducts or other structures that cannot be protected.

- **Lung Tumor Ablation**[13-15]
  - Ablative techniques may be considered alone or in conjunction with resection for resectable disease. All original sites of disease need to be amenable to thermal ablation or resection.
  - Image-guided thermal ablation can also be considered when unresectable and amenable to complete thermal ablation.
  - Image-guided thermal ablation may be considered in selected surgical candidates with small tumors that can be completely ablated with margins.
  - Image-guided thermal ablation may be considered for recurrences after surgery or prior ablation as long as all visible disease is amenable to thermal ablation.

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PRINCIPLES OF SURGERY AND LOCOREGIONAL THERAPIES

- **Arterially Directed Embolic Therapy**
  - Hepatic transarterial radioembolization (TARE) with yttrium-90 (Y-90) microspheres[16,17]
    - ◊ Y-90 radioembolization (radiation lobectomy approach) can be considered instead of portal vein embolization when hepatic metastatic disease is not optimally resectable based on insufficient remnant liver volume or when there is borderline resectable disease that would benefit from tumor downsizing and remnant hypertrophy.
    - ◊ Hepatic TARE with Y-90 microspheres can be considered in selected patients with chemotherapy-resistant/refractory disease and with predominant hepatic metastases.
    - ◊ Radiation segmentectomy approach can be used for limited volume liver disease that is not amenable to resection or thermal ablation.
  - Transarterial chemoembolization (TACE)[18]
    - ◊ TACE involves hepatic artery catheterization to locally deliver chemotherapy in combination with arterial embolization.
    - ◊ TACE for hepatic metastatic tumors can be considered in highly selected cases with chemotherapy-resistant/refractory disease, preserved liver function, and predominant hepatic metastases.
    - ◊ The most commonly accepted variation for the treatment of metastatic CRC involves the use of drug-eluting bead TACE (DEB-TACE) using irinotecan as the chemotherapeutic agent (DEBIRI).
    - ◊ DEBIRI can be used along with irinotecan-based chemotherapy for unresectable liver-dominant disease.

- **External Beam Radiation Therapy (EBRT)**
  - EBRT to the metastatic site can be considered in appropriately selected cases in which the patient has a limited number of metastases, including the liver or lung or other select locations; or the patient is symptomatic; or in the setting of a clinical trial.
  - The possible techniques include three-dimensional CRT (3D-CRT), intensity-modulated RT (IMRT), and stereotactive body RT (SBRT).
    - ◊ SBRT is an advanced technique of hypofractionated RT with photons that delivers large ablative doses of radiation. SBRT in the management of liver or lung metastases can be an alternative to ablation/embolization techniques or when these therapies have not been successful or are contraindicated.[19-24]
    - ◊ SBRT or other hypofractionated regimens with biologically effective dose (BED) 10 >100 Gy are preferred in the context of oligometastatic disease to provide durable local control. Final dosing should also take into account adjacent normal organs (see REC-E).

- **Hepatic Arterial Infusion (HAI)**
  - Eligibility:
    - ◊ Multidisciplinary experience with HAI therapy
    - ◊ Candidate for major surgery
    - ◊ Unresectable colorectal liver metastases or resectable colorectal liver metastases at high risk for recurrence
    - ◊ Treated with at least one line of systemic chemotherapy
    - ◊ No extrahepatic disease; primary tumor may be in place
    - ◊ Suitable hepatic arterial anatomy
    - ◊ No portal hypertension
    - ◊ No active viral hepatitis
    - ◊ Direct bilirubin ≤1.5 mg/dL, alkaline phosphatase <2 times the upper limit of normal (ULN)
    - ◊ HAI chemotherapy cannot be delivered with concurrent bevacizumab or other vascular endothelial growth factor (VEGF) inhibitors
    - ◊ No prior radiation to the liver

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PRINCIPLES OF SURGERY AND LOCOREGIONAL THERAPIES
CRITERIA FOR RESECTABILITY OF METASTASES AND LOCOREGIONAL THERAPIES WITHIN SURGERY

Local therapies for liver and lung oligometastases should be selected through multidisciplinary review based on technical feasibility, patient comorbidities, and disease distribution.

Liver

- Hepatic resection is the treatment of choice for resectable liver metastases from CRC.[25]
- Complete resection must be feasible based on anatomic grounds and the extent of disease; maintenance of adequate hepatic function is required.[26,27]
- There should be no unresectable extrahepatic sites of disease.[28-30]
- Partial debulking (R1/R2 resection) is not recommended.
- Patients with resectable metastatic disease and primary tumor in place should have both sites resected with curative intent.
  - These can be resected in one operation or as a staged approach, depending on the complexity of the hepatectomy or colectomy, comorbid diseases, surgical exposure, and surgeon expertise.
  - Staged procedures can be performed as liver-first or primary-first approaches.[31]
- In the setting of neoadjuvant therapy, placement of pre-treatment fiducial marker(s) in smaller lesions may be considered.
- Re-resection and re-ablation can be considered in selected patients.[32]
- When hepatic metastatic disease is not optimally resectable based on insufficient remnant liver volume, approaches using preoperative portal vein embolization or staged liver resections can be considered.
- At the time of surgery, ablative techniques may be considered alone or in conjunction with resection.[25] All original sites of disease should be amenable to thermal ablation or resection.
- Thermal ablation may be considered in selected surgical candidates with small tumors that can be completely ablated with margins.
- Arterially directed catheter therapy, and in particular Y-90 microsphere radioembolization, is an option in selected patients with chemotherapy-resistant/-refractory disease and with predominant hepatic metastases.
- Ablative EBRT may be considered in selected cases or in the setting of a clinical trial and should not be used indiscriminately in patients with potentially surgically resectable disease or disease that can be percutaneously ablated with margins.
- For liver metastases, in general, resection may be prioritized over local thermal ablation. However, for select patients with small lesions (≤3 cm), thermal ablation should be prioritized, when feasible, due to equivalent outcomes and reduced toxicity compared to surgery.[33] SBRT can be considered in select cases where other local therapy options are not possible or in patients who prefer a noninvasive approach.

Lung

- Complete resection based on the anatomic location and extent of disease with maintenance of adequate function is required.[34-37]
- The primary tumor must have been resected for cure (R0).
- Resectable extrapulmonary metastases do not preclude resection.[38-41]
- Re-resection and re-ablation can be considered in selected patients.[42]
- Ablative techniques may be considered alone or in conjunction with resection for resectable disease. All original sites of disease need to be amenable to thermal ablation or resection.
- Ablative techniques can also be considered when unresectable and amenable to complete thermal ablation.
- Thermal ablation may be considered in selected surgical candidates with small tumors that can be completely ablated with margins.
- Patients with resectable synchronous metastases can be resected synchronously or using a staged approach.
- Ablative EBRT may be considered in selected cases or in the setting of a clinical trial and should not be used indiscriminately in patients with potentially surgically resectable disease or disease that can be percutaneously ablated.
- For lung metastases, surgery, thermal ablation, and SBRT can be considered based on multidisciplinary discussion and patient preference.

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PRINCIPLES OF SURGERY AND LOCOREGIONAL THERAPIES
CRITERIA FOR RESECTABILITY OF METASTASES AND LOCOREGIONAL THERAPIES WITHIN SURGERY

Evaluation for Conversion to Resectable or Ablatable Disease

- Re-evaluation for resection and/or ablation should be considered in otherwise unresectable patients after 2 months of preoperative chemotherapy and every 2 months thereafter.[43-46]
- Metastatic tumor(s) with a higher likelihood of being converted to resectable and/or ablatable are those in which the initial disease is confined to limited sites.
- When considering whether disease has been converted to resectable and/or ablatable, all original sites need to be amenable to treatment.[47] Preoperative chemotherapy regimens with high response rates should be considered for patients with potentially convertible disease.[48]

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PRINCIPLES OF SURGERY AND LOCOREGIONAL THERAPIES
ENDOSCOPIC SUBMUCOSAL DISSECTION (ESD)

Therapeutic Principles

- ESD is a minimally invasive, organ-preserving procedure that can provide curative resection for early rectal cancers by removing these lesions en bloc.
- ESD usually involves using an injection with submucosal dissection of the lesion to achieve complete en bloc resection.
- ESD can provide either curative resection after en bloc removal or, if not curative, can provide an accurate pathologic staging of disease.

Pre-ESD Endoscopic Evaluation

- Successful curative resection of rectal lesions requires a thorough assessment of the lesion, including subclassification of laterally spreading tumor (LST), as well as vessel, surface, and pit pattern.[49]
- The morphology of all lesions should be described using the Paris classification, which has been shown to correlate to the likelihood of invasive cancer.[50]
- ESD should be performed at a high-volume center by an experienced endoscopist or surgeon.[49]
- EUS or MRI should be used in the rectum prior to resection when suspicious features of deep submucosal invasion are present.[49]

Criteria for Resection

- ESD with en bloc resection should be considered for rectal lesions at risk for submucosally invasive cancer[51,52]
  - Type V Kudo pit pattern
  - Depressed component (Paris 0–IIc)
  - Complex morphology (0–Is or 0–IIa+Is)
  - Nongranular, LST ≥20 mm in size
  - Granular, laterally spreading tumor ≥30 mm in size
- Residual or recurrent colorectal adenomas
- Rectal lesions that have surface features (vascular or pit pattern) suggestive of advanced dysplasia or early submucosally invasive carcinoma

Curative Resection

- Based on both the Japanese[53] and European [49] guidelines, resection of rectal lesions can be considered curative when:
  - Negative circumferential and deep vertical tumor margins
  - Submucosal invasion depth <1000 µm
  - Absence of lymphovascular invasion
  - Absent or grade 1 (low-grade) tumor budding
  - Well (G1) to moderately (G2) differentiated tumor histology
    - ◊ If a removed lesion does not meet the above criteria, a multidisciplinary team including an endoscopist, surgeon, and pathologist should use a shared decision-making process to decide whether to proceed with surgery versus intensive surveillance.

Surveillance After ESD[51]

- Risk of lymph node metastasis after curative resection of superficial T1 rectal cancer with submucosal invasion <1000 µm is estimated to be 3%–6%.
- Flexible sigmoidoscopy is recommended 3–6 months after ESD assuming prior colonoscopy.
- Second follow-up endoscopy is recommended 3–6 months after first surveillance exam or colonoscopy if 1 year from ESD.
- Surveillance using flexible sigmoidoscopy is recommended every 6 months for a total of 5 years from ESD.
- EUS or pelvis MRI with contrast is recommended every 3–6 months for 2 years, then every 6 months to complete 5 years.

Surgery vs. ESD

- Prior retrospective studies have found that ESD and TEM/TAMIS do not differ in terms of outcomes, including local recurrence, R0 resection rate, and adverse events.[54-56]
- The decision between surgery and ESD should be determined based on the patient's candidacy for surgery, availability of expertise in ESD, and patient preference.

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PRINCIPLES OF PERIOPERATIVE THERAPY

Not every patient with rectal cancer requires trimodality treatment—trials with adaptive designs have demonstrated some patients will have favorable outcomes with selective usage of radiation or selective usage of surgery, based on reassessment of response during therapy.[1,2]
The regimens used in patients who will undergo or have undergone surgery include both concurrent chemotherapy/RT and chemotherapy alone. Perioperative treatment is recommended for up to a total of 3 to 6 months.

Perioperative Chemotherapy:

- Modified FOLFOX6[3,4,5]
  Oxaliplatin 85 mg/m^2 IV, Day 1,[a] Leucovorin 400 mg/m^2 IV Day 1,[b] Fluorouracil 400 mg/m^2 IV bolus on Day 1, followed by 1200 mg/m^2/day x 2 days (total 2400 mg/m^2 over 46–48 hours) continuous infusion. Repeat every 2 weeks to a total of 6 months perioperative therapy.
- CAPEOX[6,7]
  Oxaliplatin 130 mg/m^2 IV Day 1.[a] Capecitabine 1000 mg/m^2 PO twice daily for 14 days every 3 weeks. Repeat every 3 weeks to a total of 6 months perioperative therapy.
- FOLFIRINOX[8,c]
  Oxaliplatin 85 mg/m² IV on Day 1,[a] Leucovorin 400 mg/m² IV over 2 hours on Day 1,[b] Irinotecan 180 mg/m² IV over 30–90 minutes on Day 1, Fluorouracil 400 mg/m² IV push Day 1, Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46 hours) continuous infusion. Repeat every 2 weeks.
- Modified FOLFIRINOX[9,c]
  Oxaliplatin 85 mg/m² IV on Day 1,[a] Leucovorin 400 mg/m² IV over 2 hours on Day 1,[b] Irinotecan 150 mg/m² IV over 30–90 minutes on Day 1, Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46 hours) continuous infusion. Repeat every 2 weeks.

Dosing Schedules for Concurrent Chemotherapy/RT:

- Fluorouracil (continuous infusion)[10] + RT
  Fluorouracil 225 mg/m^2 IV over 24 hours daily on Days 1–5 or Days 1–7 for 5 weeks with RT
- Capecitabine[11,12] + RT
  Capecitabine 825 mg/m^2 PO BID, Monday–Friday, on days of radiation treatment only, throughout the duration of RT (typically 28–30 treatment days)

[a] Oxaliplatin may be given either over 2 hours, or may be infused over a shorter time at a rate of 1 mg/m^2/min. Leucovorin infusion should match infusion time of oxaliplatin. Cercek A, Park V, Yaeger R, et al. Faster FOLFOX: oxaliplatin can be safely infused at a rate of 1 mg/m^2/min. J Oncol Pract 2016;12:e548-553.
[b] Leucovorin 400 mg/m^2 is the equivalent of levoleucovorin 200 mg/m^2.
[c] FOLFIRINOX is recommended instead of FOLFOXIRI because FOLFOXIRI uses a high dose of fluorouracil (3200 mg/m² over 48 hours). Patients in the United States have been shown to have greater toxicity with fluorouracil. The dose of fluorouracil (2400 mg/m² over 46 hours) is a starting dose consistent with the dose recommended in FOLFOX or FOLFIRI and should be strongly considered for U.S. patients.

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PRINCIPLES OF RADIATION THERAPY

General Principles

- Chemotherapy with a fluoropyrimidine in oral or continuous venous infusion form should be delivered concurrently with conventionally fractionated RT.
- Endorectal brachytherapy can be considered as a means of dose escalation to achieve organ preservation in select patients at institutions with experience with this treatment technique.[1,2]
- In patients with a limited number of liver or lung metastases, ablative radiotherapy to the metastatic site can be considered in selected cases. Ablative radiotherapy can be considered for patients with unresectable metastasis or in patients preferring a nonoperative approach. Radiotherapy should be delivered in a highly conformal manner. The techniques can include 3D-conformal RT, IMRT, or SBRT.

Treatment Information

- Image-guided RT (IGRT) with kilovoltage (kV) imaging or cone-beam CT imaging should be routinely used during the course of treatment with IMRT and SBRT.
- IMRT is preferred for reirradiation of previously treated patients with recurrent disease, patients treated postoperatively due to increased acute or later toxicity,[3] or in unique anatomical situations (eg, coverage of external iliac lymph nodes for T4 tumors invading anterior pelvic organs or inguinal lymph nodes for low-lying tumors involving the anal canal or avoidance of small bowel).
- In patients with locally recurrent disease after prior pelvis RT, consider use of hyperfractionated pelvic re-irradiation or proton beam therapy if re-treatment is planned.[4,5]
- IORT, if available, may be considered for very close or positive margins after resection, as an additional boost, especially for patients with T4 or recurrent cancers.
- Arterially directed catheter therapy, and in particular Y-90 microsphere selective internal radiation, is an option in highly selected patients with chemotherapy-resistant/-refractory disease and with predominant hepatic metastases.
- SBRT or other hypofractionated regimens with BED10 >100 Gy are preferred in the context of oligometastatic disease to provide durable local control. Final dosing should also take into account adjacent normal organs. SBRT can be considered for larger lesions or more extensive disease, if there is sufficient uninvolved liver/lung and liver/lung radiation tolerance can be respected. There should be no other systemic disease or it should be minimal and addressed in a comprehensive management plan.

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PRINCIPLES OF RADIATION THERAPY

Treatment Information

- **Target Volumes**
  - Target volume definition should be performed per International Commission on Radiation Units and Measurements (ICRU) 50 recommendations.
  - Gross tumor volume (GTV) should include all primary tumor and involved lymph nodes, using information from physical examination, endoscopic findings, diagnostic imaging, and the simulation planning study for delineation. Clinical target volume (CTV) should include the GTV plus areas at risk for microscopic spread from the primary tumor and at-risk nodal areas. A consensus atlas may be helpful to review when defining elective nodal CTVs.[6]
  - At-risk nodal regions include mesorectal, presacral, posterior obturator nodes, and internal iliac nodes. The external iliac nodes should also be included for T4 tumors involving anterior structures. Consider including the inguinal nodes for low-lying tumors involving the anal canal.
  - Fusion of the pelvis MRI is strongly recommended to optimally define gross disease.
  - If using 3D-conformal RT, multiple RT fields should be used (generally a 3- or 4-field technique). Prone positioning, full bladder, and other techniques to minimize the volume of small bowel in the fields are encouraged.
  - For postoperative patients treated by abdominoperineal resection, the perineal wound should be included within the fields.
- **RT Dosing**
  - 45–54 Gy in 25–30 fractions to the pelvis. Alternatively, a radiation plan using a simultaneous integrated boost can be used to deliver 45 Gy to the pelvis and 50 Gy to the tumor bed in 25 fractions or 45 Gy to the pelvis and 54 Gy to the tumor bed in 27 fractions.
    - ◊ For resectable cancers, after 45 Gy a tumor bed boost with a 2-cm margin of 5.4 to 9.0 Gy in 3 to 5 fractions could be considered for preoperative radiation.
    - ◊ Small bowel max point dose should be limited to Dmax 55 Gy, V45 Gy should be ≤150 cc, or V50 should be ≤30 cc for individual small bowel loops.[7]
    - ◊ For unresectable cancers, doses higher than 54 Gy may be required, if technically feasible.
  - Short-course RT (25 Gy in 5 fractions) can also be considered for patients for preoperative radiation.
    - ◊ For high-risk rectal cancer (clinical tumor stage cT4a or cT4b, EMVI, clinical nodal stage cN2, involved MRF [tumor or lymph node 1 mm or less from the MRF], or enlarged lateral lymph nodes considered to be metastatic), the 5-year follow-up of the RAPIDO trial now indicates a statistically higher locoregional recurrence rate (10%) in the experimental arm of short-course RT → chemotherapy → surgery versus control arm (6%) of chemoRT → surgery → adjuvant chemotherapy.[8]
    - ◊ Small bowel max point dose should be limited to D0.03 cc 26.5 Gy, and V20 Gy should be 50 cc for individual small bowel loops. Depending upon anatomy and clinical circumstances, it may be reasonable to accept a D0.03cc up to 28.75 Gy.

Supportive Care

- Patients should be considered for vaginal dilators and instructed on the symptoms of vaginal stenosis.
- Patients of childbearing potential should be counseled about the effects of premature menopause and consideration should be given to referral for discussion of hormone replacement strategies.
- Patients of childbearing potential should be counseled that an irradiated uterus cannot carry a fetus to term.
- Patients should be counseled on sexual dysfunction, potential for future low testosterone levels, and infertility risks and given information regarding sperm banking or oocyte, egg, or ovarian tissue banking, as appropriate, prior to treatment.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

• General Principles ▸ Continuum of Care ▸ Toxicity Risk Assessment ▸ Checkpoint Inhibitor Immunotherapy	REC-F 1 of 17
• Treatment for *KRAS* WT/*NRAS* WT/*BRAF* WT	REC-F 2 of 17
• Treatment for (*KRAS* mutation or *NRAS* mutation) and *BRAF* WT	REC-F 3 of 17
• Treatment for *KRAS* WT/*NRAS* WT/*BRAF* V600E mutation	REC-F 4 of 17
• Treatment for dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype (eg, TMB >50 mut/Mb)	REC-F 5 of 17
• Tumor-Agnostic Therapies	REC-F 6 of 17
• Footnotes	REC-F 7 of 17
• Regimens and Dosing	REC-F 8 of 17
Note: If both dMMR/MSI-H and *BRAF* V600E are positive, treatment should start with a checkpoint inhibitor, if feasible.	

Note: All recommendations are category 2A unless otherwise indicated.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

General
- An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.
- For chemotherapy dosing and references, see Chemotherapy Regimens and References (REC-F [8 of 17]).

Continuum of Care
- Discontinuation of oxaliplatin should be strongly considered after 3 to 4 months of therapy (or sooner for unacceptable neurotoxicity) while maintaining other agents until time of progression. Oxaliplatin may be reintroduced if it was discontinued for neurotoxicity rather than for disease progression.
- A single-agent fluoropyrimidine is not recommended after progression on a fluoropyrimidine-containing regimen.
- Bevacizumab is the preferred anti-angiogenic agent and can be continued into the second line of therapy and beyond.
- EGFR inhibitors include cetuximab and panitumumab. An EGFR inhibitor should not be continued beyond progression of disease. There is no efficacy benefit in changing the EGFR inhibitor at the time of progression.
- Ziv-aflibercept and ramucirumab have only shown activity when given in conjunction with FOLFIRI in FOLFIRI-naïve patients. There are no data to suggest activity of FOLFIRI-ziv-aflibercept or FOLFIRI-ramucirumab in a patient whose disease has progressed on FOLFIRI-bevacizumab, or vice versa.
- Arterially directed catheter therapy, and in particular Y-90 microsphere selective internal radiation, is an option in highly selected patients with chemotherapy resistant/-refractory disease and with predominant hepatic metastases.[a]
- C/A/P CT with contrast or chest CT and abdomen/pelvis MRI with contrast are recommended to monitor progress of therapy. FDG-PET/CT should not be used. See Principles of Imaging (REC-A).
- If patients had therapy stopped for reasons other than progression (eg, cumulative toxicity, elective treatment break, patient preference), rechallenge is an option at time of progression.

Toxicity Risk Assessment
- For information on *DPYD* testing and fluoropyrimidine-associated toxicity, see the NCCN Guidelines for Colon Cancer.
- Irinotecan should be used with caution in patients with Gilbert syndrome or elevated serum bilirubin. There is a commercially available test for UGT1A1. Guidelines for use in clinical practice have not been established.
- Anti-angiogenic therapy should be held for at least 28 days prior to elective surgery; do not administer for 28 days following major surgery and until adequate wound healing.
- For infection risk, monitoring, and prophylaxis recommendations for targeted therapies, see INF-A in the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections.

Checkpoint Inhibitor Immunotherapy[b]
- Nivolumab and hyaluronidase-nvhy is not approved for concurrent use with IV ipilimumab; however, for nivolumab monotherapy, nivolumab and hyaluronidase-nvhy subcutaneous injection may be substituted for IV nivolumab. Nivolumab and hyaluronidase-nvhy has different dosing and administration instructions compared to IV nivolumab.
- Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.
- Atezolizumab and hyaluronidase-tqjs subcutaneous injection may be substituted for IV atezolizumab. Atezolizumab and hyaluronidase-tqjs has different dosing and administration instructions compared to IV atezolizumab.
- If disease response on checkpoint inhibitor, consider discontinuing checkpoint inhibitor after 2 years of treatment.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

pMMR/MSS (or dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb] that is ineligible for or progressed on checkpoint inhibitor immunotherapy)

***KRAS* WT/*NRAS* WT/*BRAF* WT[c,d]**	
Initial Therapy	**Less Intensive Therapy (any line of therapy, if not previously given)**
• CAPEOX (Capecitabine/Oxaliplatin) ± Bevacizumab • CAPEOX + (Cetuximab or Panitumumab) • FOLFIRI (Leucovorin/Fluorouracil/Irinotecan) ± Bevacizumab • FOLFIRI + (Cetuximab or Panitumumab) • FOLFIRINOX (Leucovorin/Fluorouracil/Irinotecan/Oxaliplatin) ± Bevacizumab • FOLFOX (Leucovorin/Fluorouracil/Oxaliplatin) ± Bevacizumab • FOLFOX + (Cetuximab or Panitumumab)	• Capecitabine ± Bevacizumab • Cetuximab or Panitumumab[e] • Fluorouracil/Leucovorin ± Bevacizumab • (Pertuzumab or Tucatinib) + Trastuzumab (HER2 [*ERBB2*] overexpression/amplification)[d,f] • Tumor-Agnostic therapy (REC-F 6 of 17) • Best supportive care[g]
Subsequent Therapy (if not previously given)	
Prior Oxaliplatin	**Prior Oxaliplatin and Irinotecan or Beyond Second Line**
• FOLFIRI + (Cetuximab or Panitumumab) • FOLFIRI ± (Bevacizumab [preferred] or Ziv-aflibercept or Ramucirumab) • Irinotecan + (Cetuximab or Panitumumab) • Irinotecan ± (Bevacizumab [preferred] or Ziv-aflibercept or Ramucirumab)	• Fam-trastuzumab deruxtecan-nxki (HER2 overexpression [IHC 3+])[d,h] • (Pertuzumab or Tucatinib) + Trastuzumab (HER2 [*ERBB2*] overexpression/amplification)[d,f] • Tumor-Agnostic therapy (REC-F 6 of 17) • For disease that has progressed through all available regimens: ▸ Trifluridine and Tipiracil ± Bevacizumab (Bevacizumab combo preferred) ▸ Fruquintinib ▸ Regorafenib • Best supportive care[g]
Prior Irinotecan	
• CAPEOX ± Bevacizumab • CAPEOX + (Cetuximab or Panitumumab) • FOLFOX ± Bevacizumab • FOLFOX + (Cetuximab or Panitumumab) • Irinotecan + (Cetuximab or Panitumumab)	

Footnotes

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

pMMR/MSS (or dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb] that is ineligible for or progressed on checkpoint inhibitor immunotherapy)

<table>
<tr><th colspan="2">(KRAS mutation or NRAS mutation) and BRAF WT[c,d]</th></tr>
<tr><th>Initial Therapy</th><th>Less Intensive Therapy (any line of therapy, if not previously given)</th></tr>
<tr><td>• CAPEOX ± Bevacizumab
• FOLFIRI ± Bevacizumab
• FOLFIRINOX ± Bevacizumab
• FOLFOX ± Bevacizumab</td><td>• Capecitabine ± Bevacizumab
• Fluorouracil/Leucovorin ± Bevacizumab
• Adagrasib or Sotorasib monotherapy (KRAS G12C mutation)
• Tumor-Agnostic therapy (REC-F 6 of 17)
• Best supportive care[g]</td></tr>
<tr><th colspan="2">Subsequent Therapy (if not previously given)</th></tr>
<tr><th>Prior Oxaliplatin</th><th>Prior Oxaliplatin and Irinotecan or Beyond Second Line</th></tr>
<tr><td>• FOLFIRI ± (Bevacizumab [preferred] or Ziv-aflibercept or Ramucirumab)
• Irinotecan ± (Bevacizumab [preferred] or Ziv-aflibercept or Ramucirumab)</td><td rowspan="3">• Fam-trastuzumab deruxtecan-nxki (HER2 overexpression [IHC 3+])[d,h]
• (Adagrasib or Sotorasib) + (Cetuximab or Panitumumab) (KRAS G12C mutation)
• Tumor-Agnostic therapy (REC-F 6 of 17)
• For disease that has progressed through all available regimens:
▸ Trifluridine and Tipiracil ± Bevacizumab (Bevacizumab combo preferred)
▸ Fruquintinib
▸ Regorafenib
• Best supportive care[g]</td></tr>
<tr><th>Prior Irinotecan</th></tr>
<tr><td>• FOLFOX ± Bevacizumab
• CAPEOX ± Bevacizumab</td></tr>
</table>

Footnotes

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

pMMR/MSS (or dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb] that is ineligible for or progressed on checkpoint inhibitor immunotherapy)

<table>
<tr><th colspan="2">*KRAS* WT/*NRAS* WT/*BRAF* V600E mutation[c,d,i]</th></tr>
<tr><th>Initial Therapy</th><th>Less Intensive Therapy (any line of therapy, if not previously given)</th></tr>
<tr><td>Preferred
• FOLFIRI/Encorafenib + (Cetuximab or Panitumumab)
• FOLFOX/Encorafenib + (Cetuximab or Panitumumab)[j]

Other Recommended
• CAPEOX ± Bevacizumab
• CAPEOX/Encorafenib + (Cetuximab or Panitumumab)
• FOLFIRI ± Bevacizumab
• FOLFIRINOX ± Bevacizumab
• FOLFOX ± Bevacizumab</td><td>• Capecitabine ± Bevacizumab
• Encorafenib + (Cetuximab or Panitumumab)
• Fluorouracil/Leucovorin ± Bevacizumab
• Tumor-Agnostic therapy (REC-F 6 of 17)
• Best supportive care[g]</td></tr>
<tr><th colspan="2">Subsequent Therapy (if not previously given)</th></tr>
<tr><th>Prior Oxaliplatin</th><th>Prior Oxaliplatin and Irinotecan or Beyond Second Line</th></tr>
<tr><td>• FOLFIRI ± (Bevacizumab [preferred] or Ziv-aflibercept or Ramucirumab)
• Irinotecan ± (Bevacizumab [preferred] or Ziv-aflibercept or Ramucirumab)</td><td rowspan="3">• Fam-trastuzumab deruxtecan-nxki (HER2 overexpression [IHC 3+])[d,h]
• Tumor-Agnostic therapy (REC-F 6 of 17)
• For disease that has progressed through all available regimens:
▸ Trifluridine and Tipiracil ± Bevacizumab (Bevacizumab combo preferred)
▸ Fruquintinib
▸ Regorafenib
• Best supportive care[g]</td></tr>
<tr><th>Prior Irinotecan</th></tr>
<tr><td>• CAPEOX ± Bevacizumab
• FOLFOX ± Bevacizumab</td></tr>
</table>

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype (eg, TMB >50 mut/Mb)

Checkpoint Inhibitor Therapy for Any Line of Therapy[*,b,k] (no prior immunotherapy received)		
PD-1	**PD-L1**	**PD-1/CTLA-4**
• Cemiplimab-rwlc • Dostarlimab-gxly • Nivolumab • Pembrolizumab • Penpulimab-kcqx • Retifanlimab-dlwr • Tislelizumab-jsgr • Toripalimab-tpzi	• Atezolizumab • Avelumab • Cosibelimab-ipdl • Durvalumab	• Ipilimumab + Nivolumab

Prior Immunotherapy Received
• Systemic therapy (see REC-F) • Ipilimumab + Nivolumab (if checkpoint inhibitor monotherapy was previously received)

Footnotes

[*] Patients should be followed closely for 10 weeks to assess for response. Some of these agents have not been specifically studied in CRC but are included in this Guideline based on the presumed class effect from data demonstrated in other cancer types. See Discussion section for more information.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE

Tumor-Agnostic Therapies	
HER2 overexpression (IHC 3+)	• Fam trastuzumab deruxtecan[d,h]
***NTRK 1/2/3* gene fusion**	• Entrectinib • Larotrectinib • Repotrectinib[l]
***RET* gene fusion**	• Selpercatinib

Footnotes

Note: The table above includes certain FDA-approved tumor-agnostic therapy options. There are additional tumor-agnostic approvals that are not currently recommended by the Panel for CRC.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – FOOTNOTES

[a] Principles of Surgery and Locoregional Therapies (REC-C).
[b] NCCN Guidelines for Management of Immune Checkpoint Inhibitor-Related Toxicities.
[c] Patients with BRAF mutations other than V600E may be considered for anti-EGFR therapy.
[d] Principles of Pathologic Review (REC-B).
[e] Cetuximab or panitumumab are recommended in combination with irinotecan-based therapy or as single-agent therapy for patients who cannot tolerate irinotecan.
[f] If no previous treatment with HER2 inhibitor.
[g] NCCN Guidelines for Palliative Care.
[h] Some activity was seen after previous treatment with a HER2-targeted regimen. May not be indicated in patients with underlying lung issues due to lung toxicity (drug-related deaths from interstitial lung disease were reported in 3.5% of patients on the DESTINY-CRC01 trial).
[i] Patients with BRAF V600E-mutant cancers should receive an encorafenib containing regimen during one line of therapy. A chemotherapy containing option is preferred in the first-line metastatic setting.
[j] BRAF V600E regimen may be given with FOLFOX as subsequent-line therapy if no previous treatment with oxaliplatin or BRAF-targeting regimen.
[k] Combination programmed death ligand 1 (PD-L1)/cytotoxic T-lymphocyte–associated antigen 4 (CTLA-4) blockade has been associated with improved survival in unresectable or metastatic CRC, but also higher toxicity.
[l] On the TRIDENT-1 trial, repotrectinib showed activity in both NTRK TKI-naïve and NTRK TKI-pretreated patients.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – REGIMENS

Modified FOLFOX6[1,2,3]
Oxaliplatin 85 mg/m² IV Day 1[m]
Leucovorin 400 mg/m² IV Day 1[n]
Fluorouracil 400 mg/m² IV bolus on Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

Modified FOLFOX 7[4]
Oxaliplatin 85 mg/m² IV Day 1[m]
Leucovorin 400 mg/m² IV Day 1[n]
Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

FOLFOX + Bevacizumab[5,o]
Bevacizumab 5 mg/kg IV, Day 1
Repeat every 2 weeks

FOLFOX + Panitumumab[6] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFOX + Cetuximab[7] (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² IV over 2 hours first infusion, followed by 250 mg/m² IV over 60 minutes weekly
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks (preferred for every 2 weeks)

CAPEOX[8]
Oxaliplatin 130 mg/m² IV Day 1[m]
Capecitabine 1000[p] mg/m² twice daily PO for 14 days
Repeat every 3 weeks

CAPEOX + Bevacizumab[8,o]
Oxaliplatin 130 mg/m² IV Day 1[m]
Capecitabine 1000[p] mg/m² PO twice daily for 14 days
Bevacizumab 7.5 mg/kg IV Day 1
Repeat every 3 weeks

CAPEOX + Cetuximab[9-11] (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² IV over 2 hours first infusion, followed by 250 mg/m² IV over 60 minutes weekly
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks (preferred for every 2 weeks)

CAPEOX + Panitumumab[9-11] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFIRI[12,13]
Irinotecan 180 mg/m² IV over 30–90 minutes, Day 1
Leucovorin[n] 400 mg/m² IV infusion to match duration of Irinotecan infusion, Day 1
Fluorouracil 400 mg/m² IV bolus Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) continuous infusion
Repeat every 2 weeks

FOLFIRI + Bevacizumab[14,o]
Bevacizumab 5 mg/kg IV, Day 1
Repeat every 2 weeks

FOLFIRI + Cetuximab (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² IV over 2 hours first infusion, followed by 250 mg/m² IV over 60 minutes weekly[15]
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks[16] (preferred for every 2 weeks)

[m] Oxaliplatin may be given either over 2 hours, or may be infused over a shorter time at a rate of 1 mg/m²/min. Leucovorin infusion should match infusion time of oxaliplatin. Cercek A, Park V, Yaeger R, et al. Faster FOLFOX: oxaliplatin can be safely infused at a rate of 1 mg/m²/min. J Oncol Pract 2016;12:e548-553.
[n] Leucovorin 400 mg/m² is the equivalent of levoleucovorin 200 mg/m².
[o] Bevacizumab may be safely given at a rate of 0.5 mg/kg/min (5 mg/kg over 10 minutes and 7.5 mg/kg over 15 minutes).
[p] The majority of safety and efficacy data for this regimen have been developed in Europe, where a capecitabine starting dose of 1000 mg/m² twice daily for 14 days, repeated every 21 days, is standard. Evidence suggests that North American patients may experience greater toxicity with capecitabine (as well as with other fluoropyrimidines) than European patients, and may require a lower dose of capecitabine.

References

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – REGIMENS

FOLFIRI + Panitumumab[17] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFIRI + Ziv-aflibercept[18]
Ziv-aflibercept 4 mg/kg IV over 60 minutes, Day 1
Repeat every 2 weeks

FOLFIRI + Ramucirumab[19]
Ramucirumab 8 mg/kg over 60 minutes, Day 1
Repeat every 2 weeks

FOLFIRINOX[20,q]
Oxaliplatin 85 mg/m² IV on Day 1,[m] Leucovorin 400 mg/m² IV over 2 hours on Day 1, Irinotecan 165–180 mg/m² IV over 30–90 minutes on Day 1, Fluorouracil 400 mg/m² IV push Day 1, Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46 hours) continuous infusion.
Repeat every 2 weeks

Modified FOLFIRINOX[21-23,q]
Oxaliplatin 85 mg/m² IV on Day 1,[m] Leucovorin 400 mg/m² IV over 2 hours on Day 1, Irinotecan 150 mg/m² IV over 30–90 minutes on Day 1, Fluorouracil 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46 hours) continuous infusion.
Repeat every 2 weeks

FOLFIRINOX or Modified FOLFIRINOX + Bevacizumab[24,o]
Bevacizumab 5 mg/kg IV, Day 1
Repeat every 2 weeks

Bolus or infusional Fluorouracil/Leucovorin
Roswell Park regimen[25]
Leucovorin 500 mg/m² IV over 2 hours, Days 1, 8, 15, 22, 29, and 36
Fluorouracil 500 mg/m² IV bolus 1 hour after start of Leucovorin, Days 1, 8, 15, 22, 29, and 36
Repeat every 8 weeks

Simplified biweekly infusional Fluorouracil/Leucovorin (sLV5FU2)[12]
Leucovorin[n] 400 mg/m² IV over 2 hours on Day 1, followed by Fluorouracil bolus 400 mg/m² followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) continuous infusion
Repeat every 2 weeks

Weekly
Leucovorin 20 mg/m² IV over 2 hours on Day 1, Fluorouracil 500 mg/m² IV bolus injection 1 hour after the start of Leucovorin
Repeat weekly[26]
or
Fluorouracil 2600 mg/m² by 24-hour infusion plus Leucovorin 500 mg/m²
Repeat every week[26]

Bolus or infusional Fluorouracil + Bevacizumab[o]
Bevacizumab 5 mg/kg IV on Day 1
Repeat every 2 weeks

Capecitabine[27,p]
Capecitabine 850–1250 mg/m² PO twice daily for 14 days
Repeat every 3 weeks

[m] Oxaliplatin may be given either over 2 hours, or may be infused over a shorter time at a rate of 1 mg/m²/min. Leucovorin infusion should match infusion time of oxaliplatin. Cercek A, Park V, Yaeger R, et al. Faster FOLFOX: oxaliplatin can be safely infused at a rate of 1 mg/m²/min. J Oncol Pract 2016;12:e548-553.
[n] Leucovorin 400 mg/m² is the equivalent of levoleucovorin 200 mg/m².
[o] Bevacizumab may be safely given at a rate of 0.5 mg/kg/min (5 mg/kg over 10 minutes and 7.5 mg/kg over 15 minutes).
[q] FOLFIRINOX is recommended instead of FOLFOXIRI because FOLFOXIRI uses a high dose of fluorouracil (3200 mg/m² over 48 hours). Patients in the United States have been shown to have greater toxicity with fluorouracil. The dose of fluorouracil (2400 mg/m² over 46 hours) is a starting dose consistent with the dose recommended in FOLFOX or FOLFIRI and should be strongly considered for U.S. patients.

References

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – REGIMENS

Capecitabine + Bevacizumab[28,o]
Bevacizumab 7.5 mg/kg IV, Day 1
Repeat every 3 weeks

Irinotecan
Irinotecan 125 mg/m² IV over 30–90 minutes, Days 1 and 8
Repeat every 3 weeks[29,30]
or Irinotecan 180 mg/m² IV over 30–90 minutes, Day 1
Repeat every 2 weeks
or Irinotecan 300–350 mg/m² IV over 30–90 minutes, Day 1
Repeat every 3 weeks

Irinotecan + Cetuximab (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² first infusion, followed by 250 mg/m² IV weekly[31]
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks[16]
(preferred for every 2 weeks)

Irinotecan + Panitumumab[17,32] (*KRAS*/*NRAS*/*BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes every 2 weeks

Irinotecan + Bevacizumab[33,o]
Irinotecan 180 mg/m² IV, Day 1
Bevacizumab 5 mg/kg IV, Day 1
Repeat every 2 weeks
or
Irinotecan 300–350 mg/m² IV, Day 1
Bevacizumab 7.5 mg/kg IV, Day 1
Repeat every 3 weeks

Irinotecan + Ramucirumab[19]
Ramucirumab 8 mg/kg IV over 60 minutes every 2 weeks

Irinotecan + Ziv-aflibercept
Irinotecan 180 mg/m² IV, Day 1
Ziv-aflibercept 4 mg/kg IV, Day 1
Repeat every 2 weeks

Cetuximab (*KRAS/NRAS/BRAF* WT)
Cetuximab 400 mg/m² first infusion, followed by 250 mg/m² IV weekly[31]
or Cetuximab 500 mg/m² IV over 2 hours, Day 1, every 2 weeks[16]
(preferred for every 2 weeks)

Panitumumab[34] (*KRAS/NRAS/BRAF* WT)
Panitumumab 6 mg/kg IV over 60 minutes every 2 weeks

Regorafenib
Regorafenib 160 mg PO daily on Days 1–21[35]
or
First cycle: Regorafenib 80 mg PO daily on Days 1–7, followed by 120 mg PO daily on Days 8–14, followed by 160 mg PO daily on Days 15–21[36]
Subsequent cycles: Regorafenib 160 mg PO daily on Days 1–21
Repeat every 28 days

Trifluridine and Tipiracil ± Bevacizumab[37,38]
Trifluridine and Tipiracil 35 mg/m² up to a maximum dose of 80 mg per dose (based on the Trifluridine component)
PO twice daily Days 1–5 and 8–12
Bevacizumab 5 mg/kg on Days 1 and 15
Repeat every 28 days

[o] Bevacizumab may be safely given at a rate of 0.5 mg/kg/min (5 mg/kg over 10 minutes and 7.5 mg/kg over 15 minutes).
[p] The majority of safety and efficacy data for this regimen have been developed in Europe, where a capecitabine starting dose of 1000 mg/m² twice daily for 14 days, repeated every 21 days, is standard. Evidence suggests that North American patients may experience greater toxicity with capecitabine (as well as with other fluoropyrimidines) than European patients, and may require a lower dose of capecitabine.

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – REGIMENS

Pembrolizumab[39] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Pembrolizumab 2 mg/kg IV every 3 weeks
or Pembrolizumab 200 mg IV every 3 weeks
or Pembrolizumab 400 mg IV every 6 weeks
or Pembrolizumab 4 mg/kg IV every 6 weeks

Nivolumab[40] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Nivolumab 3 mg/kg every 2 weeks
or Nivolumab 240 mg IV every 2 weeks
or Nivolumab 480 mg IV every 4 weeks
or Nivolumab 6 mg/kg IV every 4 weeks

Ipilimumab + Nivolumab[41] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Nivolumab 3 mg/kg (30-minute IV infusion) and Ipilimumab 1 mg/kg (30-minute IV infusion) once every 3 weeks for four doses, followed by Nivolumab 3 mg/kg IV or Nivolumab 240 mg IV every 2 weeks or Nivolumab 480 mg IV every 4 weeks

Dostarlimab-gxly[42] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
Dostarlimab-gxly 500 mg IV every 3 weeks for 4 doses followed by 1000 mg IV every 6 weeks

Cemiplimab-rwlc[43,44] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
350 mg IV on Day 1
Repeat every 3 weeks

Retifanlimab-dlwr[45,46] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
500 mg IV on Day 1
Repeat every 4 weeks

Tislelizumab-jsgr[47-50] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
200 mg IV on Day 1
Repeat every 3 weeks

Toripalimab-tpzi[51] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
3 mg/kg IV on Day 1
Repeat every 2 weeks

Penpulimab-kcqx[52] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
200 mg IV Day 1
Repeat every 2 weeks

Atezolizumab[53] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
840 mg IV Day 1 every 14 days
1200 mg IV Day 1 every 21 days
1680 mg IV Day 1 every 28 days

Avelumab[54,55] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
800 mg IV Day 1
Every 14 days

Cosibelimab-ipdl[56] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
1200 mg IV Day 1
Every 21 days

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – REGIMENS

Durvalumab[57] (dMMR/MSI-H or *POLE/POLD1* mutation with ultra-hypermutated phenotype [eg, TMB >50 mut/Mb])
1500 mg IV Day 1
Every 28 days

Pertuzumab + Trastuzumab[58]
(HER2 [*ERBB2*] overexpression/amplification and *KRAS, NRAS,* and *BRAF* WT)
Trastuzumab 8 mg/kg IV loading dose on Day 1 of cycle 1, followed by 6 mg/kg IV every 21 days
Pertuzumab 840 mg IV loading dose on Day 1 of cycle 1, followed by 420 mg IV every 21 days

Tucatinib + Trastuzumab[59]
(HER2 [*ERBB2*] overexpression/amplification and *KRAS, NRAS,* and *BRAF* WT)
Trastuzumab 8 mg/kg IV loading dose on Day 1 of cycle 1, followed by 6 mg/kg IV every 21 days
Tucatinib 300 mg PO twice daily

Fam-trastuzumab deruxtecan-nxki[60] (HER2 overexpression, IHC 3+)
Fam-trastuzumab deruxtecan-nxki 5.4 mg/kg IV on Day 1
Repeat every 21 days

Encorafenib + Cetuximab[61-63]
(*BRAF* V600E mutation)
Encorafenib 300 mg PO daily
Cetuximab 400 mg/m² IV followed by 250 mg/m² IV weekly or Cetuximab 500 mg/m² IV every 2 weeks

Encorafenib + Panitumumab[62-64]
(*BRAF* V600E mutation)
Encorafenib 300 mg PO daily
Panitumumab 6 mg/kg IV every 14 days

FOLFOX/Encorafenib + Cetuximab[64]
(*BRAF* V600E mutation)
Encorafenib 300 mg PO daily
Cetuximab 500 mg/m² IV Day 1
Oxaliplatin 85 mg/m² IV Day 1
Leucovorin 400 mg/m² IV Day 1
Fluorouracil 400 mg/m² IV bolus on Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

FOLFOX/Encorafenib + Panitumumab[64]
(*BRAF* V600E mutation)
Encorafenib 300 mg PO daily
Panitumumab 6mg/kg IV every Day 1
Oxaliplatin 85 mg/m² IV Day 1
Leucovorin 400 mg/m² IV Day 1
Fluorouracil 400 mg/m² IV bolus on Day 1, followed by 1200 mg/m²/day x 2 days (total 2400 mg/m² over 46–48 hours) IV continuous infusion
Repeat every 2 weeks

FOLFIRI/Encorafenib + Cetuximab (*BRAF* V600E mutation)[65,66]
Encorafenib 300 mg PO daily
Cetuximab 500 mg/m² IV every 14 days
Irinotecan 180 mg/m² IV over 30–90 minutes, Day 1
Leucovorin 400 mg/m² IV infusion to match duration of Irinotecan infusion, Day 1
Fluorouracil 400 mg/m² IV bolus, 2400 mg/m² IV continuous infusion (over 46–48 hours) every 14 days

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PRINCIPLES OF SYSTEMIC THERAPY FOR ADVANCED OR METASTATIC DISEASE – REGIMENS

FOLFIRI/Encorafenib + Panitumumab (*BRAF* V600E mutation)[65,66]
Encorafenib 300 mg PO daily
Panitumumab 6 mg/kg IV every 14 days
Irinotecan 180 mg/m² IV over 30–90 minutes, Day 1
Leucovorin 400 mg/m² IV infusion to match duration of Irinotecan infusion, Day 1
Fluorouracil 400 mg/m² IV bolus, 2400 mg/m² IV continuous infusion (over 46–48 hours) every 14 days

CAPEOX/Encorafenib + Cetuximab
Encorafenib 300 mg PO daily
Cetuximab 500 mg/m² IV every 14 days
Oxaliplatin 130 mg/m² IV, Capecitabine 1000 mg/m² twice daily PO for 14 days; repeat every 3 weeks

CAPEOX/Encorafenib + Panitumumab
Encorafenib 300 mg PO daily
Panitumumab 6 mg/kg IV every 14 days
Oxaliplatin 130 mg/m² IV, Capecitabine 1000 mg/m² twice daily PO for 14 days; repeat every 3 weeks

Larotrectinib[67] (*NTRK 1/2/3* gene fusion)
100 mg PO twice daily

Entrectinib[68] (*NTRK 1/2/3* gene fusion)
600 mg PO once daily

Repotrectinib[69] (*NTRK 1/2/3* gene fusion)
160 mg PO daily for first 14 days,
Then increase to 160 mg PO twice daily

Selpercatinib[70] (*RET* gene fusion)
- Patients ≥50 kg: 160 mg PO twice daily
- Patients <50 kg: 120 mg PO twice daily

Adagrasib (*KRAS* G12C mutation)
600 mg PO BID

Sotorasib[71] (*KRAS* G12C mutation)
960 mg PO daily

Adagrasib + Cetuximab[72] (*KRAS* G12C mutation)
Adagrasib 600 mg PO BID
Cetuximab 500 mg/m² IV every 2 weeks

Adagrasib + Panitumumab (*KRAS* G12C mutation)
Adagrasib 600 mg PO BID
Panitumumab 6 mg/kg IV every 2 weeks

Sotorasib + Cetuximab (*KRAS* G12C mutation)
Sotorasib 960 mg PO daily
Cetuximab 500 mg/m² IV every 2 weeks

Sotorasib + Panitumumab[73] (*KRAS* G12C mutation)
Sotorasib 960 mg PO daily
Panitumumab 6 mg/kg IV every 2 weeks

Fruquintinib[74]
5 mg PO daily on Days 1–21
Repeat every 28 days

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PRINCIPLES OF SURVIVORSHIP – COLORECTAL LONG-TERM FOLLOW-UP CARE

Colorectal Cancer Surveillance

- See REC-10.
- Long-term surveillance should be carefully managed with routine good medical care and monitoring, including cancer screening, routine health care, and preventive care.
- Routine CEA monitoring and routine CT scanning are not recommended beyond 5 years.

Survivorship Care Planning

The oncologist and primary care provider should have defined roles in the surveillance period, with roles communicated to the patient.[1]

- Develop survivorship care plan that includes:
  - Overall summary of treatment, including all surgeries, radiation treatments, and chemotherapy received
  - Description of possible expected time to resolution of acute toxicities, long-term effects of treatment, and possible late sequelae of treatment
  - Surveillance recommendations
  - Delineation of appropriate timing of transfer of care with specific responsibilities identified for primary care physician and oncologist
  - Health behavior recommendations
  - Fertility counseling

Management of Late/Long-Term Sequelae of Disease or Treatment[2-6]

- For issues related to distress, pain, neuropathy, fatigue, or sexual dysfunction, see NCCN Guidelines for Survivorship.
- Bowel function changes: chronic diarrhea, incontinence, stool frequency, stool clustering, urgency, and cramping
  - Consider anti-diarrheal agents, bulk-forming agents, diet manipulation, pelvic floor rehabilitation, and protective undergarments.
  - Management of an ostomy
    - ◊ Consider participation in an ostomy support group or coordination of care with a health care provider specializing in ostomy care (ie, ostomy nurse).
    - ◊ Screen for distress around body changes (NCCN Guidelines for Distress Management) and precautions around involvement with physical activity (SPA-A in the NCCN Guidelines for Survivorship).
- For oxaliplatin-induced neuropathy
  - Consider duloxetine for painful neuropathy only, not effective for numbness, tingling, or cold sensitivity.[7]
  - Refer to pain management specialist for refractory cases.
  - Pregabalin or gabapentin are not recommended.
- Urogenital dysfunction after resection and/or pelvic radiation[8,9]
  - Screen for sexual dysfunction, erectile dysfunction, dyspareunia, and vaginal dryness.
  - Screen for urinary incontinence, frequency, and urgency.
  - Consider referral to urologist or gynecologist for persistent symptoms.
- Potential for pelvic fractures/decreased bone density after pelvic radiation
  - Consider bone density monitoring.

Counseling Regarding Healthy Lifestyle and Wellness[10]

(NCCN Guidelines for Survivorship)

- Undergo all age- and gender-appropriate cancer and preventive health screenings as per national guidelines.
- Maintain a healthy body weight throughout life.
- Adopt a physically active lifestyle. Discuss and facilitate options to achieve a goal of at least 150 minutes of moderate to vigorous physical activity per week. Activity recommendations may require modification based on treatment sequelae (ie, ostomy, neuropathy).
- Consume a healthy diet with an emphasis on plant sources. Diet recommendations may be modified based on severity of bowel dysfunction.
- Consider daily aspirin 325 mg for secondary prevention.
- Drink alcohol sparingly, if at all.
- Seek smoking cessation counseling as appropriate (NCCN Guidelines for Smoking Cessation).

Additional health monitoring and immunizations should be performed as indicated under the care of a primary care physician. Survivors are encouraged to maintain a therapeutic relationship with a primary care physician throughout their lifetime.

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PRINCIPLES OF NONOPERATIVE MANAGEMENT

To provide nonoperative management (NOM) for patients with rectal cancer, the multidisciplinary team's diagnostic skills are crucial. They must accurately assess clinical, radiologic, and pathologic findings, determining patient eligibility for NOM and closely monitoring progress. The team's expertise extends to tracking treatment responses, identifying surgical needs promptly, and adjusting the management plan as necessary. Additionally, the team should maintain a comprehensive understanding of the watchful waiting literature and surveillance methodology, adeptly treating patients with complete or near-complete clinical responses and regularly monitoring for potential tumor recurrence or progression. Given this, NOM is recommended only at centers with experienced multidisciplinary teams and for patients committed to intensive surveillance.

Criteria for Complete Clinical Response

- High-definition flexible endoscopy[1]
  - Pale smooth scar with or without telangiectasia
  - No ulceration, nodularity, or mucosal irregularities
  - No stricture
- DRE[1]
  - Smooth, flat scar
  - No nodularity
- Diffusion-weighted MRI[2]
  - Fibrotic, linear scar with low signal intensity on T2-weighted images
  - No diffusion restriction
  - No suspicious lymph nodes
- All of the criteria must be satisfied in order to define a complete clinical response
- Biopsy offers no added diagnostic value if the criteria are met[3,4]
- ctDNA has no proven role in the NOM of patients

Timing of Assessment for Complete Clinical Response

- For patients treated with chemotherapy first followed by radiation (induction chemotherapy), assessment should be performed no earlier than 8 weeks after completion of radiotherapy to allow time for delayed response to radiation.[5]
- For patients treated with radiation first followed by chemotherapy (consolidation chemotherapy), assessment should be completed within a month of completion of chemotherapy.

Near Complete Response[6,7]

- If the patient has had a nCR and wishes to avoid surgery, then an additional 8 weeks of observation followed by reassessment can be considered.
  - An nCR is defined by:
    - ◊ Smooth induration or superficial minor mucosal irregularity on DRE
    - ◊ Endoscopic appearance with irregular small mucosal nodules, superficial ulceration, or mild persistent erythema
    - ◊ T2-weighted MRI with downstaging with or without residual fibrosis, small area of residual signal, and complete or partial regression of lymph nodes
    - ◊ Diffusion-weighted MRI with small area of residual high signal intensity

Indications for Surgery

- Radical surgery is indicated for patients who do not ultimately achieve a complete clinical response based on the above criteria or patients who have tumor regrowth after a clinical response.
- If residual tumor or regrowth is suspected at the time of assessment, it is not necessary to perform biopsies. False-negative biopsies are common in this scenario and a high degree of suspicion for tumor is sufficient as an indication for surgery.[8]

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American Joint Committee on Cancer (AJCC) TNM Staging Classification for Rectal Cancer 8th ed., 2017

Table 1. **Definitions for T, N, M**

T	**Primary Tumor**
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma *in situ*, intramucosal carcinoma (involvement of lamina propria with no extension through muscularis mucosae)
T1	Tumor invades the submucosa (through the muscularis mucosa but not into the muscularis propria)
T2	Tumor invades the muscularis propria
T3	Tumor invades through the muscularis propria into pericolorectal tissues
T4	Tumor invades* the visceral peritoneum or invades or adheres** to adjacent organ or structure
T4a	Tumor invades* through the visceral peritoneum (including gross perforation of the bowel through tumor and continuous invasion of tumor through areas of inflammation to the surface of the visceral peritoneum)
T4b	Tumor directly invades* or adheres** to adjacent organs or structures

N	**Regional Lymph Nodes**
NX	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	One to three regional lymph nodes are positive (tumor in lymph nodes measuring ≥0.2 mm), or any number of tumor deposits are present and all identifiable lymph nodes are negative
N1a	One regional lymph node is positive
N1b	Two or three regional lymph nodes are positive
N1c	No regional lymph nodes are positive, but there are tumor deposits in the subserosa, mesentery, or nonperitonealized pericolic, or perirectal/mesorectal tissues
N2	Four or more regional lymph nodes are positive
N2a	Four to six regional lymph nodes are positive
N2b	Seven or more regional lymph nodes are positive

M	**Distant Metastasis**
M0	No distant metastasis by imaging, etc.; no evidence of tumor in distant sites or organs. (This category is not assigned by pathologists)
M1	Metastasis to one or more distant sites or organs or peritoneal metastasis is identified
M1a	Metastasis to one site or organ is identified without peritoneal metastasis
M1b	Metastasis to two or more sites or organs is identified without peritoneal metastasis
M1c	Metastasis to the peritoneal surface is identified alone or with other site or organ metastases

* Direct invasion in T4 includes invasion of other organs or other segments of the colorectum as a result of direct extension through the serosa, as confirmed on microscopic examination (for example, invasion of the sigmoid colon by a carcinoma of the cecum) or, for cancers in a retroperitoneal or subperitoneal location, direct invasion of other organs or structures by virtue of extension beyond the muscularis propria (i.e., respectively, a tumor on the posterior wall of the descending colon invading the left kidney or lateral abdominal wall; or a mid or distal rectal cancer with invasion of prostate, seminal vesicles, cervix, or vagina).

** Tumor that is adherent to other organs or structures, grossly, is classified cT4b. However, if no tumor is present in the adhesion, microscopically, the classification should be pT1-4a depending on the anatomical depth of wall invasion. The V and L classification should be used to identify the presence or absence of vascular or lymphatic invasion whereas the PN prognostic factor should be used for perineural invasion.

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American Joint Committee on Cancer (AJCC)
TNM Staging System for Rectal Cancer 8th ed., 2017

Table 2. **Prognostic Groups**

	T	N	M
Stage 0	Tis	N0	M0
Stage I	T1, T2	N0	M0
Stage IIA	T3	N0	M0
Stage IIB	T4a	N0	M0
Stage IIC	T4b	N0	M0
Stage IIIA	T1-T2	N1/N1c	M0
	T1	N2a	M0
Stage IIIB	T3-T4a	N1/N1c	M0
	T2-T3	N2a	M0
	T1-T2	N2b	M0
Stage IIIC	T4a	N2a	M0
	T3-T4a	N2b	M0
	T4b	N1-N2	M0
Stage IVA	Any T	Any N	M1a
Stage IVB	Any T	Any N	M1b
Stage IVC	Any T	Any N	M1c

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ABBREVIATIONS

3D-CRT	three-dimensional conformal radiation therapy
ASCO	American Society of Clinical Oncology
BED	biologically effective dose
C/A/P	chest/abdomen/pelvis
CAP	College of American Pathologists
CBC	complete blood count
cCR	clinical complete response
CEA	carcinoembryonic antigen
CLIA	Clinical Laboratory Improvement Amendments
CLIA-88	clinical Laboratory Improvement Amendments of 1988
CRC	colorectal cancer
CRM	circumferential resection margin
ctDNA	circulating tumor DNA
CTLA-4	cytotoxic T-lymphocyte–associated antigen 4
CTV	clinical target volume
DEB-TACE	drug-eluting bead transarterial chemoembolization
dMMR	mismatch repair deficient
DRE	digital rectal examination
DWI	diffusion-weighted imaging
EBRT	external beam radiation therapy
ED	exonuclease domain
EMVI	extramural vascular invasion
ESD	endoscopic submucosal dissection
EUS	endoscopic ultrasound
FAP	familial adenomatous polyposis
FISH	fluorescence in situ hybridization
GBCA	gadolinium-based contrast agent
GFR	glomerular filtration rate
GTV	gross tumor volume
HAI	hepatic arterial infusion
H&E	hematoxylin and eosin
ICRU	International Commission on Radiation Units and Measurements
IGRT	image-guided radiation therapy
IHC	immunohistochemistry
IMRT	intensity-modulated radiation therapy
IORT	intraoperative radiation therapy
ITC	isolated tumor cluster
LS	Lynch syndrome
LST	laterally spreading tumor
MEN2	multiple endocrine neoplasia type 2
MGPT	multigene panel testing
MMR	mismatch repair
MRF	mesorectal fascia
MSI	microsatellite instability
MSI-H	microsatellite instability-high
MSS	microsatellite stable
nCR	near complete response
NGS	next-generation sequencing
NOM	nonoperative management
PCR	polymerase chain reaction
PD-1	programmed cell death protein 1
PD-L1	programmed death ligand 1
pMMR	proficient mismatch repair
PNI	perineural invasion
PPAP	polymerase proofreading-associated polyposis
PV	pathogenic variant
SBRT	stereotactic body radiation therapy
SNV	single nucleotide variant

Continued

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ABBREVIATIONS

TACE	**transarterial chemoembolization**
TAMIS	**transanal minimally invasive surgery**
TARE	**transarterial radioembolization**
TEM	**transanal endoscopic microsurgery**
TMB	**tumor mutational burden**
TMB-H	**tumor mutational burden-high**
TME	**total mesorectal excision**
TNT	**total neoadjuvant therapy**
ULN	**upper limit of normal**
VEGF	**vascular endothelial growth factor**
WT	**wild-type**

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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Discussion

This discussion corresponds to the NCCN Guidelines for Rectal Cancer. Last updated October 31, 2025

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Discussion update in progress

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Overview

Colorectal cancer (CRC) is the fourth most frequently diagnosed cancer and the second leading cause of cancer death in the United States. The majority of the large bowel is referred to as the colon, while the most distal portion of approximately 15 cm (6 inches) is called the rectum. Approximately a third of CRCs occur in the rectum. In 2025, an estimated 46,950 new cases of rectal cancer will occur in the United States (27,950 cases in males; 19,000 cases in females). During the same year, it is estimated that 52,900 people will die from rectal and colon cancers combined.[1] Despite these high numbers, the incidence of colon and rectal cancers per 100,000 people decreased from 60.5 in 1976 to 46.4 in 2005 and, more recently, 38.7 in 2016.[2,3] In addition, mortality from CRC has been decreasing for decades (since 1947 in females and since 1980 in males) and is currently down by >50% from peak mortality rates.[1,3] These improvements in incidence of and mortality from CRC are thought to be a result of cancer prevention and earlier diagnoses through screening and of better treatment modalities. Recent data show continued rapid declines in incidence among those aged ≥65 years, with a decrease of 3.3% annually between 2011 and 2016.[3] CRC incidence and mortality rates vary by race and ethnicity with the highest rates in non-Hispanic Black individuals and lowest in Asian Americans/Pacific Islanders.[3] The magnitude of disparity in mortality rates is double that of incidence rates. Reasons for these racial disparities include differences in risk factor prevalence, access to health care and other social determinants of health, comorbidities, and tumor characteristics.

Conversely, incidence has increased among those <65 years, with a 1% annual increase in those aged 50 to 64 years and 2% annual increase in those <50 years. CRC death rates also showed age-dependent trends, declining by 3% annually for those ≥65 years of age, compared to a 0.6% annual decline for individuals aged 50 to 64 years and a 1.3% annual increase for individuals <50 years.[3] A retrospective cohort study of the SEER CRC registry demonstrated the rising incidence of CRC in patients <50 years.[4] The authors estimate that the incidence rates for colon and rectal cancers will increase by 90.0% and 124.2%, respectively, for patients 20 to 34 years by 2030. The cause of this trend is currently unknown. One review suggests that CRC that occurs in patients <45 years may be clinicopathologically and genetically different from CRC in adults ≥45 years, although this has not been confirmed broadly. If cancer in this population is different, there would be a need to develop specific treatment strategies for this population.[5] In a cohort study of 1959 patients with metastatic CRC (mCRC), patients who developed mCRC at a younger age (<50 years) showed worse survival outcomes and unique adverse event (AE) profiles, which the authors partially attribute to distinct genetic profiles.[6] On the other hand, other studies show no difference between early- and late-onset CRC, so more research is needed.[7,8]

This Discussion summarizes the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Rectal Cancer. These guidelines begin with the clinical presentation of the patient to the primary care physician or gastroenterologist and address diagnosis, pathologic staging, surgical management, perioperative treatment, management of recurrent and metastatic disease, patient surveillance, and survivorship. These guidelines overlap considerably with the NCCN Guidelines® for Colon Cancer, especially in the treatment of metastatic disease. To avoid repetition, this discussion may refer to the NCCN Guidelines for Colon Cancer for these sections, which would otherwise be identical. NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

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Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of the NCCN Guidelines for Rectal Cancer, an electronic search of the PubMed database was performed to obtain key literature in CRC published since the previous Guidelines update, using the search terms: "colon cancer, colorectal cancer, rectal cancer." The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[9]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[10] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Risk Assessment

Approximately 20% of cases of CRC are associated with familial clustering, and first-degree relatives of patients with colorectal adenomas or invasive CRC are at increased risk for CRC.[11-15] Genetic susceptibility to CRC includes well-defined inherited syndromes, such as Lynch syndrome (also known as hereditary nonpolyposis CRC [HNPCC]) and familial adenomatous polyposis (FAP).[16-18] Therefore, it is recommended that all patients with CRC be queried regarding their family history and considered for risk assessment, as detailed in the NCCN Guidelines for Colorectal Cancer Screening (available at www.NCCN.org). Similar risk factors exist for colon cancer and rectal cancer. Please refer to *Risk Assessment* in the NCCN Guidelines for Colon Cancer (available at www.NCCN.org) for a detailed discussion of familial syndromes and other factors associated with CRC risk.

TNM Staging

The NCCN Guidelines for Rectal Cancer adhere to the current TNM (tumor, node, metastases) staging system of the American Joint Committee on Cancer (AJCC) Cancer Staging Manual (Table 1 of the guidelines).[19] The TNM categories reflect very similar survival outcomes

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for rectal and colon cancers; these diseases therefore share the same staging system.

In the 8th edition of the AJCC Cancer Staging Manual, T1 tumors involve the submucosa; T2 tumors penetrate through the submucosa into the muscularis propria; T3 tumors penetrate through the muscularis propria; T4a tumors directly penetrate to the surface of the visceral peritoneum; and T4b tumors directly invade or are adherent to other organs or structures.[19]

Regional lymph node classification includes N1a (1 positive lymph node); N1b (2–3 positive lymph nodes); N2a (4–6 positive nodes); and N2b (≥7 positive nodes). In addition, tumor deposit(s) in the subserosa, mesentery, or non-peritonealized pericolic or perirectal tissues without regional nodal metastasis (ie, satellite tumor nodules) have been classified as N1c. Within each T stage, survival is inversely correlated with N stage (N0, N1a, N1b, N2a, and N2b).[19]

In rectal cancer, T stage has more prognostic value than N stage: patients with stage IIIA disease (T1–2) have longer rectal cancer-specific survival than patients with stage IIA (T3), IIB (T4a), and IIC (T4b) rectal cancer.[20]

Metastatic disease is classified as M1a when metastases are to only one site/solid organ (including to lymph nodes outside the primary tumor regional drainage area). M1b is used for metastases to multiple distant sites or solid organs, exclusive of peritoneal carcinomatosis. The 8th edition of the AJCC Cancer Staging Manual includes the M1c category for peritoneal carcinomatosis with or without blood-borne metastasis to visceral organs.[19] Patients with peritoneal metastases have a shorter progression-free survival (PFS) and overall survival (OS) than those without peritoneal involvement.[21]

The prefixes “p” and “yp” used in TNM staging denote pathologic staging and pathologic staging following neoadjuvant therapy, respectively.[19]

Pathology

Pathologic staging information is provided by examination of the surgical specimen. Some of the information that should be detailed in the report of the pathologic evaluation of rectal cancer includes: 1) gross description of the tumor and specimen; 2) grade of the cancer; 3) depth of penetration and extension to adjacent structures (T); 4) number of regional lymph nodes evaluated; 5) number of positive regional lymph nodes (N); 6) the presence of distant metastases to other organs or sites including non-regional lymph nodes (M); 7) the status of proximal, distal, circumferential (radial), and mesenteric margins[22-26]; 8) neoadjuvant treatment effect[27,28]; 9) lymphovascular invasion (LVI)[29]; 10) perineural invasion (PNI)[30-32]; and 11) the number of tumor deposits.[33-37]

Margins

The 8th edition of the AJCC Cancer Staging Manual includes the suggestion that the surgeon mark the area of the specimen with the deepest tumor penetration so that the pathologist can directly evaluate the status of the resection margins.[19]

The circumferential margin or circumferential resection margin (CRM) is an important pathologic staging parameter in rectal cancer.[38] The radial margin for resected segments of the colon that are completely encased by a peritonealized (serosal) surface is also referred to as the peritoneal margin. The CRM is very important in segments of the colon or rectum that are either not encased or only partially encased in peritoneum.[38] The CRM is the closest radial margin between the deepest penetration of the tumor and the edge of resected soft tissue around the rectum (ie, the retroperitoneal or subperitoneal aspect of the tumor) or from the edge of a lymph node and should be measured in millimeters. Identification of the

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CRM is determined through evaluation of the outer circumference of the rectal and mesorectal specimen that often requires inking of the outer surfaces and "bread-loaf" slicing of the specimen.[39] The Panel defines an involved or threatened CRM as tumor within 1 mm from the resected margin.[24,26,40,41] This definition differs slightly from the recommendations of the 2016 European Society of Gastrointestinal and Abdominal Radiology (ESGAR) consensus meeting in that ESGAR defined the mesorectal fascia as "involved" when the distance between the mesorectal fascia and the tumor is ≤1 mm, while in their template, "threatened/involved" is listed as ≤2 mm.[42]

Accurate pathologic assessment of the CRM of resected rectal tumor specimens is crucial, because the CRM has been shown to be a strong predictor of both local recurrence and OS,[38,40,43-45] including in patients undergoing neoadjuvant therapy,[25,46] and is an important consideration when postoperative treatment decisions are made. Furthermore, in a retrospective study of >17,000 patients with rectal cancer, CRM was found to be a better predictor of local recurrence for patients undergoing surgery as initial therapy than for those who had received preoperative therapy.[25] CRM positivity based solely on intranodal tumor should be noted as such; some studies have shown that positive intranodal CRM is associated with lower recurrence rates than a positive CRM by direct tumor extension. Additional components of the pathologic evaluation of the surgical specimen following a total mesorectal excision (TME) are described under *Surgical Approaches*, below.

Lymph Nodes

The AJCC and College of American Pathologists (CAP) recommend evaluation of 12 lymph nodes to accurately identify early-stage CRCs.[19,38,47] The number of lymph nodes that can be retrieved varies with age and gender of the patient and on tumor grade or site.[48] The literature lacks consensus regarding the minimal number of lymph nodes needed to accurately identify early-stage rectal cancer.[49] Most of these studies have combined rectal and colon cancers with surgery as the initial treatment. Two studies confined only to rectal cancer have reported 14 and >10 lymph nodes as the minimal number to accurately identify stage II rectal cancer.[50,51] A more recent analysis of patients with stage I or II rectal cancer in the SEER database found that OS improved with greater numbers of lymph nodes retrieved.[52] Furthermore, the mean number of lymph nodes retrieved from rectal cancers treated with neoadjuvant therapy was significantly less than those treated by surgery alone (13 vs. 19, $P < .05$; 7 vs. 10, $P \leq .0001$).[53-55] In fact, retrieval of fewer lymph nodes may be a marker of a higher tumor response and better prognosis following neoadjuvant treatment.[56-58]

Results of studies evaluating the sentinel node for micrometastatic disease through use of hematoxylin and eosin (H&E) staining to identify small foci of tumor cells and the identification of particular tumor antigens through immunohistochemical analysis have been reported.[59,60] Although results of some of these studies seem promising, there is no uniformity in the definition of "true" clinically relevant metastatic carcinoma. Some studies have considered detection of single cells by immunohistochemistry or by H&E, so-called isolated tumor cells (ITCs), to be micrometastasis.[60,61] In addition, results of one study demonstrated that, following neoadjuvant radiotherapy for rectal cancer, the sensitivity for the sentinel node procedure was only 40%.[62] Furthermore, in a recent study involving 156 patients with colon cancer and 44 patients with rectal cancer, this "ultrastaging" of lymph nodes only changed the staging for 1% of patients.[63] Others have noted that micrometastasis found in node-negative disease did not predict outcome.[64] In contrast, a recent meta-analysis found that the presence of micrometastases increases the likelihood of disease recurrence, whereas the presence of ITCs does not.[65]

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There is also potential benefit of assessing regional lymph nodes for ITCs. One study of 312 consecutive patients with pN0 disease found that positive cytokeratin staining was associated with a higher risk of recurrence.[66] Relapse occurred in 14% of patients with positive nodes compared to 4.7% of those with negative nodes (hazard ratio [HR], 3.00; 95% CI, 1.23–7.32; *P* = .013). A recent systematic review and meta-analysis reached a similar conclusion, finding decreased survival in patients with pN0 disease with immunohistochemical or reverse transcriptase polymerase chain reaction (RT-PCR) evidence of tumor cells in regional nodes.[67] The 8th edition of the AJCC Cancer Staging Manual notes that micrometastases have been defined as clusters of 10 to 20 tumor cells or clumps of tumor ≥0.2 mm in diameter and recommends that these micrometastases be considered as standard positive nodes.[19]

Response to Treatment

The most recent CAP Guidelines require that the pathology report comment on treatment effects of neoadjuvant therapy.[68] The tumor response should be graded on a scale of 0 (complete response [CR] – no viable cancer cells observed) to 3 (poor response – minimal or no tumor kill; extensive residual cancer).[27,28,68,69]

Perineural Invasion

Several studies have demonstrated that the presence of PNI is associated with a significantly worse prognosis.[30-32,70-72] For example, one retrospective analysis of 269 consecutive patients who had colorectal tumors resected at one institution found a 4-fold greater 5-year survival in patients without PNI versus patients whose tumors invaded nearby neural structures.[31] Multivariate analysis of patients with stage II rectal cancer showed that patients with PNI have a significantly worse 5-year disease-free survival (DFS) compared to those without PNI (29% vs. 82%; *P* = .0005).[32] Similar results were seen for patients with stage III disease.[30] A meta-analysis that included 58 studies and 22,900 patients also found that PNI is associated with a worse 5-year OS (relative risk [RR], 2.09; 95% CI, 1.68–2.61) and 5-year DFS (RR, 2.35; 95% CI, 1.66–3.31).[71] PNI is therefore included as a high-risk factor for systemic recurrence.

Tumor Deposits

Tumor deposits, also called extranodal tumor deposits, peritumoral deposits, or satellite nodules, are irregular discrete tumor deposits in the perirectal fat that are away from the leading edge of the tumor and show no evidence of residual lymph node tissue, but that are within the lymphatic drainage of the primary tumor. They are not counted as lymph nodes replaced by tumor. Most of these tumor deposits are thought to be due to LVI or occasionally PNI. The number of tumor deposits should be recorded in the pathology report, since they have been shown to be associated with reductions in DFS and OS.[33-37,72] Multivariate survival analysis in one study showed that patients with pN0 tumors without satellite nodules had a 91.5% 5-year survival rate compared to 37.0% for patients with pN0 tumors and the presence of satellite nodules (*P* < .0001).[37] Another retrospective study found a similar difference in 5-year OS rates (80.3% vs. 34.9%, respectively; *P* < .001).[73] The association of tumor deposits with decreased survival also holds in patients with rectal cancer who had neoadjuvant chemoradiation (chemoRT).[74-76] Tumor deposits are classified as pN1c.[19]

Tumor Budding

Tumor budding is defined as the presence of a single cell or a cluster of ≤4 neoplastic cells as detected by H&E staining at the advancing edge of an invasive carcinoma. As specified by the 2016 International Tumor Budding Consensus Conference (ITBCC), the total number of buds should be reported from a selected hot spot measuring 0.785 mm^2.[77] Budding is separated into three tiers: low (0–4 buds), intermediate (5–9 buds), and high (≥10 buds).

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Several studies have shown that high-grade tumor budding in pT1 CRC or malignant polyps is associated with an increased risk of lymph node metastasis, although the methodologies for assessing tumor budding were not uniform.[78-82] Studies have also supported tumor budding as an independent prognostic factor for rectal cancer, both before and after neoadjuvant therapy.[83-86] A meta-analysis of 11 studies showed that high-grade tumor budding increased the risk of poor outcomes in patients with rectal cancer after neoadjuvant therapy (poor 5-year DFS, $P < .00001$; poor 5-year OS, $P = .003$; local recurrence, $P = .007$; and distant metastasis, $P < .0001$).[84] Other studies have agreed with these results, with one multivariate analysis finding that tumor budding was superior to ypT and ypN status in predicting worse outcomes.[86] A retrospective analysis of pre-treatment rectal biopsies showed that tumor budding also predicted a poor pathologic response to neoadjuvant chemoradiotherapy (higher ypT stage, $P = .032$; lymph node involvement, $P = .018$) and poor outcomes (lower 5-year DFS, $P < .001$; lower cancer-specific 5-year survival, $P = .021$).[85] No patients with tumor budding exhibited a grade 1 tumor regression or complete pathologic response. Tumor budding is therefore included as a high-risk factor for recurrence and may inform treatment decision-making.

Clinical Presentation and Treatment of Nonmetastatic Disease

Management of Malignant Polyps

A malignant rectal polyp is defined as an adenoma that harbors a focus of cancer invading through the muscularis mucosae and into the submucosa (pT1).[87] Conversely, polyps classified as carcinoma in situ (pTis) have not penetrated into the submucosa and are therefore incapable of regional nodal metastasis.[38] Before making a decision about formal surgical resection for an endoscopically resected pedunculated or sessile malignant polyp, physicians should review the pathology[88] and consult with the patient. The Panel recommends marking the malignant polyp site at the time of colonoscopy or within 2 weeks if deemed necessary by the surgeon. All patients with a malignant polyp should undergo mismatch repair (MMR) or microsatellite instability (MSI) testing at diagnosis.

In patients with pedunculated polyps (adenomas), no additional surgery is required if the polyp has been completely removed endoscopically with favorable histologic features.[88,89] Favorable histologic features include lesions of grade 1 or 2 without angiolymphatic invasion and with a negative resection margin.[88] There is controversy as to whether malignant colorectal polyps with a sessile configuration can be successfully treated by endoscopic removal. The literature seems to indicate that endoscopically removed sessile malignant polyps have a significantly greater incidence of adverse outcomes (residual disease, recurrent disease, mortality, and hematogenous metastasis, but not lymph node metastasis) than do pedunculated malignant polyps. However, when one closely looks at the data, configuration by itself is not a significant variable for adverse outcomes and endoscopically removed malignant sessile polyps with grade I or II histology, negative margins, and no lymphovascular invasion can be successfully treated with endoscopic polypectomy alone. Also see the section on *Endoscopically Removed Malignant Polyps* in *Principles of Pathologic and Molecular Review* in the algorithm. Rectal surgery is also an option for these patients. Endoscopic ultrasound (EUS) or MRI of the pelvis is recommended prior to surgical treatment for sessile polyps.

Rectal surgery is also recommended for patients with malignant polyps with unfavorable histologic features or when the specimen is fragmented or margins cannot be assessed. A complete workup is recommended prior to surgery for patients with malignant polyps showing these characteristics since more extensive disease is more likely in this situation (see section on *Clinical Evaluation/Staging* under *Management of Localized Rectal Cancer*). Unfavorable histologic features for adenomas are grade 3 or 4,

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angiolymphatic invasion, or a positive/nonassessable margin of resection. In such cases, risk of nodal involvement is higher. It should be noted that no consensus currently exists as to the definition of what constitutes a positive margin of resection. A positive margin for an endoscopically removed polyp has been defined as the presence of tumor within 1 to 2 mm from the transected margin or by the presence of tumor cells within the diathermy of the transected margin.[88,90-92] In addition, several studies have shown that tumor budding is an adverse histologic feature associated with adverse outcome and may preclude polypectomy as an adequate treatment of endoscopically removed malignant polyps.[93-96]

Rectal surgery consists of either a transanal local excision, if appropriate, or a transabdominal resection. In patients with unfavorable pathologic features, transabdominal resection should be considered in order to include lymphadenectomy. All patients who have malignant polyps removed by transanal local excision or transabdominal resection should undergo total colonoscopy to rule out other synchronous polyps and should undergo surveillance as described in the guidelines.

Management of Localized Rectal Cancer

Rectal cancer is a cancerous lesion in the rectum, which lies below a virtual line from the sacral promontory to the upper edge of the symphysis as determined by MRI (see Figure 1). The rectum ends at the superior border of the functional anal canal, defined as the palpable upper border of the anal sphincter and puborectalis muscles of the anorectal ring.

The determination of an optimal treatment plan for an individual patient with rectal cancer is a complex process. In addition to decisions relating to the intent of rectal cancer surgery (ie, curative or palliative), consideration must also be given to the likely functional results of treatment, including the probability of maintaining or restoring normal bowel function/anal continence and preserving genitourinary functions. For patients with distal rectal cancer, in particular, the simultaneous achievement of the goals of cure and of minimal impact on quality of life can be challenging.[97] Furthermore, the risk of pelvic recurrence is higher in patients with rectal cancer compared to those with colon cancer, and locally recurrent rectal cancer is associated with a poor prognosis.[98-100] Careful patient selection with respect to particular treatment options and the use of sequenced multimodality therapy that combines chemoRT, systemic therapy, and operative treatment for most patients is recommended.[101] However, newer approaches have been developed that allow select patients to avoid the morbidities of surgery and/or radiation by omitting one or both.

Clinical Evaluation/Staging

The initial clinical workup of patients with rectal cancer provides important preoperative information on the clinical stage of disease. Since the clinical stage is used to direct decisions regarding choice of primary treatment, including surgical intent (eg, curative or palliative) and the recommended preoperative treatment approach, the implications of either clinically understaging or overstaging rectal cancer can be substantial. Based on this, a multidisciplinary team evaluation is recommended, including a formal surgical evaluation. A discussion of infertility risks and counseling on fertility preservation, if appropriate, should be carried out prior to the start of treatment.

Patients who present with rectal cancer appropriate for resection require a complete staging evaluation, which includes total colonoscopy to evaluate for synchronous lesions or other pathologic conditions of the colon and rectum. Proctoscopy can be useful in determining the distance of the cancer from the anal verge and length and, therefore, is a consideration. Patients with rectal cancer also require a complete physical examination, including carcinoembryonic antigen (CEA) determination and assessment of performance status to determine operative risk.

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Clinical staging is also based on histopathologic examination of the specimen obtained via biopsy or local excision (eg, excised polyps). Endoscopic biopsy specimens of the lesion should undergo careful pathology review for evidence of invasion into the muscularis mucosa. If removal of the rectum is contemplated, early consultation with an enterostomal therapist is recommended for preoperative marking of the site and patient teaching purposes. All patients with rectal cancer should undergo MMR or MSI testing at diagnosis to aid in the diagnosis of Lynch syndrome and to determine eligibility for checkpoint inhibitors as neoadjuvant therapy. Those with loss of MMR proteins and/or MSI should be referred for genetic counseling and testing. *PIK3CA* testing may also be considered for those determined to have stage II–III disease, as a mutation in this gene could inform adjuvant use of aspirin to improve outcomes.

Imaging also plays a critical role in preoperative evaluation, for evaluation of the primary tumor and regional adenopathy, and to assess for the presence of distant metastases. Preoperative imaging for rectal cancer includes chest/abdominal CT and pelvic MRI or chest CT and abdominal/pelvic MRI, as described below. Endorectal ultrasound may be recommended if MRI is contraindicated or inconclusive, or for superficial lesions.

Preoperative Pelvic Imaging in Rectal Cancer

The accessibility of rectal cancer to evaluation by pelvic MRI with contrast makes possible preoperative assessments of depth of tumor penetration and the presence of local lymph nodal metastases.[102,103] Pelvic MRI has the ability to provide accurate images of soft tissue structures in the mesorectum, including the mesorectal fascia, so as to provide information useful in the prediction of the CRM prior to radical surgery.[104-109] The CRM by MRI is measured at the closest distance of the tumor to the mesorectal fascia. The Panel defines a clear CRM as >1 mm from mesorectal fascia and levator muscles and not invading into the intersphincteric plane. An involved or threatened CRM, in contrast, is within 1 mm of mesorectal fascia; or, for lower third rectal tumors, within 1 mm from levator muscle.[41] Published 5-year follow-up results of the MERCURY trial show that high-resolution MRI can accurately assess the CRM preoperatively, differentiating patients with low- and high-risk disease.[110] Patients with MRI-clear CRM had a 5-year OS of 62.2% compared with 42.2% in patients with MRI-involved CRM (HR, 1.97; 95% CI, 1.27–3.04; $P < .01$). The preoperative MRI imaging also predicted DFS (HR, 1.65; 95% CI, 1.01–2.69; $P < .05$) and local recurrence (HR, 3.50; 95% CI, 1.53–8.00; $P < .05$). MRI has also been shown to be accurate for the prediction of T and N stage.[111] Moreover, radiomics based on preoperative MRI imaging has shown potential in predicting treatment response, which extends the preoperative MRI from clinical staging to therapeutic decision-making for neoadjuvant treatment.[112] ESGAR has developed consensus guidelines for standardized imaging of rectal cancer by MRI.[42]

Only a limited number of studies using CT for the purpose of T-staging have been performed, and it is not currently considered to be an optimal method for staging the extent of tumor penetration.[104,107,113] In addition, CT has poor sensitivity for the prediction of CRM status.[114] Furthermore, CT has lower sensitivity and specificity for the prediction of lymph node involvement than MRI (CT, 55% and 74%; MRI, 66% and 76%).[113] Therefore, pelvic CT is not recommended for rectal staging.

A 2004 meta-analysis showed that EUS and MRI have similar sensitivities and specificities for evaluation of lymph nodes (EUS, 67% and 78%; MRI, 66% and 76%).[113] However, newer data suggest that EUS is not very accurate for rectal cancer staging.[115] Furthermore, EUS cannot fully image high or bulky rectal tumors nor regions beyond the immediate area of the primary tumor (eg, tumor deposits, vascular invasion).[104] Another disadvantage of EUS is a high degree of operator dependence.[113] At this

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time, the Panel recommends that EUS may be used to evaluate the pelvis if MRI is contraindicated (eg, because of a pacemaker) or inconclusive, or it may be considered as an alternative for superficial lesions.

Preoperative Imaging for Distant Metastases

Additional information regarding the occurrence of distant metastases should be determined preoperatively through chest and abdominal imaging. Chest imaging should be by CT scan, whereas imaging of the abdomen can be performed with CT or MRI. Lung metastases occur in approximately 4% to 9% of patients with colon and rectal cancer,[116-118] and studies have shown that 20% to 34% of patients with CRC present with synchronous liver metastases.[119,120]

The consensus of the Panel is that a PET scan is not indicated for preoperative staging of rectal cancer. PET/CT, if done, does not supplant a contrast-enhanced diagnostic CT scan. PET/CT should only be used to evaluate an equivocal finding on a contrast-enhanced CT scan or in patients with a strong contraindication to IV contrast.

Restaging/Assessing Treatment Response

Restaging after neoadjuvant treatment is done to detect distant metastases that would change the treatment strategy, to plan the surgical approach, and, increasingly, to determine if additional therapy or resection can be avoided for select patients. MRI, CT, and EUS have been used for restaging after neoadjuvant treatment, but the accuracy of these techniques for determining T stage and lymph node involvement is limited.[121-129] As with initial staging, the Panel recommends pelvic MRI for restaging with chest and abdominal imaging to assess for distant disease. Abdominal/pelvic CT has been shown to identify resectable liver metastases in 2.2% (95% CI, 0.8%–5.1%) of patients during restaging, with false-positive findings that could cause unnecessary treatment in 1.3% (95% CI, 0.3%–3.9%) of patients.[130] In this study, the use of restaging abdominal/pelvic CT was at the physician's discretion, and no difference was seen in relapse-free survival (RFS) for those who had an abdominal/pelvic CT before resection compared with those who did not.

Advanced functional MRI techniques (eg, dynamic contrast-enhanced MRI, diffusion-weighted MRI) allow for the measurement of microcirculation, vascular permeability, and tissue cellularity and thus may be useful for determining response to neoadjuvant treatment and restaging patients with rectal cancer.[128,131-133] FDG-PET/CT is also being investigated for its ability to accurately determine response to neoadjuvant treatment.[132,134]

At this time, the Panel recommends chest CT, abdominal CT or MRI, and pelvic MRI for restaging.

Surgical Approaches

A variety of surgical approaches, depending on the location and extent of disease, are used to treat primary rectal cancer lesions.[45,135,136] These methods include local procedures, such as polypectomy, transanal local excision, endoscopic submucosal dissection (ESD), and transanal endoscopic microsurgery (TEM), and more invasive procedures involving a transabdominal resection (eg, low anterior resection [LAR], proctectomy with TME and coloanal anastomosis, and abdominoperineal resection [APR]).[135,136]

Transanal Local Excision

Transanal local excision is only appropriate for selected T1, N0 early-stage cancers. Small (<3 cm), well to moderately differentiated tumors that are within 8 cm of the anal verge and limited to <30% of the rectal circumference and for which there is no evidence of nodal involvement can be approached with transanal local excision with negative margins.[137] In addition, full-thickness excision must be feasible.

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TEM can facilitate excision of small tumors through the anus when lesions can be adequately identified in the rectum. TEM may be technically feasible for more proximal lesions. Although data are limited, a 2015 meta-analysis found that TEM may achieve superior oncologic outcomes compared with transanal local excision.[138] A small prospective, single-blind, randomized trial compared laparoscopic surgery with laparoscopy combined with TEM in 60 patients with rectal cancer.[139] The TEM group had shorter operation times and hospital stays, and no local nor distant recurrences were seen in either group after a median follow-up of 28 months.

Both transanal local excision and TEM involve a full-thickness excision performed perpendicularly through the bowel wall into the perirectal fat. Negative (>3 mm) deep and mucosal margins are required, and tumor fragmentation should be avoided. The locally excised specimen should be oriented and pinned before fixation and brought to the pathologist by the surgeon to facilitate an oriented histopathologic evaluation of the specimen. If pathologic examination reveals adverse features such as positive margins, LVI, poor differentiation, or invasion into the lower third of the submucosa (sm3 level),[140,141] a more radical resection is recommended.

Advantages of a local procedure include minimal morbidity (eg, a sphincter-sparing procedure) and mortality and rapid postoperative recovery.[97,142] Limitations of a local excision include the absence of pathologic staging of nodal involvement. Further, evidence indicates that lymph node micrometastases are both common in early rectal lesions and unlikely to be identified by endorectal ultrasound.[143] These observations may underlie the findings that patients undergoing local excision have a higher local recurrence rate than those undergoing radical resection.[142,144,145] A retrospective study of 282 patients undergoing either transanal local excision or radical resection for T1 rectal cancer from 1985 to 2004 showed respective local recurrence rates of 13.2% and 2.7% for these two groups (P = .001).[145] A similar retrospective study of 2124 patients showed local recurrence rates of 12.5% and 6.9% for patients undergoing local excision versus standard resection, respectively (P = .003).[142] More recently, an analysis of >164,000 individuals from the National Cancer Database (NCDB) with resected, invasive, nonmetastatic rectal cancer diagnosed from 1998 to 2010 found that positive margins were more likely after local excision compared to transabdominal excision in both the T1 and T2 populations (95% vs. 76% in T1/T2 combined; P < .001).[146] In the T1, N0 population, a small but significant decrease in OS was also noted in the local excision group. Interestingly, limited data suggest that TEM might have superior oncologic outcomes in patients with stage I rectal cancer compared with radical resection,[144,147] although not all studies have seen such results.[148]

Thus, careful patient selection for local excision of T1, N0 rectal cancer is important, as is the careful examination of the resection specimen with subsequent transabdominal resection in patients found to have T2 disease or high-risk features, as described above.

There has been increasing interest in local excision techniques for patients with higher-risk rectal cancers that achieve a clinical CR (cCR) after neoadjuvant treatment. This strategy is attractive due to its potential for reduced surgical extent, lower morbidity, and improved quality of life compared to TME. However, applying results from relevant trials to this particular scenario is complicated by various factors, including inclusion criteria (which often involve T1/T2 N0 cases), different neoadjuvant treatment regimens, criteria for subsequent local excision, and staging/restaging techniques. The following summarize recent trials that are investigating this approach.

The GRECCAR-2 trial, which involved 145 patients with cT2/T3 N0–N1 rectal cancers ≤4 cm in diameter showing good response to neoadjuvant

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chemoRT compared local excision to TME.[149] This study did not demonstrate the superiority of local excision over TME. Notably, 26 of the 74 patients in the local excision group later underwent completion TME due to the identification of unfavorable features at the time of local excision. This led to the conclusion that better selection criteria are needed to reduce the necessity for completion TME. However, a recent 5-year follow-up report confirmed no differences in oncologic outcomes between local excision and TME for carefully selected patients.[150]

The CARTS study, a multicenter phase II trial, examined 47 patients who showed a good response (defined as ycT0–2) after neoadjuvant chemoRT and were treated with local excision.[151] Of these, 12 patients underwent completion TME following local excision due to poor response or unfavorable features. The 5-year actuarial rates for local recurrence, DFS, and OS were 7.7%, 81.6%, and 82.8%, respectively. Notably, the rates of LAR syndrome were very high, suggesting the impact of radiation therapy (RT) and possibly the excision of larger or deeper residual lesions.

In the TAUTEM study, 173 patients with T2/T3N0M0 rectal cancers were randomized into two groups: one receiving TME alone and the other undergoing neoadjuvant chemoRT followed by local excision via TEM.[152] In the CRT-TEM group, 81 patients completed the prescribed treatment. However, 14 of these patients (17.3%) subsequently needed radical surgery due to factors such as early disease progression, poor pathologic features, and positive surgical margins. The CRT-TEM group experienced significantly fewer postoperative complications (20.7%) compared to the TME group (50.6%, $P < .0001$). Longer-term results of this trial showed that the chemoRT/local excision approach achieved similar outcomes compared to TME for disease recurrence, OS, and DFS rates.[153]

ReSARCh is a prospective observational trial involving patients who achieved either a major clinical response or cCR after neoadjuvant chemoRT.[154] These patients were subsequently treated with either local excision or a NOM approach. The study's preliminary short-term results reveal that within the local excision group, 26 patients (26.5%) were identified as needing TME based on predetermined criteria, such as a stage beyond ypT1 and poor histologic features. Of these 26 patients, only 11 consented to undergo TME. Among those who underwent TME, 10 patients were found to have no residual disease, and 1 patient had a positive mesorectal lymph node. Clavien-Dindo grade 3 or higher morbidity occurred in 3 patients. The long-term outcomes of the study are still pending.

NEO (CCTG CO.28) is a phase II non-randomized trial focused on stage cT1–T3b N0 rectal cancers that are considered endoscopically resectable by the study surgeon.[155] Patients receive 3 months of oxaliplatin-based chemotherapy, with those showing a response proceeding to TEM. Of the 58 patients enrolled, all started chemotherapy, with 56 proceeding to TEM surgery. A total of 33 patients experienced downstaging to ypT0/1 N0/X, resulting in an intention-to-treat organ preservation rate of 57%. The remaining 23 patients were advised to undergo TME surgery, with 13 declining and 10 undergoing the procedure. Among those who had TME, 7 out of 10 showed no evidence of residual disease. Specifically, in the cT3 subset (13 patients), the organ preservation rate was 8 out of 13 (62%), similar to the rates for cT1 (63%) and cT2 (62%) categories.

When considering local excision in this setting, several crucial factors must be considered. Firstly, there's a question of whether a response at the primary tumor site reflects a similar response in the lymph nodes. Historical data from studies of radical surgery suggest that the rate of mesorectal lymph node positivity in patients with ypT0 and ypT1 are in the range of 9% to 14% and 17% to 28%, respectively.[156-158] However, the GRECCAR-2 trial noted 0% lymph node positivity in patients with ypT0 and ypT1 disease who underwent TME surgery after exhibiting a good response.[149]

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Another concern involves potentially worse oncologic and surgical outcomes following TME after neoadjuvant chemoRT and local excision. Completion TME has been associated with a higher rate of major complications, as well as reduced bowel function and quality of life, compared to TME alone.[159,160] The GRECCAR-2 trial observed increased major morbidity/side effects in the group that received TME following local excision versus the TME or local excision groups alone.[149] Additionally, subsequent resections for local recurrences after neoadjuvant chemoRT and local excision have reported concerning outcomes, including a high rate of positive CRM and only a 60% 2-year local recurrence-free rate.[161]

Considering this evidence, selecting local excision following neoadjuvant therapy requires well-informed and careful patient selection. Local excision could be a viable option for patients who have a strong but incomplete response and are unable or unwilling to undergo standard TME surgery. The most suitable candidates would be those meeting the near CR criteria as outlined in the *Principles of Nonoperative Management* in the algorithm. Informed consent should cover discussions about potential recurrence; the need for proctectomy and stoma; increased risks of complexity, recurrence, and complications; as well as the possibility of worsened bowel function or quality of life after subsequent radical surgery. The efficacy of local excision for patients who have achieved a cCR is not fully established. According to current data, cCR typically correlates with a durable response.[162] Therefore, implementing local excision might introduce unnecessary risks without clear benefits. Given this uncertainty and potential for added risk, local excision is not routinely recommended for patients who have achieved a clear cCR following neoadjuvant therapy.

Endoscopic Submucosal Dissection

ESD is a technique that was first described in Japan as an alternative to endoscopic mucosal resection (EMR) of early gastric cancers.[163,164] ESD has been performed worldwide for >20 years and in the United States for more than a decade.[163] As advances in hands-on educational opportunities have increased in the United States, ESD is now a procedure offered at many U.S. medical centers.[165]

ESD is a minimally invasive, organ-preserving procedure that can provide curative resection for early rectal cancers through the removal of large complex polyps in an en-bloc manner.[164] ESD involves using a submucosal injection to lift the polyp, followed by a circumferential incision of the mucosa using an endoscopic knife, and submucosal dissection underneath the lesion, above the muscularis propria, to fully resect the lesion en-bloc.[164,166] The benefit of ESD as compared to EMR is that ESD can offer higher rates of curative resection and the intact specimen produced by ESD allows for more accurate pathologic and oncologic assessment.[163,164] Curative resection is often achieved after R0 resection in conjunction with other favorable criteria, including well to moderately differentiated histology, <1000 μm invasion into the submucosa, and lack of lymphovascular invasion.[163,164] Furthermore, when ESD is noncurative, the pathology findings can help drive appropriate subsequent therapy. Last, en-bloc resection with ESD is safe and effective for lesions in the rectum because part of the rectum is below the peritoneal reflection, making ESD in the rectum relatively less technically challenging and safer than in other regions of the colon.[164] Furthermore, ESD can reach lesions in the proximal rectum that may be challenging to reach surgically because of their location.[167,168] A recently published large prospective study in North America found that among 188 rectal lesions removed, 88.8% were removed en bloc, 85.6% achieved R0 resection, and 79.8% were deemed curative.[167] The AE rate for rectal lesions in this study was 5.9% and included delayed bleeding and perforation; however, none of the patients required surgery for an AE after rectal ESD. Also, 70% of these patients were discharged on the same day and among those admitted, the average length of stay was 1.13 days.

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Transanal local excision using advanced techniques, such as TEM or transanal minimally invasive surgery (TAMIS), are also well-established surgical procedures for the resection of early rectal cancers. As compared to radical resection, these approaches help to preserve function and reduce the morbidity associated with radical resection approaches.[169] While there are no published prospective studies comparing ESD to TEM/TAMIS, there is an ongoing randomized controlled trial (RCT) in the Netherlands comparing TAMIS to ESD[170] and several retrospective studies comparing TAMIS to ESD.[171-173] A systematic review and meta-analysis of these studies found that ESD and surgical techniques do not differ in terms of outcomes, including local recurrence, en bloc/R0 resection rate, and AEs.[174]

Local excision of rectal cancer has also been associated with an increased risk of local recurrence, with some studies showing rates of 1.1% to 6.3%, which are higher than observed with early colon cancers (0%–1.9%).[175-177] Intraluminal recurrence after en bloc, R0 ESD of rectal neoplasia is rare with rates ≤2.5%.[178-181] However, if rectal cancer does recur, it can be distant and appear after 3 to 5 years. For these reasons, after curative resection of a T1 rectal adenocarcinoma with favorable prognostic factors, a flexible sigmoidoscopy is recommended at 3 to 6 months post ESD and every 6 months after for a total of 5 years from ESD.[163] EUS or pelvic MRI with contrast is recommended every 3 to 6 months for 2 years, then every 6 months for a total of 5 years.

With the increasing availability of ESD in North America, and the evidence supporting its use, the NCCN Panel supports ESD as a treatment option for both surgical and non-surgical candidates with T1, N0 rectal cancer, for institutions with the expertise needed to perform this technique. Please see the section on ESD within the *Principles of Surgery and Locoregional Therapies* in the algorithm for more information on pre-ESD endoscopic evaluation, criteria for resection with ESD, guidelines for when resection can be considered curative, surveillance after ESD, and decision-making between surgery and ESD. Understaging of rectal cancers is common and, therefore, thorough staging should be carried out prior to considering ESD as a treatment option. Furthermore, if unfavorable tumor characteristics are discovered following ESD (eg, lymphovascular invasion), the patient should consult with a multidisciplinary team including an endoscopist, surgeon, and pathologist in a shared decision-making process to decide whether to proceed with surgery.

Transabdominal Resection

Patients with rectal cancer who do not meet requirements for local surgery should be treated with a transabdominal resection. Organ-preserving procedures that maintain sphincter function are preferable, but not possible in all cases. Preoperative therapy may result in tumor downsizing and a decrease in tumor bulk (see section on *Neoadjuvant Therapy for Resectable Nonmetastatic Disease*, below); sphincter preservation may become possible in cases where initial tumor bulk prevented consideration of such surgery and exposure to the tumor is improved by neoadjuvant treatment.

In transabdominal resections, TME is recommended. A TME involves an en bloc removal of the mesorectum, including associated vascular and lymphatic structures, fatty tissue, and mesorectal fascia as a “tumor package” through sharp dissection and is designed to spare the autonomic nerves.[97,136,182] The lymphatic drainage regions of rectal tumors are influenced by their position in the rectum. More distal tumors are more likely to be characterized by both upward and lateral lymphatic drainage, whereas the likelihood of only upward mesorectal drainage is much higher for more proximal tumors.[183] The TME approach is designed to radically remove lymphatic drainage regions of tumors located above the level of the levator muscles.[184] The Panel does not recommend extension of nodal dissection beyond the field of resection (eg, into the distribution of iliac

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lymph nodes) unless these nodes are clinically suspicious. In cases where anal function is intact and distal clearance is adequate, the TME may be followed by creation of a coloanal anastomosis.

For lesions in the mid to upper rectum, an LAR extended 4 to 5 cm below the distal edge of the tumor using TME, followed by creation of a colorectal anastomosis, is the treatment of choice. Where creation of an anastomosis is not possible, colostomy is required. Wide TME is recommended in order to facilitate adequate lymphadenectomy and improve the probability of achieving negative circumferential margins.

An APR with TME should be performed when the tumor directly involves the anal sphincter or the levator muscles. An APR is also necessary in cases where a margin-negative resection of the tumor would result in loss of anal sphincter function and incontinence. An APR involves en bloc resection of the rectosigmoid, the rectum, and the anus, as well as the surrounding mesentery, mesorectum (TME), and perianal soft tissue, and it necessitates creation of a colostomy.[185] In the NSABP R-04 trial, patients who had an APR reported worse body image, worse micturition symptoms, and less sexual enjoyment at 1-year post surgery than those who had sphincter-sparing surgery.[186] An extralevator APR may have benefits over a conventional APR approach, including lower rates of intraoperative perforation, CRM involvement, and local recurrence, although inconsistencies are seen between studies.[187,188]

Pathologists play a key role in evaluating the surgical specimen, including a macroscopic assessment of both its external appearance/completeness and the CRM.[189,190] The Panel defines an involved or threatened CRM as tumor within 1 mm from the resected margin (see *Pathology*, above).[24,26,40,41] Detailed descriptions of how the quality of the mesorectal specimens should be scored were provided in the Dutch Rectal Cancer Trial, and these guidelines are endorsed by the NCCN Panel.[24]

Recent retrospective comparisons of the outcomes of patients undergoing an APR versus an LAR in the treatment of rectal cancer have shown that those treated with an APR have worse local control and OS.[191,192] Whether these differences can be attributed to the surgical procedure alone, to patient- and tumor-related characteristics, or some combination of these factors is presently unclear. However, results from a recent retrospective study of 3633 patients with T3–4 rectal cancer tumors included in five large European trials suggest that there is an association between the APR procedure itself and the increased risks of recurrence and death.[191] Importantly, quality of life between patients with or without a permanent colostomy appears to be fairly comparable.[193,194]

For patients with T4 tumors or recurrent cancers or if margins are very close or positive, intraoperative RT (IORT),[195-199] which involves direct exposure of tumors to RT during surgery while removing normal structures from the field of treatment, may be considered as an additional boost to facilitate resection.

Laparoscopic Resection

Data from randomized studies evaluating use of laparoscopic surgery in the treatment of patients with rectal cancer have matured in recent years.[200-203] One large prospective multicenter study, which included 4405 patients with rectal cancer but was not randomized, found no differences in recurrence or survival, although complications and other measures of quality indicated a benefit to the laparoscopic approach.[204] The phase III COLOR II trial, powered for non-inferiority, randomized patients with localized rectal cancer to laparoscopic or open surgery. Short-term secondary endpoints were met, with patients in the laparoscopic arm losing less blood, having shorter hospital stays, and having a quicker return of bowel function, but with longer operation times.[205] No differences were seen in completeness of resection, percentage of patients with a positive CRM, morbidity, or mortality between the arms. The primary

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endpoint of locoregional recurrence at 3 years was identical in the two groups, at 5.0%, and no statistically significant differences were seen in DFS or OS.[200]

In the CLASICC trial comparing laparoscopically assisted resection to open resection, nearly half of the 794 patients were diagnosed with rectal cancer.[206] No significant differences in local recurrence, DFS, or OS were observed between the two groups of patients with colon or rectal cancer based on surgical approach. A 5-year follow-up of the CLASICC trial showed that this lack of difference in local recurrence, DFS, or OS was maintained for patients with rectal cancer, despite a trend towards better 5-year OS after laparoscopic surgery (52.9% and 60.3% for open and laparoscopic surgery, respectively; *P* = .132).[207]

The COREAN trial randomized patients with stage II or III low- to mid-rectal cancer to an open or laparoscopic resection, with short-term benefits seen with the laparoscopic approach.[208] The primary endpoint, 3-year DFS, did not differ between the two groups at 72.5% (95% CI, 65.0%–78.6%) for open surgery and 79.2% (95% CI, 72.3%–84.6%) for the laparoscopic group.[201] Factors that may confound conclusions drawn from randomized studies comparing open surgery to laparoscopically assisted surgery for CRC have been described,[209] and longer-term outcomes from laparoscopic rectal surgery have not been reported.

Two other trials, ACOSOG Z6051 and ALaCaRT, have reported pathologic outcomes.[202,203] In Z6051, the primary endpoint was a composite of CRM >1 mm, negative distal margin, and TME completeness.[202] No significant differences were observed between the arms in these three measures or in the composite of successful resection. For example, complete or nearly complete TME was achieved in 92.1% (95% CI, 88.7%–95.5%) in the laparoscopic resection arm and 95.1% (95% CI, 92.2%–97.9%) in the open resection arm, for a difference of −3.0 (95% CI, -7.4 to 1.5; *P* = .20). However, the criteria for non-inferiority of the laparoscopic approach were not met in these initial results. Follow-up results of Z6051 reported similar 2-year DFS rates between laparoscopic (79.5%) and open resection (83.2%).[210] Locoregional and distant recurrence rates were also found to be similar between laparoscopic and open resection (4.6% vs. 4.5% for locoregional recurrences and 14.6% vs. 16.7% for distant recurrences). In ALaCaRT, the primary endpoint was also a composite of resection quality measures.[203] Successful resections were achieved in 82% of the laparoscopic resection arm and 89% of the open resection arm, for a difference of -7.0% (95% CI, -12.4% to infinity). A negative CRM was achieved in 93% and 97%, respectively (risk difference, -3.7%; 95% CI, -7.6% to 0.1%; *P* = .06). Follow-up results for ALaCaRT showed similar recurrence, DFS, and OS rates for laparoscopic versus open resection after 2 years.[211] Two-year locoregional recurrence rates were 5.4% and 3.1%, 2-year DFS rates were 80% and 82%, and 2-year OS rates were 94% and 93% for laparoscopic resection and open resection, respectively. As in Z6051, the criteria for non-inferiority of the laparoscopic approach were not met in the initial ALaCaRT report, but the techniques were found to not differ significantly after longer follow-up with oncologic outcomes.

An analysis of results from >18,000 individuals in the NCDB undergoing LAR for rectal cancer found short-term oncologic outcomes to be similar between the open and laparoscopic approaches.[212] In addition, older reviews and meta-analyses consistently found the laparoscopic approach to be safe and feasible,[201,213-226] even though a meta-analysis published in 2017 found that the risk for a non-complete mesorectal excision is significantly higher in patients receiving a laparoscopic resection compared with those receiving an open resection.[227] Several studies have also compared outcomes of robotic-assisted resection to conventional laparoscopic resection.[228-233] Comparable results are generally seen between the approaches in conversion to open resection, TME quality, postoperative complications, and quality of life.

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In conclusion, some studies have shown that laparoscopy is associated with similar short- and long-term outcomes when compared to open surgery,[200,201] whereas other studies have shown the laparoscopic approach to be associated with higher rates of CRM positivity and incomplete TME.[202,203] The Panel defined principles by which minimally invasive resection of rectal cancer can be considered: the procedure can be considered by an experienced surgeon, should include thorough abdominal exploration, and should be limited to lower-risk tumors, as outlined in the guidelines. An international group of experts has defined standards for the technical details of laparoscopic TME.[234]

Neoadjuvant Therapy for Resectable Nonmetastatic Disease

Neoadjuvant therapy for stage II (T3–4, node-negative disease with tumor penetration through the muscle wall) or stage III (node-positive disease without distant metastasis) rectal cancer usually includes locoregional treatment due to the relatively high risk of locoregional recurrence. This risk is associated with the close proximity of the rectum to pelvic structures and organs, the absence of a serosa surrounding the rectum, and technical difficulties associated with obtaining wide surgical margins at resection. In contrast, adjuvant treatment of colon cancer is more focused on preventing distant metastases since this disease is characterized by lower rates of local recurrence.

Although RT has been associated with decreased rates of local recurrence of rectal cancer, it is also associated with increased toxicity (eg, radiation-induced injury, hematologic toxicities) relative to surgery alone.[39,235,236] It has been suggested that some patients with disease at lower risk of local recurrence (eg, proximal rectal cancer staged as T3, N0, M0, characterized by clear margins and favorable prognostic features) may be adequately treated with surgery and adjuvant chemotherapy.[39,237,238] However, 22% of 188 patients clinically staged with T3, N0 rectal cancer by either EUS or MRI who subsequently received preoperative chemoRT had positive lymph nodes following pathologic review of the surgical specimens according to results of a retrospective multicenter study,[239] suggesting that many patients are understaged and would benefit from chemoRT. Therefore, the guidelines recommend preoperative treatment for most patients with T3, N0 disease. Patients with T3, N0 low-risk high rectal tumors may be treated more similarly to colon cancer, with upfront transabdominal resection.

Combined-modality therapy consisting of surgery, concurrent fluoropyrimidine-based chemotherapy with ionizing radiation to the pelvis (chemoRT), and chemotherapy is recommended for the majority of patients with stage II or stage III rectal cancer. Use of perioperative pelvic RT in the treatment of patients with stage II/III rectal cancer continues to evolve.

The Total Neoadjuvant Therapy Approach

A treatment approach for stage II or III rectal cancer, including courses of both chemoRT and chemotherapy given as neoadjuvant therapy before transabdominal resection, has been gaining prominence. This approach, called total neoadjuvant therapy (TNT), was first tested in several small, phase II trials, but more recently has been supported by phase III trial data.[240-245] The apparent benefits of the TNT approach include higher rates of pathologic CR (pCR) and longer DFS,[246-251] minimizing the length of time patients need an ileostomy,[247] facilitating resection, and improving the tolerance and completion rates of chemotherapy.[242,246,250,251] For some patients, surgery may be avoided if a cCR is achieved as a result of neoadjuvant therapy. Hence, the NCCN Panel recommends TNT as the preferred approach for stage II–III rectal cancer.

In the Spanish GCR-3 randomized phase II trial, patients were randomized to receive CAPEOX (capecitabine/oxaliplatin) either before chemoRT or after surgery.[242,252] Similar pCR rates and 5-year DFS and OS were seen, and induction chemotherapy appeared to be less toxic and better

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tolerated. The GCR-3 trial provided the rationale for RAPIDO and demonstrated that the TNT approach increased adherence, lowered acute toxicity, and yielded similar outcomes compared to the traditional approach. A pooled analysis of two phase II trials, EXPERT and EXPERT-C, assessed the safety and efficacy of neoadjuvant chemotherapy followed by chemoRT and surgery.[253] Of the 269 patients who were included, 91.1% completed chemotherapy, 88.1% completed chemoRT, and 89.2% underwent curative surgery. Five-year PFS and OS rates were 66.4% and 73.3%, respectively. Another phase II trial comparing response rates in patients with stage II–III rectal cancer treated with chemoRT alone or chemoRT followed by increasing durations of FOLFOX (5-fluorouracil [5-FU]/leucovorin [LV]/oxaliplatin) prior to resection found that delivery of FOLFOX was independently associated with higher rates of pCR, with the highest CR rate (38%) following six cycles of neoadjuvant FOLFOX and the lowest CR rate (18%) in the group that received chemoRT alone.[246] However, it is difficult to determine if the higher pCR rate with FOLFOX was due to the increased duration of FOLFOX, the longer duration of time between chemoRT and surgery, or some combination of the two.

More recently, the TNT approach has been tested in phase III trials. RAPIDO, a randomized phase III trial, compared a standard treatment approach (chemoRT, followed by TME, then optional adjuvant chemotherapy with CAPEOX or FOLFOX) to an experimental TNT approach (short-course RT, followed by chemotherapy before TME) in 912 patients with locally advanced rectal cancer.[250] At 3 years after randomization, the rate of disease-related treatment failure was 23.7% with TNT compared to 30.4% with standard treatment (HR, 0.75; 95% CI, 0.60–0.95; P = .019). No differences were found in the secondary endpoint of OS. Serious AEs occurred in 38% of the TNT group and 34% of the standard treatment group. Comparing the pCR rates between the arms of the RAPIDO study, 28% on the TNT arm achieved pCR compared to 14% for standard treatment and 5-year oncologic outcomes were similar between the two arms for those with pCR of disease.[254]

Another randomized phase III trial, UNICANCER-PRODIGE 23, compared a neoadjuvant therapy approach including both FOLFIRINOX (5-FU/LV/irinotecan/oxaliplatin) and chemoRT prior to TME to a standard approach of neoadjuvant chemoRT alone followed by TME for 461 patients with locally advanced rectal cancer.[251] Both arms followed TME by adjuvant FOLFOX, although the duration of adjuvant treatment was shorter in the group that had received neoadjuvant chemotherapy. After a median follow-up of 46.5 months, 3-year DFS was 76% in the group that received neoadjuvant chemotherapy, compared to 69% in the standard treatment group (HR, 0.69; 95% CI, 0.49–0.97; P = .034). During the whole treatment period, serious AEs occurred in 11% of patients in the neoadjuvant chemotherapy group and 23% in the standard-of-care group (P = .0049). No postoperative deaths occurred within 30 days in the neoadjuvant group, while five deaths occurred in the standard treatment group (4 from cardiac or vascular events, 1 from suicide). Long-term results from PRODIGE 23 reported similar results, with 7-year DFS of 67.6% for the neoadjuvant therapy group versus 62.5% for the group receiving standard treatment.[255] Seven-year OS was 81.9% and 76.1%, respectively.

These results have also been supported by systemic review and meta-analyses showing a higher pCR rate with TNT.[248,249] In a single-institution retrospective cohort analysis of patients with T3/4 or node-positive rectal cancer, patients in the TNT group received a greater percentage of the planned chemotherapy dose than those in the adjuvant chemotherapy group.[247] The CR rates were 36% and 21% in the TNT and adjuvant chemotherapy groups, respectively.

A few trials have investigated the use of FOLFIRINOX or FOLFOXIRI (5-FU/LV/irinotecan/oxaliplatin) as neoadjuvant chemotherapy for locally

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advanced rectal cancer. One of these trials was the randomized, phase III UNICANCER-PRODIGE 23 study, which was described above.[251] The prospective, single-arm, phase II FORTUNE study investigated the use of FOLFOXIRI as initial therapy for patients with stage II or III rectal cancer.[256] After initial chemotherapy, patients were treated with either surgery or RT/chemoRT followed by surgery, depending on the response to initial FOLFOXIRI. Of 103 patients who completed neoadjuvant therapy, pCR and tumor downstaging rates were 20.4% and 42.7%, respectively. Another phase II trial of patients with node-positive, cT4, or high-risk T3 rectal cancer investigated the use of induction FOLFOXIRI plus bevacizumab followed by capecitabine-based chemoRT with bevacizumab.[257] Surgery was performed 8 weeks after completion of the chemoRT. Of 49 enrolled patients, 44 completed surgery and 2-year DFS was 80%. While the NCCN Panel recommends induction chemotherapy with FOLFIRINOX as an option for T4, node-positive rectal cancer, the addition of targeted agents (such as bevacizumab) is not currently recommended in this setting. While UNICANCER-PRODIGE 23 enrolled patients with cT3 and cT4, node-negative tumors,[251] the NCCN Panel only recommends the use of FOLFIRINOX for the cT4, N+ tumors due to the higher toxicity of FOLFIRINOX compared to FOLFOX or CAPEOX and the results observed with CAPEOX or FOLFOX in RAPIDO, which enrolled patients at higher risk of recurrence.[250] It is important to note that the trials evaluating TNT with FOLFIRINOX or FOLFOXIRI compared the TNT regimen to a standard preoperative chemoRT approach, not to a TNT strategy using FOLFOX; therefore, there are insufficient data to compare FOLFOX to FOLFIRINOX in this setting.

The TNT approach has demonstrated benefits including the early prevention or eradication of micrometastases; higher rates of pCR and longer PFS[246-251]; minimizing the length of time patients need an ileostomy[247]; facilitating resection; and improving the tolerance and completion rates of chemotherapy.[242,246,250,251] For some patients, surgery may be avoided if a CR is achieved as a result of neoadjuvant therapy (see *Watch-and-Wait Nonoperative Approach for Clinical Complete Responders*, below). Based on this, the NCCN Panel recommends TNT for stage II–III rectal cancer.

Sequencing of Therapy for TNT

It is not established whether it is better to start with chemotherapy, then follow with chemoRT, or vice versa when following a TNT approach. Results from the phase II Organ Preservation in Rectal Adenocarcinoma (OPRA) trial suggest that initiating treatment with chemoRT may improve TME-free survival, though was not powered to directly compare the two TNT strategies.[258,259] The randomized phase II CAO/ARO/AIO-12 study also looked at this question, comparing TNT approaches using either induction chemotherapy with FOLFOX followed by 5-FU/oxaliplatin chemoRT or chemoRT followed by consolidation chemotherapy.[260] This trial reported that upfront chemoRT led to higher completion rates for chemoRT, but lower completion rates for chemotherapy compared to upfront chemotherapy. pCR was observed in 17% of those who received upfront chemotherapy and 25% of those who received upfront chemoRT. In both OPRA and CAO/ARO/AIO-12 the time between radiation and assessment for CR was substantially shorter in the arms that gave systemic chemotherapy first, and this may confound interpretation of the differences. A secondary analysis reporting long-term (median, 43 months) results from the CAO/ARO/AIO-12 study showed similar long-term outcomes between the two groups, including 3-year DFS (73% for both groups; HR, 0.95; 95% CI, 0.63–1.45), 3-year incidence of local recurrence (6% vs. 5%), and distant metastases (18% vs. 16%).[261] Chronic toxicity of grade 3 or higher occurred in 11.8% of patients who received chemotherapy first compared to 9.9% who received chemoRT first. Collectively, these data suggest that the TNT approach of chemoRT followed by chemotherapy results in a higher rate of pCR while showing no significant differences in DFS, locoregional recurrence, distant

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metastases, or toxicities. Similar to other studies, pCR is not a validated surrogate endpoint for survival outcomes. While the NCCN Panel continues to monitor the data for consideration in future updates, the induction and consolidation chemotherapy approaches are currently considered as equivalent options for TNT in the guidelines.

Regimens for Concurrent ChemoRT

A number of randomized trials have established the benefit of adding chemotherapy (most often 5-FU/LV or capecitabine) to RT for treatment of localized rectal cancer. Putative benefits of the addition of chemotherapy concurrent with RT include local RT sensitization and systemic control of disease (ie, eradication of micrometastases). Preoperative chemoRT also has the potential to increase rates of pCR and sphincter preservation.

In a study of patients with T3–4 rectal cancer without evidence of distant metastases who were randomly assigned to receive either preoperative RT alone or preoperative concurrent chemoRT with 5-FU/LV, no difference in OS or sphincter preservation was observed in the two groups, although patients receiving chemoRT were significantly more likely to exhibit a pCR (11.4% vs. 3.6%; $P < .05$) and grade 3/4 toxicity (14.6% vs. 2.7%; $P < .05$) and less likely to exhibit local recurrence of disease (8.1% vs. 16.5%; $P < .05$).[262]

Preliminary results of a phase III trial that included an evaluation of the addition of chemotherapy to preoperative RT in patients with T3–4 resectable rectal cancer demonstrated that use of 5-FU/LV chemotherapy enhanced the tumoricidal effect of RT when the two approaches were used concurrently.[263] Significant reductions in tumor size, pTN stage, lymphatic invasion, vascular invasion, and PNI rates were observed.[263] More mature results from this trial reported that no significant differences in OS were associated with adding 5-FU–based chemotherapy preoperatively or postoperatively.[264]

The conclusions of these trials have been supported in a 2009 systematic review that included four RCTs.[265] In addition, a recent Cochrane review of six RCTs found that chemotherapy added to preoperative radiation in patients with stage III, locally advanced rectal cancer reduced the risk of local recurrence, but had no effect on OS, 30-day mortality, sphincter preservation, and late toxicity.[266] Similarly, a separate Cochrane review of stage II and III resectable disease found that the addition of chemotherapy to preoperative radiation enhances pathologic response and improves local control, but has no effect on DFS or OS.[267] Another recent meta-analysis of five RCTs comparing neoadjuvant chemoRT with neoadjuvant radiotherapy reached similar conclusions.[236]

With respect to the type of chemotherapy administered concurrently with RT,[238] the equivalence of bolus 5-FU/LV and infusional 5-FU in concurrent chemoRT for rectal cancer is supported by the results of a phase III trial (median follow-up of 5.7 years) in which similar outcomes with respect to OS and RFS were observed when an infusion of 5-FU or bolus 5-FU plus LV was administered concurrently with postoperative RT, although hematologic toxicity was greater in the group of patients receiving bolus 5-FU.[268] However, results from an earlier trial from the North Central Cancer Treatment Group (NCCTG) showed that postoperative administration of infusional 5-FU during pelvic irradiation was associated with longer OS when compared to bolus 5-FU.[269] Most of the patients in this study had node-positive disease. The Panel considers bolus 5-FU/LV/RT as an option for patients not able to tolerate capecitabine or infusional 5-FU.

Recent studies have shown that capecitabine is equivalent to 5-FU in perioperative chemoRT therapy.[270,271] The randomized NSABP R-04 trial compared the preoperative use of infusional 5-FU with or without oxaliplatin to capecitabine with or without oxaliplatin in 1608 patients with stage II or III rectal cancer.[271,272] No differences in locoregional events, DFS, OS, complete pathologic response, sphincter-saving surgery, or

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surgical downstaging were seen between the regimens, while toxicity was increased with the inclusion of oxaliplatin.

Similarly, a phase III randomized trial in which 401 patients with stage II or III rectal cancer received capecitabine– or 5-FU–based chemoRT either pre- or postoperatively showed that capecitabine was non-inferior to 5-FU with regard to 5-year OS (capecitabine 75.7% vs. 5-FU 66.6%; $P = .0004$), with capecitabine showing borderline significance for superiority ($P = .053$).[270] Furthermore, in this trial capecitabine demonstrated a significant improvement in 3-year DFS (75.2% vs. 66.6%; $P = .034$).[270] Because of these studies, capecitabine given concurrently with RT is listed in the guidelines as an acceptable alternative to infusional 5-FU in those patients who are able to manage the responsibilities inherent in self-administered, oral chemotherapy.

Addition of Oxaliplatin: In attempts to improve on the outcomes achieved with neoadjuvant 5-FU/RT or capecitabine/RT, several large randomized phase III trials (ACCORD 12, STAR-01, R-04, CAO/ARO/AIO-04, FOWARC, and PETACC 6) addressed the addition of oxaliplatin to the regimens. In a planned interim report of primary tumor response in the STAR-01 trial, grade 3 and 4 AEs occurred more frequently in patients receiving infusional 5-FU/oxaliplatin/RT than in those receiving infusional 5-FU/RT (24% vs. 8%; $P < .001$), while there was no difference in pathologic response between the study arms (16% pCR in both arms).[273] Results of the NSABP R-04 trial also showed that the addition of oxaliplatin did not improve clinical outcomes including the endpoints of locoregional events, DFS, OS, pCR, sphincter-saving surgery, and surgical downstaging, while it increased toxicity.[271,272]

Similar results were seen in the ACCORD 12/0405-Prodige 2 trial, in which capecitabine/RT (45 Gy) was compared to CAPEOX/RT (50 Gy) and the primary endpoint was pCR.[274] The pCR rates were similar at 19.2% and 13.9% ($P = .09$) for the oxaliplatin-containing arm and the control arm, respectively. Although patients treated with oxaliplatin and the higher radiation dose in the ACCORD 12 trial had an increased rate of minimal residual disease at the time of surgery (39.4% vs. 28.9%; $P = .008$), this did not translate to improved local recurrence rates, DFS, or OS at 3 years. The results did not change after longer term follow-up.[275] The PETACC 6 trial also investigated whether the addition of oxaliplatin to pre- and postoperative capecitabine would improve DFS for locally advanced rectal cancer.[276] Similar to other trials, oxaliplatin was found to impair tolerability without improving efficacy.

Results of the German CAO/ARO/AIO-04 trial have been published.[277,278] This trial also assessed the addition of oxaliplatin to a 5-FU/RT regimen. In contrast to STAR-01, R-04, and ACCORD 12, higher rates of pCR were seen in the oxaliplatin arm (17% vs. 13%, $P = .038$),[278] but this result could be because of differences in the 5-FU schedule between the arms.[279] The primary endpoint of this trial, the 3-year DFS rate, was 75.9% (95% CI, 72.4%–79.5%) in the oxaliplatin arm versus 71.2% (95% CI, 67.6%–74.9%) in the control group ($P = .03$).[277] Importantly, oxaliplatin was also added to the adjuvant therapy in the AIO-04 trial but not in the other trials, so cross-trial comparisons are limited.

In line with CAO/ARO/AIO-04, early results from the Chinese FOWARC phase III, open-label, multicenter trial, which randomized patients with locally advanced rectal cancer to neoadjuvant treatment consisting of infusional 5-FU/LV-RT, FOLFOX-RT, or FOLFOX, found that FOLFOX-RT resulted in higher rates of pCR and downstaging than the other regimens.[280] However, final results from FOWARC showed no significant improvement in 3-year DFS, local recurrence rates, or OS for FOLFOX with or without RT compared to 5-FU/LV-RT.[281]

Another randomized, multicenter, phase III trial looked at the addition of oxaliplatin during concurrent capecitabine chemoRT in the adjuvant setting for pathologic stage II/III disease.[282] Interim analysis showed no significant

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difference in 3-year DFS, OS, local recurrences, or distant metastases, with an increase in grade 3/4 acute toxicity in the CAPEOX-RT group.

Based on these results, the addition of oxaliplatin to neoadjuvant chemoRT is not recommended.

Addition of Targeted Agents: The randomized phase II EXPERT-C trial assessed CR rate with the addition of cetuximab to radiation treatment in 165 patients.[283] Patients in the control arm received CAPEOX followed by capecitabine/RT, then surgery followed by CAPEOX. Patients randomized to the cetuximab arm received the same therapy with weekly cetuximab throughout all phases. A significant improvement in OS was seen in patients with *KRAS* exon 2/3 wild-type tumors treated with cetuximab (HR, 0.27; 95% CI, 0.07–0.99; *P* = .034). However, the primary endpoint of CR rate was not met, and other phase II trials have not shown a clear benefit to the addition of cetuximab in this setting.[284,285] Further evaluation of this regimen is warranted.

The randomized, multicenter, phase II SAKK 41/07 trial evaluated the addition of panitumumab to preoperative capecitabine-based chemoRT in patients with locally advanced, *KRAS* wild-type rectal cancer.[286] The primary endpoint was pathologic near-complete plus complete tumor response, which occurred in 53% (95% CI, 36%–69%) of patients in the panitumumab arm versus 32% (95% CI, 16%–52%) in the control arm. Patients receiving panitumumab experienced increased rates of grade 3 or greater toxicity.

Another phase II study, RaP Study/STAR-03, also assessed the potential role of panitumumab in neoadjuvant chemoRT in patients with *KRAS* wild-type, cT3, N0 or cT2–3, N1–2, mid to low rectal cancer with a predicted negative CRM.[287] All patients were treated with panitumumab-chemoRT followed by resection and adjuvant FOLFOX. The primary endpoint of pCR was observed in 10.9% (95% CI, 4.7%–17.1%) of participants, not meeting the pre-specified level of 16%.

A phase II study of 57 patients with resectable T3/T4 rectal cancer evaluated preoperative treatment with capecitabine, oxaliplatin, bevacizumab, and RT, followed by surgery 8 weeks later and adjuvant FOLFOX/bevacizumab.[288] The 5-year OS rate was 80%, and the 5-year RFS rate was 81%. However, the primary endpoint of pCR was not met, significant toxicities were observed, and adherence to adjuvant therapy was low. Other randomized trials have also investigated the use of targeted therapies (eg, bevacizumab, ziv-aflibercept) within neoadjuvant therapy for localized rectal cancer with mixed conclusions.[257,289-293]

At this time the Panel does not endorse the use of bevacizumab, cetuximab, panitumumab, irinotecan, or oxaliplatin with concurrent radiotherapy for rectal cancer.

Neoadjuvant Systemic Therapy Without ChemoRT or RT

Although RT has been associated with decreased rates of local recurrence of rectal cancer, it is also associated with increased toxicity (eg, radiation-induced injury, hematologic toxicities) relative to surgery alone.[39,235,236] Therefore, there has been clinical interest in techniques that selectively omit RT from TNT protocols.

A small, single-center, phase II pilot trial treated patients with stage II or III rectal cancer with induction FOLFOX/bevacizumab chemotherapy followed by chemoRT only in those with stable or progressive disease and resection in all patients.[294] All 32 of the participants had an R0 resection, and the 4-year DFS was 84% (95% CI, 67%–94%). Another phase II trial, which included 60 patients with stage II/III rectal cancer (excluding cT4b) from eight institutions, assessed the R0 resection rate after FOLFOX plus either bevacizumab or cetuximab.[295] An R0 resection was achieved in 98.3% of the participants, and the pCR rate was 16.7%.

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The phase III FOWARC trial, discussed above, compared neoadjuvant therapy with and without radiation (without additional therapy for those with stable or progressive disease) and found that neoadjuvant FOLFOX without radiation gave lower rates of pCR than regimens that included radiation (6.6% vs. 14.0% for 5-FU-RT and 27.5% for FOLFOX-RT).[280] The rate of downstaging in the FOLFOX group was similar to the 5-FU-RT group but lower than the FOLFOX-RT group (35.5% vs. 37.1% for 5-FU-RT and 56.4% for FOLFOX-RT). However, final results from FOWARC showed no significant improvement in DFS, local recurrence rates, or OS for FOLFOX with or without RT compared to 5-FU/LV-RT.[281] Three-year DFS was 72.9%, 77.2%, and 73.5% (P = .709); 3-year local recurrence rate after resection was 8.0%, 7.0%, and 8.3% (P = .873); and 3-year OS was 91.3%, 89.1%, and 90.7% (P = .971) for 5-FU/LV-RT, FOLFOX-RT, and FOLFOX without RT, respectively.

PROSPECT was another phase III randomized study comparing neoadjuvant chemoRT to neoadjuvant chemotherapy (FOLFOX) with selective use of chemoRT based on response in patients who were candidates for sphincter-sparing surgery, and had either T2, N1–2 or T3, N0–2 rectal cancer (≤4 nodes, ≤1 cm).[296] In this trial, 1128 patients started treatment, 585 in the FOLFOX group and 543 in the chemoRT group. After a median follow-up of 58 months, DFS was similar between the two groups (HR, 0.92; 95% CI, 0.74–1.14; P = .005 for noninferiority). Five-year DFS was 80.8% in the FOLFOX group compared to 78.6% in the chemoRT group. OS and local recurrence rates were also similar. Notably, 98% of patients on this trial were free of local recurrence at 5 years, a rate that the authors attribute to careful selection of patients without high-risk features in the trial protocol. In the FOLFOX group, only 9.1% of patients went on to receive preoperative chemoRT and 1.4% received postoperative chemoRT. Patient-reported outcomes on the PROSPECT trial noted worse short-term AEs of anxiety, appetite, constipation, depression, dysphagia, dyspnea, edema, fatigue, mucositis, nausea, neuropathy, and vomiting during neoadjuvant treatment with FOLFOX compared to chemoRT, while 12-month long-term outcomes were better with FOLFOX than chemoRT in regards to fatigue, neuropathy, and sexual function.[297] Finally, initial results from the ongoing phase III CONVERT trial, comparing neoadjuvant chemotherapy with CAPEOX to neoadjuvant chemoRT, are also looking promising for this strategy.[298]

Finally, while it did not include upfront chemotherapy, the prospective OCUM study investigated whether neoadjuvant chemoRT could be restricted to those at high risk of locoregional recurrence without compromising outcomes.[299] High-risk was defined by the T-stage; location of the tumor in the rectum; and the distance between the tumor, suspicious lymph nodes or tumor deposits, and mesorectal fascia. Patients with disease that had a ≤1 mm distance and/or cT3-4 tumors in the lower-third of the rectum were classified as high-risk and received neoadjuvant chemoRT prior to TME, while those with a distance >1 mm were classified low-risk and underwent upfront TME. Of the patients on the study, a total of 530 (60%) underwent up-front surgery, and 354 (40%) had neoadjuvant chemoRT followed by surgery. 5-year local recurrence rates were 4.1% for all patients treated per protocol, 2.9% after upfront TME, and 5.7% with neoadjuvant chemoRT followed by TME. The 5-year rate of distant metastases was 15.9% and 30.5%, respectively. OCUM concluded that patients with rectal cancer that was classified as low-risk were able to safely avoid neoadjuvant chemoRT without compromising outcomes. OCUM also concluded that for high-risk disease neoadjuvant therapy should be intensified to improve outcomes.

Based on these results, selective omission of chemoRT following favorable response to neoadjuvant chemotherapy may be considered as an option for patients who meet the inclusion criteria of the PROSPECT trial and wish to avoid the long-term effects of RT. The Guidelines recommend restaging with sigmoidoscopy, with or without MRI, following

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12 to 16 weeks of chemotherapy. If tumor regression is >20% the treatment plan may omit chemoRT and proceed directly to surgery, while short-course RT or chemoRT is recommended prior to surgery if tumor regression is ≤20%. Per the NCCN Panel, the adjuvant chemotherapy that was given on some of these trials is not necessary and would not be recommended as TNT is now the preferred approach. For neoadjuvant therapy without RT for MMR deficient (dMMR)/MSI-high (MSI-H) disease, see *Checkpoint Inhibitor Immunotherapy for dMMR/MSI-H Nonmetastatic Disease*, below.

Technical Aspects of Radiation Therapy

Multiple RT fields should include the tumor or tumor bed with a 2- to 5-cm margin, the mesorectum, the presacral nodes, and the internal iliac nodes. The external iliac nodes should also be included for T4 tumors involving anterior structures; inclusion of the inguinal nodes for tumors invading into the distal anal canal can also be considered. Recommended doses of radiation are typically 45 to 50 Gy in 25 to 28 fractions to the pelvis using three or four fields. Positioning and other techniques to minimize radiation to the small bowel are encouraged. The Radiation Therapy Oncology Group (RTOG) has established normal pelvic contouring atlases (available online at https://seor.es/wp-content/uploads/2014/03/RTOGAnorectalContouringGuidelines-1.pdf).[300] Intensity-modulated RT (IMRT) should be considered for clinical situations such as re-irradiation of previously treated recurrent disease, patients treated postoperatively due to increased acute or late toxicity, T4 primary tumors given the more anterior field changes with coverage of the external iliac nodes, which includes more small bowel, or unique anatomical situations where IMRT facilitates the delivery of recommended target volumes while respecting accepted normal tissue dose-volume constraints.[301] Ablative stereotactic body radiotherapy (SBRT) should only be used in the setting of a clinical trial or in the setting of oligometastasis to the lung, liver, or an abdominopelvic node when other modalities are not appropriate.

Coordination of preoperative chemoRT and surgery is important. Although longer intervals from completion of chemoRT to surgery have been shown to be associated with an increase in pCR rates,[302-307] it is unclear whether such longer intervals are associated with clinical benefit. Results of one NCDB analysis suggest that an interval of >8 weeks is associated with increased odds of pCR,[308] whereas other similar analyses concluded that an interval >56 or 60 days (8–8.5 weeks) is associated with higher rates of positive margins, lower rates of sphincter preservation, and/or shorter survival.[309,310] A pooled analysis of seven randomized trials concluded that the best time to achieve pCR was at 10 weeks following neoadjuvant chemoRT, with 95% of pCR events occurring within that time period.[311]

The GRECCAR-6 phase III, multicenter, randomized, open-label, parallel-group controlled trial randomized patients with stage II/III rectal cancer treated with chemoRT to a 7- or 11-week interval before surgery.[312] The pCR rate was not different between the groups (15.0% vs. 17.4%; *P* = .60), but the morbidity (44.5% vs. 32%; *P* = .04), medical complications (32.8% vs. 19.2%; *P* = .01), and rate of complete mesorectal resection (78.7% vs. 90%; *P* = .02) were worse in the 11-week group. The rate of anastomotic leaks and the mean length of hospital stay were similar between the groups. Three-year survival results from the GRECCAR-6 trial showed no difference in 3-year OS (*P* = .8868), DFS (*P* = .9409), distant recurrence (*P* = .7432), or local recurrence (*P* = .3944) between the 7- or 11-week interval groups.[313]

The CRONOS study divided 1506 patients with locally advanced rectal cancer into three groups with different time intervals between completion of neoadjuvant therapy and surgery: short (≤8 weeks), intermediate (>8 and ≤12 weeks), and long (>12 weeks).[314] No differences were noted between the groups for pCR rates, although the long-interval group had a

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significantly lower risk of poor response (TRG 2–3), systemic recurrence, a higher conversion risk, minor postoperative complications, and incomplete mesorectum compared with the intermediate-interval group, indicating an increased surgical complexity with a longer interval.

Short-Course Radiation

Several European studies have looked at the efficacy of a shorter course of preoperative RT (25 Gy over 5 days), not combined with chemotherapy, for the treatment of rectal cancer. The results of the Swedish Rectal Cancer Trial evaluating the use of short-course RT administered preoperatively for resectable rectal cancer showed a survival advantage and a decreased rate of local recurrence with this approach compared with surgery alone.[315] However, a follow-up study published in 2005 showed that the patients with short-course preoperative RT had increased RR for postoperative hospitalization due to bowel obstructions and other GI complications.[316] A number of other studies also investigating the effectiveness of preoperative short-course RT in patients with rectal cancer staged as T1–3 have demonstrated that OS was not significantly affected despite improvements in local control of disease.[317-319] A more recent multicenter, randomized study of 1350 patients with rectal cancer compared 1) short-course preoperative RT and no postoperative treatment with 2) no preoperative RT and a postoperative approach that included chemoRT in selected patients (ie, those with a positive CRM following resection) and no RT in patients without evidence of residual disease following surgery.[320] Results indicated that patients in the preoperative RT arm had significantly lower local recurrence rates and a 6% absolute improvement in 3-year DFS (P = .03), although no difference in OS was observed between the study arms.[320,321]

Long-term (12 years) follow-up of one of the short-course RT trials (the Dutch TME trial[318]) was reported.[322] The analysis showed that 10-year survival was significantly improved in patients with stage III disease and a negative CRM in the RT plus surgery group compared to the group that received surgery alone (50% vs. 40%; P = .032).[322] However, this long follow-up showed that secondary malignancies and other non-rectal cancer causes of death were more frequent in the RT group than in the control group (14% vs. 9% for secondary malignancies), negating any survival advantage in the node-negative subpopulation.

A few studies have compared short-course RT to long-course chemoRT. One randomized study of 312 patients in Poland directly compared preoperative short-course RT and more conventional preoperative long-course chemoRT and found no differences in local recurrence or survival.[323] Similarly, an Australian/New Zealand trial (Trans-Tasman Radiation Oncology Group [TROG] trial 01.04) that randomized 326 patients to short-course RT or long-course chemoRT found no differences in local recurrence and OS rates.[324] In addition, rates of late toxicity, distant recurrence, and RFS were not significantly different between the arms. Patients in the long-course arm were more likely to experience serious AEs (eg, radiation dermatitis rates, 0% vs. 5.6%; P = .003), whereas patients in the short-course arm were more likely to have a permanent stoma (38.0% vs. 29.8%; P = .13).[325] However, no overall difference was seen in health-related quality of life between the groups.[326] Finally, a trial compared short-course RT with long-course chemoRT with delayed surgery in both groups.[327] Although the long-course arm experienced greater tumor downsizing and downstaging compared with short-course treatment, no differences were seen in the R0 resection rates or postoperative morbidity. The 3-year DFS was better in the long-course arm than in the short course arm (75% vs. 59%; P = .022), with no difference in OS.[328]

The randomized phase III Polish II study randomized patients with cT3/cT4 rectal cancer to either preoperative short-course radiation followed by FOLFOX4 or preoperative long-course chemoRT with bolus 5-

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FU/LV and oxaliplatin.[329] Of 515 patients eligible for analysis, preoperative acute treatment toxicity was lower with short-course RT (P = .006). No differences in local efficacy or 3-year DFS were observed between the groups, although 3-year OS was higher for the short-course group (73% vs. 65%, P = .046). However, long-term results of this trial showed no difference in 8-year OS (49% for both groups).[330] The rate of late complications was also similar between the two groups.

The randomized phase III RAPIDO trial compared standard treatment (chemoRT, followed by TME, then optional adjuvant chemotherapy with CAPEOX or FOLFOX) to an experimental TNT approach (short-course RT, followed by chemotherapy before TME) in 912 patients with locally advanced rectal cancer. Early results of 901 evaluable patients showed a high percentage of patients who completed at least 75% of their prescribed chemotherapy (84% for the short-course arm compared to 57% in the long-course arm).[331] While considerable toxicity did occur during preoperative therapy, there were no significant differences noted in the surgical procedures performed or postoperative complications between the two arms. At 3 years after randomization, the rate of disease-related treatment failure was 23.7% with TNT compared to 30.4% with standard treatment (HR, 0.75; 95% CI, 0.60–0.95; P = .019).[250] No differences were found in the secondary endpoint of OS. Serious AEs occurred in 38% of the TNT group and 34% in the standard treatment group. While locoregional recurrence rates were similar between the study arms at 3 years, a 5-year follow-up of the RAPIDO trial reported an increased risk of locoregional recurrence in the experimental arm.[332] The experimental arm, consisting of treatment with short-course RT, chemotherapy, followed by surgery had a locoregional recurrence rate of 10%, while the control arm, consisting of treatment with chemoRT, surgery, followed by optional adjuvant chemotherapy had a locoregional recurrence rate of 6%, P = .027. OS after locoregional failure was comparable.

Stockholm III was another randomized, phase III study that compared short-course RT to long-course RT in 840 patients with resectable rectal cancer.[333,334] This trial included two randomizations, a two-arm randomization that compared short-course RT with immediate surgery to short-course RT with delayed surgery (described below), and a three-arm randomization that compared short-course RT with immediate surgery, short-course RT with delayed surgery, and long-course RT with delayed surgery. For the 385 patients in the three-arm randomization, the incidence of local recurrence was 2.3% for short-course with immediate surgery, 3.1% for short-course with delayed surgery, and 5.4% for long-course RT with a median follow-up of 5.7 years.[333] Median OS was 8.1, 10.3, and 10.5 years for short-course RT with immediate surgery, short-course with delayed surgery, and long-course RT, respectively. No comparisons showed statistically significant differences and long-term health-related quality of life was also similar between the groups.

STELLAR is a randomized, phase III trial that compared short-course RT followed by CAPEOX to capecitabine-based long-course chemoRT as neoadjuvant therapy in 599 patients with stage 2–3 rectal cancer.[335] Both groups received TME 6 to 8 weeks after preoperative treatment and adjuvant chemotherapy was given based on preoperative treatment. Three-year DFS was 64.5% for short-course RT and 62.3% for long-course chemoRT. There was also no significant difference in metastasis-free survival or locoregional recurrence between the two groups. Three-year OS was higher in the short-course RT group (86.5% vs. 75.1%; P = .033), but the prevalence of acute grade ≥3 toxicities during preoperative treatment was higher with short-course RT (26.5% vs. 12.6%; P < .001).

A 2014 systematic review identified 16 studies (RCTs, phase II trials, and retrospective studies) that addressed the interval between short-course RT and resection of rectal cancer.[336] Lower rates of severe acute post-radiation toxicity but higher rates of minor postoperative complications

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were seen in the immediate-surgery group (1- to 2-week interval) compared with the delayed surgery group (5- to 13-week interval). The pCR rates were significantly higher in the delayed-surgery group, with no differences in sphincter preservation and R0 resection rates. The Stockholm III trial also investigated the optimal interval between short-course radiotherapy and surgery in 455 patients within the two-arm randomization.[333,334] This trial showed similar oncologic outcomes and long-term health-related quality of life between the immediate surgery versus 4- to 8-week delay following short-course RT groups,[333] but a lower rate of postoperative complications in the group that delayed surgery following short-course RT (53% vs. 41%; OR, 0.61; 95% CI, 0.45–0.83; *P* = .001).[334]

Overall, it appears that short-course RT gives effective local control and the same OS as more conventional RT schedules, and therefore is considered as an appropriate option for patients with locally advanced rectal cancer. However, based on results from the RAPIDO trial, the NCCN Panel cautions that preoperative short-course RT may be associated with a higher risk of local recurrence. A multidisciplinary evaluation, including a discussion of the need for downstaging and the possibility of long-term toxicity, is recommended when considering short-course RT.

Response to Neoadjuvant Treatment

Fifty percent to 60% of patients are downstaged following neoadjuvant therapy, with about 20% of patients showing a pCR.[337-343] Recent studies have suggested that the response to neoadjuvant treatment correlates with long-term outcomes in patients with rectal cancer. One study of 3096 patients treated with neoadjuvant chemoRT found that any two of four AJCC/CAP tumor regression grading (TRG) categories predicted long-term survival outcomes including OS, DFS, local RFS, and distant metastasis-free survival.[344] In the MERCURY prospective cohort trial, 111 patients were assessed by MRI and pathologic staging.[345] On multivariate analysis, MRI-assessed tumor regression grade was significantly associated with OS and DFS. Patients with poor tumor regression grade had 5-year survival rates of 27% versus 72% for patients with good tumor regression grade (*P* = .001), and DFS rates were 31% versus 64% (*P* = .007). Referring to the AJCC/CAP TRG classification scheme, a new MRI-TRG system verified in 1033 cases found that the MRI-TRG system might be a surrogate for AJCC/CAP's TRG classification scheme. Particularly, as compared with other pairs, MRI-TRG 0 displayed the highest sensitivity (90.1%) and specificity (92.8%) in identifying AJCC/CAP TRG 0 category (pCR) patients, since radiotherapy induced anorectal fibrosis and edema was rare in these two subgroup patients.[346] Similarly, in the CAO/ARO/AIO-94 trial, patients with pathologic complete regression had 10-year cumulative incidence of distant metastasis and DFS of 10.5% and 89.5%, respectively, while those with poor regression had corresponding incidences of 39.6% and 63%.[347] A recent retrospective review of 725 patients with rectal cancer found similar results.[341] In this study, pathologically determined response to neoadjuvant treatment correlated with long-term outcomes. Five-year RFS rates were 90.5%, 78.7%, and 58.5% for patients with complete, intermediate, and poor responses, respectively (*P* < .001). Distant metastases and local recurrences also correlated with the level of response. Other studies have also shown a prognostic effect of response to neoadjuvant treatment.[348,349]

Watch-and-Wait Nonoperative Approach for Clinical Complete Responders

As preoperative treatment and imaging modalities have improved, it has become apparent that patients with a cCR to neoadjuvant therapy may be spared the morbidities of surgery, an approach called nonoperative management (NOM). A small prospective study included a more thorough assessment of treatment response and used very strict criteria to select 21 of 192 patients (11%) with cCR who were then observed with careful follow-up and compared to 20 patients with a complete pathologic

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response after resection.[350] Only one patient in the NOM group developed a local recurrence after a mean follow-up of 25 months; that patient underwent successful surgery. No statistical differences in long-term outcomes were seen between the groups. Short-term functional outcomes, however, were better in the NOM group, with better bowel function scores, less incontinence, and 10 patients avoiding permanent colostomy. Other prospective studies, case series, and systematic reviews added to the growing evidence that the NOM approach warranted further investigation.[162,351-356]

The International Watch & Wait Database (IWWD) aims to collect data to expand knowledge on the benefits, risks, and safety of organ preservation in rectal cancer using a large-scale registry of pooled individual patient data from multiple institutions. A 2018 analysis included data from 880 patients in the IWWD with disease that had a cCR following neoadjuvant therapy and were followed by the watch-and-wait protocol.[357] In this analysis, the 2-year incidence of local recurrence was 25.2% and 88% of local recurrences occurred in the first 2 years. Distant metastases occurred in 8% of patients, 5-year OS was 85%, and 5-year disease-specific survival was 94%. A 2021 analysis of the IWWD showed similar results.[358] This analysis included 793 patients with cCR who were followed using the watch-and-wait strategy. With a median follow-up of 55.2 months, the probability of remaining free of local recurrence for an additional 2 years was 88.1% after 1 year of DFS, 97.3% after 3 years of DFS, and 98.6% after 5 years of DFS. These same measures for distant metastasis-free survival were 93.8% for 1 year, 97.8% for 3 years, and 96.6% for 5 years. Together, current data from the IWWD suggest that disease recurrence occurs most frequently within the first 2 to 3 years following CR and a more intense surveillance schedule is recommended during this time period.[357,358]

The OPRA trial is a randomized, phase II trial of the watch-and-wait approach.[259] OPRA assessed the outcomes of 324 patients with stage II or III rectal cancer treated with TNT using either an induction chemotherapy followed by chemoRT approach or an approach using chemoRT followed by consolidation chemotherapy. Following neoadjuvant treatment, patients received either TME or observation (NOM) based on tumor response. Organ preservation was achievable in about half of patients treated with TNT on OPRA with 3-year TME-free survival of 41% in the induction chemotherapy group and 53% in the consolidation chemotherapy group. The primary endpoint of DFS was 76% for both groups, which is in line with the 75% 3-year DFS rate observed historically. No differences were observed between the groups for RFS, distant metastasis-free survival, or OS.

After a median follow-up of 5.1 years, the OPRA trial continued to show long-term organ preservation in half of the patients treated with TNT on the trial.[359] Five-year DFS was 71% in the induction chemotherapy group and 69% in the consolidation chemotherapy group. TME-free survival was 39% for induction chemotherapy and 54% for consolidation chemotherapy. Of the 81 patients with tumor regrowth, 94% occurred in the first 2 years and 99% within 3 years, highlighting the importance of close surveillance in the first 2 years. A secondary analysis of the OPRA trial suggested a 3-tier grading schema (cCR, near-CR, and incomplete response), which could be used to estimate recurrence and survival outcomes and maximize eligibility for NOM in patients who receive TNT for locally advanced rectal cancer.[360] The PKUCH-R01 trial from China reported similar results, with a 3-year organ preservation rate of 67.2% for cT2–3 low-risk rectal cancers treated with chemoRT plus consolidation CAPEOX.[361]

Despite the impressive results of prospective trials, longer follow-up, larger sample sizes, and additional careful observational studies are still needed

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to better understand long-term outcomes for those who are treated using a watch-and-wait approach.[362] Furthermore, recent studies have found that neither FDG-PET, nor MRI, nor CT can accurately determine a pCR, complicating the selection of appropriate patients for a nonoperative approach.[121-129,363] In addition, lymph node metastases are still seen in a subset of patients with pCR.[364] An analysis of the IWWD found that patients who experienced regrowth of local disease following NOM were more likely to develop distant metastases compared to those treated with TME (22.8% vs. 10.2%; $P \leq .001$).[365] Keeping these caveats in mind, the Panel believes that an NOM approach may be considered in centers with experienced multidisciplinary teams after a careful discussion with the patient about their risk tolerance.

Careful surveillance is essential for those considering a watch-and-wait approach in order to treat tumor regrowth in a timely manner. The OPRA trial included the following surveillance protocol for watch-and-wait: DRE, flexible sigmoidoscopy, and CEA every 4 months for the first 2 years, then every 6 months for years 3 to 5; MRI every 6 months for the first 2 years, then every 12 months for years 3 to 5; CT chest/abdomen/pelvis once a year for 5 years; and colonoscopy once at year 1 and again at year 5.[259] Watch-and-wait surveillance protocols are an area of active investigation and other protocols have been suggested.[357,358,366] The watch-and-wait surveillance schedule recommended by the NCCN Panel based on clinical and institutional experiences is similar to the OPRA protocol and includes DRE and proctoscopy or flexible sigmoidoscopy every 3 to 4 months for 2 years, then every 6 months for the next 3 years, and MRI of the rectum every 6 months for up to 3 years.

Checkpoint Inhibitor Immunotherapy for dMMR/MSI-H Nonmetastatic Disease

An active focus of current research is the preoperative treatment of dMMR or MSI-H locally advanced rectal cancers with immune checkpoint inhibitors. A small prospective phase II trial studied the effects of dostarlimab-gxly, an anti-programmed cell death protein (PD-1) monoclonal antibody, on 12 patients with dMMR, stage II or III rectal adenocarcinoma.[367] All 12 patients showed a cCR, with no evidence of tumor on MRI, PET/CT, endoscopic evaluation, DRE, or biopsy. At the time of study publication, with a follow-up range of 6 to 25 months, no patients had received chemoradiotherapy or surgery and no cases of progression or recurrence had been reported. No AEs of grade ≥3 were reported. A larger follow-up of this trial included two cohorts, one including patients with dMMR locally advanced rectal cancer and another including patients with dMMR non-rectal solid tumors, both treated with dostarlimab-gxly for 6 months.[368] Of the 49 patients who completed treatment in the rectal cancer cohort, all showed a cCR and elected to forgo surgery. RFS at 2 years was 96% in this cohort, with a median follow-up for recurrence at 30.2 months. Two patients in the rectal cancer cohort had disease recurrence, one at the primary tumor site, a second in the lymph nodes. No deaths had occurred in either cohort at the time of study publication. AEs occurred in 65% of patients who received at least one dose of dostarlimab across both cohorts. Grade 3–4 AEs were rare, but included diabetes, lung infection, hypothyroidism, encephalitis, and neutropenia (one patient each).

Another small phase II study investigated pembrolizumab for neoadjuvant treatment of localized dMMR/MSI-H solid tumors.[369] Thirty-five patients were enrolled in this study, of which 27 had colorectal cancer (eight with rectal adenocarcinoma). After treatment with pembrolizumab, approximately half of the patients on the study underwent surgical resection of the tumor, while half of the patients elected for NOM, continuing pembrolizumab for up to a year without resection. In the full study population, ORR was 82%, with 30% CR and 52% partial responses. Two of the eight patients with rectal cancer had disease progression: one following two cycles of pembrolizumab, and one after 9 months. Grade 3 AEs were reported by two patients (6%).

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In addition to the limited clinical trial data, multiple retrospective series suggest a high, though not 100%, cCR rate after immunotherapy for locally advanced colon or rectal cancer.[370-372]

On the other hand, data have suggested that dMMR/MSI-H rectal cancers may not respond as well to neoadjuvant chemotherapy compared to MMR proficient (pMMR)/microsatellite stable (MSS) cancers. A retrospective study of 21 patients with dMMR rectal cancer who received neoadjuvant chemotherapy (5-FU/oxaliplatin) was compared with 63 patients with pMMR rectal cancer who received the same treatment. Of the patients with dMMR rectal cancer, six had disease progression compared to none of the patients with pMMR cancer (29% vs. 0%, $P = .0001$).[373] Responses to neoadjuvant chemoRT were comparable between the dMMR and pMMR cancers.

Based on the positive results of neoadjuvant treatment with checkpoint inhibitors, the NCCN Panel recommends neoadjuvant treatment with either dostarlimab-gxly, nivolumab, or pembrolizumab as the preferred treatment option for stage II–III dMMR/MSI-H rectal cancer. A TNT approach using neoadjuvant chemoRT and chemotherapy may be considered for patients who are not candidates for immunotherapy.

Adjuvant Chemotherapy for Resected Nonmetastatic Disease

With increasing data supporting the TNT approach to perioperative treatment of nonmetastatic rectal cancer, current adjuvant treatment recommendations are limited to disease that was upstaged to stage II or III following transabdominal resection, but had not received neoadjuvant therapy prior to surgery. For these patients, adjuvant treatment with chemotherapy (generally FOLFOX or CAPEOX), either alone or with induction or consolidation chemoRT, is recommended. Few studies have evaluated the effect of adjuvant therapy in patients with rectal cancer, with most of these studying the effect of adjuvant therapy for patients who had received neoadjuvant chemoRT, and, therefore, its role is not well-defined.[374-376] Adjuvant chemotherapy may also be considered in rare circumstances where a stage II–III dMMR/MSI-H rectal tumor does not respond to checkpoint inhibitor immunotherapy (persistent disease at 6 months) and is subsequently treated with preoperative chemoRT or RT, followed by transabdominal resection, although this recommendation is based on clinical judgment rather than data. For all other situations, the TNT approach is now the only recommended option for treatment of stage II–III rectal cancer.

Low-Dose Aspirin for Stage II–III PIK3CA Mutation-Positive CRC

Data have shown that low-dose aspirin or COX-2 inhibitor therapy after a diagnosis of CRC decreases the risk of recurrence and death.[377-383] For example, a population-based, observational, retrospective cohort study of 23,162 patients with CRC in Norway found that post-diagnosis aspirin use was associated with improved CRC-specific survival (HR, 0.85; 95% CI, 0.79–0.92) and OS (HR, 0.95; 95% CI, 0.90–1.01).[377] However, other studies have shown these differences to be non-significant.[384-386]

More recent evidence suggests that tumor mutations in *PIK3CA* may be predictive of response to low-dose aspirin. An analysis of the Nurses' Health Study and the Health Professionals Follow-up Study found that in patients with *PIK3CA*-mutated CRC, regular use of aspirin was associated with better survival outcomes.[387] This association was not found for patients with *PIK3CA* wild-type tumors. On the CALGB/SWOG 80702 trial, adjuvant celecoxib was shown to significantly improve OS in patients with *PIK3CA*-mutated stage III colon cancer, although DFS was not significantly different.[388] Similarly, the phase III randomized SAKK 41/13 trial also showed a survival benefit for patients with *PIK3CA*-mutated stage II or III colon cancer who were treated with aspirin, although results were not significant due to premature study closure.[389]

The ALASCCA trial is a randomized, multicenter trial that identified 1103 patients with either stage I–III rectal cancer or stage II–III colon cancer and

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alterations in the PI3K pathway.[390] Of these identified patients, 626 were randomized to either 160 mg of aspirin daily, started within 3 months post surgery and continued for 3 years, or placebo. Group A included those with *PIK3CA* mutations in exon 9 or 20, while Group B included those with other somatic PI3K pathway driver alterations. An abstract at the ASCO Gastrointestinal Cancers Symposium reported that, after 3 years of follow-up, both groups showed reduced recurrence rates with aspirin compared to placebo. The HRs for time to recurrence were 0.49 (95% CI; 0.24–0.98; *P* = .044) in Group A and 0.42 (95% CI; 0.21–0.83; *P* = .013) in Group B. The HRs for DFS were 0.61 (95% CI; 0.34–1.08; *P* = .091) in Group A, and 0.51 (95% CI; 0.29–0.88; *P* = .017) in Group B. Three patients experienced severe AEs from treatment with aspirin, including gastrointestinal (GI) bleeding, hematoma, and allergic reaction.

Based on these results, the NCCN Panel recommends that aspirin should be given daily for 3 years following surgery for stage II–III rectal cancer harboring a *PIK3CA* mutation. While observational data support either aspirin or celecoxib as therapeutic options to reduce risk of recurrence in patients with non-metastatic colorectal cancers with PIK3 pathway altering mutation, the committee favored use of aspirin given the data from two RCTs as well as cost considerations. The CALGB/SWOG 80702 trial evaluated the potential role of circulating tumor DNA (ctDNA) as a predictive marker for the benefit of COX-2 inhibition, but the interaction *P* value was not found to be significant and the Panel does not currently recommend the use of ctDNA for this purpose.[391]

NCCN Recommendations for Nonmetastatic Rectal Cancer

Recommendations for Patients with T1 and T2 Lesions

Several treatment options exist for node-negative T1 rectal lesions, depending on tumor characteristics, institutional experience, and patient preference. For appropriately selected patients, ESD or transanal local excision are options. Criteria for these techniques are available in *Principles of Surgery and Locoregional Therapies* in the algorithm. Another treatment option for node-negative T1 disease, especially when high-risk features are suspected, is transabdominal resection. Transabdominal resection may be done upfront, followed by adjuvant therapy, or as part of a TNT approach, where neoadjuvant therapy is done with chemoRT or short-course RT, sometimes followed by chemotherapy with FOLFOX or CAPEOX ahead of surgery. If there is a complete clinical response to neoadjuvant therapy, a watch-and-wait NOM approach may be considered. Node-negative T2 lesions are treated with transabdominal resection, since local recurrence rates of 11% to 45% have been observed for T2 lesions following local excision alone.[97,392,393] The same considerations for a TNT versus adjuvant therapy approach are applicable when treating T2 rectal tumors with transabdominal resection.

After local excision, if pathology review reveals no high-risk features, then no additional treatment is required. If, however, pathology review after local excision reveals a poorly differentiated histology, positive margins, invasion into the sm3 level, or LVI or if the tumor is restaged to pT2, additional treatment is required. The options are: 1) transabdominal resection (preferred) followed by adjuvant therapy based on pathologic stage (see *Adjuvant Treatment Recommendations for cT1–2 N0 Rectal Cancer*, below); or 2) a TNT approach followed by transabdominal resection or NOM as described above. Results of a meta-analysis suggest that transanal local excision followed by chemoRT without a transabdominal resection may be associated with higher rates of local recurrence than transanal local excision followed by transabdominal resection.[394] Careful surveillance of patients forgoing transabdominal resection in this setting is advised.

Following transabdominal resection of patients with clinical stage T1–2 N0 rectal cancer, patients should receive adjuvant therapy based on pathologic stage, if not done following a TNT approach, in which case no

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further treatment is needed (see *Adjuvant Treatment Recommendations for cT1–2 N0 Rectal Cancer*, below).

Adjuvant Treatment Recommendations for cT1–2 N0 Rectal Cancer

Patients who had a transabdominal resection for stage cT1–2 rectal cancer, without neoadjuvant therapy, are given further treatment based on the pathologic stage. Patients with tumors staged as pT1–2, N0, M0 require no further treatment. If pathology review reveals pT3, N0, M0, chemoRT, given either before or after chemotherapy, is one option. Observation can also be considered in these patients if the tumor was well-differentiated or moderately well-differentiated carcinoma invading <2 mm into the mesorectum, without lymphatic or venous vessel involvement and was located in the upper rectum.[395] Finally, chemotherapy with FOLFOX or CAPEOX alone is an option, and is especially suitable for margin-negative proximal tumors.

For patients with resected disease containing positive nodes and/or pT4 disease, adjuvant therapy may include long-course chemoRT followed by chemotherapy or vice-versa.[238,268,269] Adjuvant chemotherapy alone with FOLFOX or CAPEOX may also be considered for patients with pT1–3, N1 disease only. The Panel recommends perioperative therapy for a total duration of up to 6 months.

For all patients with disease determined to be stage II–III and harboring a *PIK3CA* mutation, the addition of aspirin is recommended for 3 years following surgery.

Recommendations for Patients with pMMR/MSS Node-Positive, T1–2 Disease, T3–4 N Any Lesions, or Locally Unresectable or Medically Inoperable Disease

For patients with higher-risk stage II or III rectal cancer, including cT3 or cT4 lesions, node-positive disease, and locally unresectable or medically inoperable disease, treatment options depend on the MMR or MSI status of the tumor. The recommendations described in this section are for pMMR/MSS disease; options for dMMR/MSI-H disease are detailed in the next section.

For higher-risk stage II or III pMMR/MSS rectal cancer, TNT is now the recommended approach for most patients. This is because TNT shows better outcomes for patients, as described in the section on *The Total Neoadjuvant Therapy Approach* above. In the TNT approach, 12 to 16 weeks of chemotherapy are followed by chemoRT or short-course RT, restaging, and transabdominal resection. Alternatively, a TNT approach may start with chemoRT or short-course RT, followed by 12 to 16 weeks of chemotherapy, then restaging and transabdominal resection. While there are some data showing better outcomes with consolidation chemotherapy (compared to induction chemotherapy), the Panel does not consider the data sufficiently definitive to prefer one approach over the other at this time. FOLFOX or CAPEOX are generally used for chemotherapy, although FOLFIRINOX may also be considered.

For those patients with a complete clinical response to neoadjuvant therapy with no evidence of residual disease on digital rectal examination, rectal MRI, and direct endoscopic evaluation, a watch-and-wait NOM approach may be considered in centers with experienced multidisciplinary teams. The degree to which risk of local and/or distant disease progression may be increased relative to standard surgical resection has not yet been adequately characterized. Decisions for NOM should involve a discussion with the patient of their risk tolerance and a careful surveillance schedule must be followed. The data supporting this approach are discussed in *Watch-and-Wait Nonoperative Approach for Clinical Complete Responders*, above.

When resection is contraindicated following primary treatment, patients should be treated with a systemic regimen for advanced disease (see discussion of *Systemic Therapy for Advanced or Metastatic Disease* in the NCCN Guidelines for Colon Cancer [available at www.NCCN.org).

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For eligible patients wishing to avoid the long-term effects of RT, selective omission of chemoRT following favorable response to neoadjuvant chemotherapy may be considered. As per the inclusion criteria of the PROSPECT trial, patients who are eligible for this approach should have non-T4 disease that is eligible for sphincter-sparing surgery. Following 12 to 16 weeks of chemotherapy with FOLFOX or CAPEOX, the tumor should be restaged with sigmoidoscopy, alone or in addition to MRI. If tumor regression is >20% the treatment plan may omit chemoRT and proceed directly to surgery, while short-course RT or chemoRT is recommended prior to surgery if tumor regression is ≤20%. See *Neoadjuvant Systemic Therapy Without ChemoRT or RT* for discussion of the considerations in choosing this approach.

For all patients with disease determined to be stage II–III and harboring a *PIK3CA* mutation, the addition of aspirin is recommended for 3 years following surgery.

Recommendations for Patients with dMMR/MSI-H Node-Positive, T1–2 Disease, T3–4 N Any Lesions, or Locally Unresectable or Medically Inoperable Disease

For higher-risk stage II–II disease that is determined to be dMMR or MSI-H, the preferred treatment approach is now neoadjuvant immunotherapy with a PD-1 inhibitor (dostarlimab, nivolumab, or pembrolizumab) for up to 6 months. Disease status should be reevaluated every 2 to 3 months to check for response to therapy. Most patients with these characteristics will show a cCR to immunotherapy, in which case surveillance is recommended, with the immunotherapy considered to be definitive treatment in these cases. If disease persists after 6 months, treatment with long-course chemoRT or short-course RT should be given. Following this treatment, the patient may proceed to either transabdominal resection or, if complete clinical response, surveillance may be considered. For those completing transabdominal resection, adjuvant chemotherapy with FOLFOX or CAPEOX may be considered. For persistent disease where resection is contraindicated, standard first-line systemic therapy may be given, although FOLFIRINOX is not recommended in this setting.

For patients who are not candidates for checkpoint inhibitor immunotherapy, a standard TNT approach may be followed. For all patients with disease determined to be stage II–III and harboring a *PIK3CA* mutation, the addition of aspirin is recommended for 3 years following surgery.

Management of Metastatic Disease

Approximately 50% to 60% of patients diagnosed with CRC will develop colorectal metastases,[396-398] and 80% to 90% of these patients have unresectable metastatic liver disease.[120,397,399-401] Metastatic disease most frequently develops metachronously after treatment for locoregional CRC, with the liver as the most common site of involvement.[402] However, 20% to 34% of patients with CRC present with synchronous liver metastases.[119,120] Some evidence indicates that synchronous metastatic colorectal liver disease is associated with a more disseminated disease state and a worse prognosis than metastatic colorectal liver disease that develops metachronously. In a retrospective study of 155 patients who underwent hepatic resection for colorectal liver metastases, patients with synchronous liver metastases had more sites of liver involvement (P = .008) and more bilobar metastases (P = .016) than patients diagnosed with metachronous liver metastases.[403]

It has been estimated that more than half of patients who die of CRC have liver metastases at autopsy, with metastatic liver disease as the cause of death in most patients.[404] Reviews of autopsy reports of patients who died from CRC showed that the liver was the only site of metastatic disease in one-third of patients.[401] Furthermore, several studies have shown rates of 5-year survival to be low in patients with metastatic liver disease not undergoing surgery.[397,405] Certain clinicopathologic factors, such as the

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presence of extrahepatic metastases, the presence of >3 tumors, and a disease-free interval of <12 months, have been associated with a poor prognosis in patients with CRC.[119,406-410]

Other groups, including the European Society for Medical Oncology (ESMO), have established guidelines for the treatment of mCRC.[411] The NCCN recommendations are discussed below. Please refer to *Surgical Management of Colorectal Metastases*, *Local Therapies for Metastases,* and *Determining Resectability* in the NCCN Guidelines for Colon Cancer (available at www.NCCN.org) for a detailed discussion of these treatment options. Specific recommendations for treatment of metastatic rectal cancer are discussed below.

Peritoneal Carcinomatosis

Approximately 17% of patients with mCRC have peritoneal carcinomatosis, with 2% having the peritoneum as the only site of metastasis. Patients with peritoneal metastases generally have a shorter PFS and OS than those without peritoneal involvement.[21,412] The goal of treatment for most abdominal/peritoneal metastases is palliative, rather than curative, and primarily consists of systemic therapy (see *Systemic Therapy for Advanced or Metastatic Disease* in the NCCN Guidelines for Colon Cancer [available at www.NCCN.org]) with palliative surgery or stenting (upper rectal lesions only) if needed for obstruction or impending obstruction.[413-415] A prospective randomized trial of 46 patients with stage IV rectal cancer and subacute large bowel obstruction found that patients who were randomized to placement of a self-expandable metal stent had a significantly lower 1-year OS rate compared with those who were randomized to primary tumor resection.[416] The Panel cautions that the use of bevacizumab in patients with colon or rectal stents is associated with a possible increased risk of bowel perforation.[417,418]

Neoadjuvant Therapy and Conversion to Resectability

The majority of patients diagnosed with mCRC have unresectable disease. However, for those with liver-limited unresectable disease that, because of involvement of critical structures, cannot be resected unless regression is accomplished, preoperative systemic therapy is being increasingly considered in highly selected cases in an attempt to downsize colorectal metastases and convert them to a resectable status. Patients presenting with large numbers of metastatic sites within the liver or lung are unlikely to achieve an R0 resection simply based on a favorable response to therapy, as the probability of complete eradication of a metastatic deposit by systemic therapy alone is low. These patients should be regarded as having unresectable disease not amenable to conversion therapy. In some highly selected cases, however, patients with disease that has had significant response to conversion systemic therapy can be converted from unresectable to resectable disease status.[419]

Please refer to *Neoadjuvant Therapy and Conversion to Resectability* and *Perioperative Therapy for Resectable Metachronous Metastatic Disease* in the NCCN Guidelines for Colon Cancer (available at www.NCCN.org) for a detailed discussion of preoperative systemic therapy options.

It is important to note that some of the treatment approaches for patients diagnosed with rectal cancer and synchronous lung or liver metastases differ relative to those for patients diagnosed with similarly staged colon cancer. In particular, initial treatment options for synchronous resectable rectal cancer that is pMMR/MSS typically include preoperative chemoRT directed toward treatment of the primary cancer; a preoperative chemotherapy regimen to target metastatic disease; and a surgical approach (ie, staged or synchronous resection of metastases and rectal lesion). Advantages of an initial chemoRT approach include a possible decreased risk of pelvic recurrence following surgery, while a disadvantage is that preoperative pelvic RT may decrease tolerance to

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systemic bevacizumab-containing adjuvant regimens, thereby limiting subsequent treatment of systemic disease. Data to guide decisions regarding optimal treatment approaches in this population of patients are very limited and specific recommendations for treatment of metastatic rectal cancer are discussed below.

Systemic Therapy for Advanced or Metastatic Disease

The current management of disseminated mCRC involves various active drugs, either in combination or as single agents. The choice of therapy is based on consideration of the goals of therapy, the type and timing of prior therapy, the mutational profile of the tumor, and the differing toxicity profiles of the constituent drugs. Although the specific regimens listed in the guideline are designated according to whether they pertain to initial therapy, therapy after first progression, or therapy after second progression, it is important to clarify that these recommendations represent a continuum of care and that these lines of treatment are blurred rather than discrete.[420] For example, if oxaliplatin is administered as part of an initial treatment regimen but is discontinued after 12 weeks or earlier for escalating neurotoxicity, continuation of the remainder of the treatment regimen would still be considered initial therapy.

Principles to consider at the start of therapy include: 1) preplanned strategies for altering therapy for patients exhibiting a tumor response or disease characterized as stable or progressive; and 2) plans for adjusting therapy for patients who experience certain toxicities. For example, decisions related to therapeutic choices after first progression of disease should be based, in part, on the prior therapies received (ie, exposing the patient to a range of cytotoxic agents). Furthermore, an evaluation of the efficacy and safety of these regimens for a patient must consider not only the component drugs, but also the doses, schedules, and methods of administration of these agents, and the potential for surgical cure and the performance status of the patient.

The continuum of care approach to the management of metastatic rectal cancer is the same as described for metastatic colon cancer. Please refer to *Systemic Therapy for Advanced or Metastatic Disease* in the NCCN Guidelines for Colon Cancer (available at www.NCCN.org) for a detailed discussion of the various options for systemic treatment. The roles of biomarkers for treatment selection in the advanced and metastatic disease setting are also discussed.

Biomarkers for Systemic Therapy

As the role of targeted therapy for treatment of advanced CRC or mCRC has become increasingly prominent, the NCCN Panel has expanded its recommendations regarding biomarker testing. Currently, determination of tumor gene status for *KRAS/NRAS* and *BRAF* mutations, as well as HER2 amplifications and MSI/MMR status (if not previously done), are recommended for patients with mCRC. Testing may be carried out for individual genes or as part of a tissue- or blood-based next-generation sequencing (NGS) panel. NGS panels have the advantage of being able to pick up rare and actionable genetic alterations, such as *NTRK* fusions. Discussion about each of these biomarkers may be found in the *Biomarkers for Systemic Therapy* section of the NCCN Guidelines for Colon Cancer (available at www.NCCN.org).

Recommendations for Treatment of Resectable Synchronous Metastases

When patients present with rectal cancer and synchronous liver- or lung-only metastases, the Panel now recommends a TNT approach including both chemotherapy and RT (short-course RT or long-course chemoRT) for pMMR/MSS tumors, using either an induction (preferred) or consolidation chemotherapy approach. For tumors that are dMMR/MSI-H or have *POLE/POLD1* mutation with ultra-hypermutated phenotype, upfront treatment with checkpoint inhibitor immunotherapy is preferred and RT may be given, either as short-course RT or long-course chemoRT after restaging. Upfront systemic treatment has the goal of early eradication of

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micrometastases, whereas the goal of short-course RT or long-course chemoRT is local control of disease prior to surgery/local therapy. Restaging should be performed before resection and surveillance may be considered if there is a complete clinical response following neoadjuvant treatment.

Resection of the primary tumor and liver can be done in a simultaneous or staged approach following neoadjuvant treatment.[421-428] Historically, in the staged approach, the primary tumor was usually resected first. However, the approach of liver resection before resection of the primary tumor is now well-accepted. In addition, emerging data suggest that chemotherapy, followed by resection of liver metastases before resection of the primary tumor, might be an effective approach in some patients, although more studies are needed.[429-431] In addition, neoadjuvant short-course radiation of T1–T3 primary rectal tumors is an option in this setting.[432] Locally ablative procedures can be considered instead of or in addition to resection in cases of liver or lung oligometastases, but resection is preferred.

The Panel acknowledges that some patients may not be candidates for systemic therapy or radiation; clinical judgment should be used in such cases.

Recommendations for Treatment of Unresectable Synchronous Metastases

Patients with unresectable synchronous liver- or lung-only metastases or who are medically inoperable are treated with intensive systemic therapy for advanced or metastatic disease to attempt to render these patients candidates for disease resection (or surveillance without resection in the case of a complete clinical response).

Chemotherapy regimens with high response rates should be considered for patients with potentially convertible disease that is pMMR/MSS.[433] These patients should be re-evaluated for resection after 2 months of chemotherapy and every 2 months thereafter while undergoing such therapy. Patients with disease that becomes resectable should receive short-course RT (preferred) or long-course chemoRT followed by immediate or delayed staged or synchronous resection and/or local therapy for metastases and resection of the rectal lesion. If there is a complete clinical response of the rectal lesion, surveillance may be considered as an alternative to immediate resection. Patients with disease that remains unresectable after initial systemic therapy should proceed to systemic therapy for advanced or metastatic disease and local therapy may be considered for select patients. Palliative RT or chemoRT can be given prior to systemic therapy if progression of the primary tumor occurred during first-line treatment.

For tumors with unresectable synchronous metastases that are dMMR/MSI-H or have *POLE/POLD1* mutation with ultra-hypermutated phenotype, treatment with checkpoint inhibitor immunotherapy is recommended. Disease status should be reevaluated every 2 to 3 months, with further treatment defined by response to therapy. If there is a cCR, surveillance may be warranted or surgery, with or without RT, may be done if disease is converted to resectable. For persistent disease, immunotherapy should be continued or a different systemic therapy may be warranted if there is disease progression.

There are conflicting data about whether resection of the primary tumor confers a survival benefit in the setting of unresectable colorectal metastases. See the NCCN Guidelines for Colon Cancer for discussion of these data (available at www.NCCN.org). Overall, the Panel believes that the risks of surgery outweigh the possible benefits of resection of asymptomatic primary tumors in this setting. Routine palliative resection of a synchronous primary lesion should therefore only be considered if the

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patient has an unequivocal imminent risk of obstruction, acute significant bleeding, perforation, or other significant tumor-related symptoms.

An intact primary tumor is not a contraindication to bevacizumab use. The risk of GI perforation in the setting of bevacizumab is not decreased by removal of the primary tumor, because large bowel perforations, in general, and perforation of the primary lesion, in particular, are rare (see *Systemic Therapy for Advanced or Metastatic Disease* in the NCCN Guidelines for Colon Cancer [available at www.NCCN.org]).

Recommendations for Treatment of Metachronous Metastases

In a single-institution, retrospective analysis of 735 patients with stage II/III rectal cancer treated with preoperative chemoRT followed by TME, the 5-year rates of liver and lung recurrences were 6.3% and 10.2%, respectively.[434] Resection of liver- and lung-only recurrences resulted in comparable survival (5.3 years and 5.1 years, respectively; $P = .39$).

On documentation of metachronous, potentially resectable, metastatic disease with dedicated contrast-enhanced CT or MRI, characterization of the disease extent using PET/CT scan should be considered in select cases if a surgical cure of M1 disease is feasible. PET/CT is used at this juncture to promptly characterize the extent of metastatic disease, and to identify possible sites of extrahepatic disease that could preclude surgery.[435,436] See the NCCN Guidelines for Colon Cancer for more details about the use of PET/CT for metachronous disease (available at www.NCCN.org).

As with other conditions in which stage IV disease is diagnosed, molecular testing of metastases or original primary tumor should be performed. Close communication between members of the multidisciplinary treatment team is recommended, including upfront evaluation by a surgeon experienced in the resection of hepatobiliary and lung metastases.

The management of metachronous metastatic disease is distinguished from that of synchronous disease through also including an evaluation of the systemic therapy history of the patient and through the absence of transabdominal resection. Patients with resectable disease are classified according to whether they have undergone previous chemotherapy (for pMMR/MSS) or previous immunotherapy (for dMMR/MSI-H or *POLE/POLD1* mutation). For patients who have resectable metastatic disease, treatment is resection with up to 6 months of perioperative therapy (pre- or postoperative or a combination of both), with choice of regimens based on dMMR/MSI status and previous therapy. Locally ablative procedures can be considered instead of or in addition to resection in cases of liver or lung oligometastases (see *Local Therapies for Metastases* in the NCCN Guidelines for Colon Cancer [available at www.NCCN.org]), but resection is preferred. For dMMR/MSI-H tumors that have a complete clinical response to checkpoint inhibitor immunotherapy, NOM is an option.

For patients with pMMR/MSS tumors and without a history of chemotherapy use, FOLFOX or CAPEOX are preferred, with capecitabine and 5-FU/LV as additional category 2B options. There are also cases when perioperative chemotherapy is not recommended in resectable metachronous disease. In particular, patients with a history of previous chemotherapy and upfront resection can be observed or may be given an active regimen for advanced disease (category 2B for the use of biologic agents in these settings). Observation is preferred if oxaliplatin-based therapy was previously administered.

Patients determined to have unresectable disease through cross-sectional imaging scan (including those considered potentially convertible) should receive an active systemic therapy regimen based on prior systemic therapy history (see *Second-Line or Subsequent Systemic Therapy* in the NCCN Guidelines for Colon Cancer [available at www.NCCN.org]). In the

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case of liver metastases only, hepatic arterial infusion chemotherapy (HAIC) with or without systemic chemotherapy (category 2B) is an option at centers with experience in the surgical and medical oncologic aspects of this procedure. Patients receiving palliative systemic therapy should be monitored with CT or MRI scans approximately every 2 to 3 months.

Post-Treatment Surveillance

After curative-intent surgery, post-treatment surveillance of patients with CRC is performed to evaluate for possible therapeutic complications, discover a recurrence that is potentially resectable for cure, and identify new metachronous neoplasms at a preinvasive stage. One study found that 95% of recurrences occurred in the first 5 years.[437]

Advantages of more intensive follow-up of patients after treatment of stage II and/or stage III disease have been demonstrated prospectively in several older studies[438-440] and in multiple meta-analyses of RCTs designed to compare low-intensity and high-intensity programs of surveillance.[441-445] In the final analysis of the Intergroup trial 0114 comparing bolus 5-FU to bolus 5-FU/LV in patients with surgically resectable rectal cancer, local recurrence rates continued to rise after 5 years.[238] Further, a population-based report indicated that long-term survival is possible in patients treated for local recurrence of rectal cancer (overall 5-year relative survival rate of 15.6%), thereby providing support for more intensive post-treatment follow-up in these patients.[446]

Results from the randomized controlled FACS trial of 1202 patients with resected stage I to III disease showed that intensive surveillance imaging or CEA screening resulted in an increased rate of curative-intent surgical treatment compared with a minimum follow-up group that only received testing if symptoms occurred, but no advantage was seen in the CEA and CT combination arm (2.3% in the minimum follow-up group, 6.7% in the CEA group, 8% in the CT group, and 6.6% in the CEA plus CT group).[447] In this study, no mortality benefit to regular monitoring with CEA, CT, or both was observed compared with minimum follow-up (death rate, 18.2% vs. 15.9%; difference, 2.3%; 95% CI, -2.6% to 7.1%). The authors concluded that any strategy of surveillance is unlikely to provide a large survival advantage over a symptom-based approach.[447] The randomized COLOFOL trial of 2509 patients with stage II or III CRC looked at follow-up testing with CT of the thorax and abdomen and CEA screening, comparing a high-frequency surveillance approach (CT and CEA at 6, 12, 18, 24, and 36 months post-surgery) to a low-frequency approach (CT and CEA at 12 and 36 months post-surgery).[448] This trial reported no significant difference in 5-year overall mortality or CRC-specific mortality between the two screening approaches.

The CEAwatch trial compared usual follow-up care to CEA measurements every 2 months, with imaging performed if CEA increases were seen twice, in 3223 patients treated for non-mCRC at 11 hospitals in the Netherlands.[449] The intensive CEA surveillance protocol resulted in the detection of more total recurrences and recurrences that could be treated with curative intent than usual follow-up, and the time to detection of recurrent disease was shorter. However, no OS or disease-specific survival benefit was seen.[450] Another randomized trial of 1228 patients found that more intensive surveillance led to earlier detection of recurrences than a less intensive program (less frequent colonoscopy and liver ultrasound and the absence of an annual chest x-ray) but also did not affect OS.[451]

The randomized phase III PRODIGE 13 trial is comparing 5-year OS after intensive radiologic monitoring (abdominal ultrasound, chest/abdomen/pelvis CT, and CEA) with a lower intensity program (abdominal ultrasound and chest x-ray) in patients with resected stage II or III colon or rectal tumors.[452] An abstract reporting results from 1995 patients on this trial concluded that the more intensive surveillance

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program did not provide any benefit in 5-year OS, but did result in more curative intent secondary surgeries for colon cancer. Surgical treatment of recurrence was performed in 40.9% of patients receiving minimal surveillance (no CT, no CEA), 66.3% of patients receiving lower intensity imaging plus CEA, 50.7% of patients receiving no CEA but higher intensity imaging, and 59.5% in the maximum surveillance group with both CEA and CT ($P = .0035$).[453] Final results from PRODIGE 13, reported in a 2022 abstract, similarly showed that intensive imaging, but not CEA screening, increased the rate of surgical treatment of recurrence.[454] While there was a trend towards better 5-year RFS with CT scanning, no survival advantage was noted with any strategy.

Meta-analyses support the conclusion that more intensive surveillance of patients with resected CRC results in earlier detection of recurrences, without any effect on survival.[442,443]

Patients who had resection of mCRC can undergo subsequent curative-intent resection of recurrent disease (see *Surgical Management of Colorectal Metastases* in the NCCN Guidelines for Colon Cancer [available at www.NCCN.org]) and therefore should undergo post-treatment surveillance. A retrospective analysis of 952 patients who underwent resection at Memorial Sloan Kettering Cancer Center showed that 27% of patients with recurrent disease underwent curative-intent resection and that 25% of those patients (6% of recurrences; 4% of the initial population) were free of disease for ≥36 months.[455]

Controversies remain regarding selection of optimal strategies for following patients after potentially curative CRC surgery, and the Panel's recommendations are based mainly on consensus. The Panel endorses surveillance as a means to identify patients who are potentially curable of metastatic disease with surgical resection. For patients with disease that shows a complete clinical response to neoadjuvant therapy and are followed with a NOM approach, careful surveillance is essential in order to treat potential tumor regrowth in a timely manner. See the *Watch-and-Wait Nonoperative Approach for Clinical Complete Responders* section, above, for a full discussion of NOM, including surveillance recommendations.

The Panel recommendations for postoperative surveillance pertain to patients who have undergone successful treatment (ie, no known residual disease following surgery) and are separated into four groups: 1) those with low-risk polyps removed by polypectomy; 2) those who received transanal local excision only; 3) patients with stage I disease and full surgical staging; and 4) patients with stage II–IV disease.

For all four groups, colonoscopy is recommended at approximately 1 year following resection (or at approximately 3–6 months post-resection if not performed preoperatively due to an obstructing lesion). Repeat colonoscopy is typically recommended at 3 years, and then every 5 years thereafter, unless follow-up colonoscopy indicates advanced adenoma (villous polyp, polyp >1 cm, or high-grade dysplasia), in which case colonoscopy should be repeated in 1 year.[456] More frequent colonoscopies may be indicated in patients who present with CRC before age 50.[456] Surveillance colonoscopies are primarily aimed at identifying and removing metachronous polyps since data show that patients with a history of CRC have an increased risk of developing second cancers,[457] particularly in the first 2 years following resection. The use of post-treatment surveillance colonoscopy has not been shown to improve survival through the early detection of recurrence of the original CRC.[456]

Proctoscopy with EUS or MRI is recommended to evaluate the rectal anastomosis for local recurrence in patients treated with transanal local excision or polypectomy only. Proctoscopy is not recommended for other patients, because isolated local recurrences are rarely found in these patients and are rarely curable. In fact, in a single-center study of 112 patients who had TME for rectal cancer, only one local recurrence occurred, and it was not identified by rectal surveillance but by CEA and

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symptoms.[458] In these 112 patients, 20 anoscopies, 44 proctoscopies, and 495 flexible sigmoidoscopies were performed.

For the stage II–IV group, history and physical examination is recommended every 3 to 6 months for 2 years, and then every 6 months for a total of 5 years; and a CEA test (also see *Managing an Increasing CEA Level*, below) is recommended at baseline and every 3 to 6 months for 2 years,[459] then every 6 months for a total of 5 years for patients who are potential candidates for aggressive curative surgery.[441,459,460] Chest, abdominal, and pelvic CT scans are recommended every 6 to 12 months for a total of 5 years for stage II–III disease and every 3 to 6 months for 2 years, and then every 6 to 12 months for a total of 5 years for stage IV disease.[441,461] The more frequent imaging cadences are category 2B recommendations, reflecting some Panel disagreement on the benefit verus risk of frequent imaging. CT scan is recommended to monitor for the presence of potentially resectable metastatic lesions, primarily in the lung and the liver. Hence, CT scan is not routinely recommended in patients who are not candidates for potentially curative resection of liver or lung metastases. A recent analysis of patients with resected or ablated colorectal liver metastases found that the frequency of surveillance imaging did not correlate with time to second procedure or median survival duration.[462] Those scanned once per year survived a median of 54 months versus 43 months for those scanned three to four times per year ($P = .08$), suggesting that annual scans may be sufficient in this population.

Routine CEA monitoring and CT scanning are not recommended beyond 5 years. In addition, use of PET/CT to monitor for disease recurrence is not recommended.[461,463] The CT that accompanies a PET/CT is usually a noncontrast CT, and therefore is not of ideal quality for routine surveillance.

Managing an Increasing CEA Level

Management of an elevated CEA level after resection should include colonoscopy; chest, abdominal, and pelvic CT scans; physical examination; and consideration of a PET/CT scan. If imaging study results are normal in the face of a rising CEA, repeat CT scans are recommended every 3 months until either disease is identified or CEA level stabilizes or declines.

In a recent retrospective chart review at Memorial Sloan Kettering Cancer Center, approximately half of elevations in CEA levels after R0 resection of locoregional CRC were false positives, with most being single high readings or repeat readings in the range of 5 to 15 ng/mL.[464] In this study, false-positive results >15 ng/mL were rare, and all results >35 ng/mL represented true positives. Following a systematic review and meta-analysis, the pooled sensitivity and specificity of CEA at a cutoff of 10 ng/mL were calculated at 68% (95% CI, 53%–79%) and 97% (95% CI, 90%–99%), respectively.[465,466] In the first 2 years post-resection, a CEA cutoff of 10 ng/mL is estimated to detect 20 recurrences, miss 10 recurrences, and result in 29 false positives.

A PET/CT scan may be considered in the scenario of an elevated CEA with negative, good-quality CT scans. A systematic review and meta-analysis found 11 studies (510 patients) that addressed the use of PET/CT in this setting.[467] The pooled estimates of sensitivity and specificity for the detection of tumor recurrence were 94.1% (95% CI, 89.4%–97.1%) and 77.2% (95% CI, 66.4%–85.9%), respectively. An analysis of outcomes of 88 patients treated for CRC under surveillance who had normal or equivocal conventional imaging results with an elevated CEA found that PET/CT had a sensitivity of 88% and a specificity of 88% for the detection of recurrences.[468]

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The Panel does not recommend a so-called blind or CEA-directed laparotomy or laparoscopy for patients whose workup for an increased CEA level is negative,[469] nor do they recommend use of anti-CEA–radiolabeled scintigraphy.

Treatment of Locally Recurrent Disease

Locally recurrent rectal cancer is characterized by isolated pelvic/anastomotic recurrence of disease. In a single-center study, Yu et al reported low rates of 5-year local recurrence (ie, 5-year locoregional control rate of 91%) for patients with rectal cancer treated with surgery and either RT or chemoRT, and 49% of recurrences occurred in the low pelvic and presacral regions with an additional 14% occurring in the mid and high pelvis.[470] In a more recent, single-institution, retrospective analysis of 735 patients with stage II/III rectal cancer treated with preoperative chemoRT followed by TME, locoregional recurrence rate at 5 years was 4.6%, occurring at a median of 24.7 months.[434]

The Panel recommends that patients with unresectable lesions be treated with systemic therapy, chemoRT, or short-course RT according to their ability to tolerate therapy. Debulking that results in gross residual cancer is not recommended. Potentially resectable isolated pelvic/anastomotic recurrence that is pMMR/MSS may be managed with neoadjuvant therapy, including chemotherapy before or after chemoRT or short-course RT, followed by resection. When following this approach, starting neoadjuvant therapy with chemotherapy is preferred. IORT or brachytherapy should be considered with resection if it can be safely delivered.[197,471-473] Alternatively, resection may be done first, followed by adjuvant chemoRT. For potentially resectable isolated pelvic/anastomotic recurrence that is dMMR/MSI-H or with *POLE/POLD1* mutation, options include checkpoint inhibitor immunotherapy (if not previously received); resection followed by long-course chemoRT; or radiation (long-course chemoRT or short-course RT) followed by resection.

A retrospective study found that re-resection was not associated with improved survival in patients with isolated locoregional recurrence (3.6 years with surgery vs. 3.2 years without surgery; $P = .353$).[434] Older studies have shown that patients with disease recurrence at the anastomotic site are more likely to be cured following re-resection than those with an isolated pelvic recurrence.[474,475] In a study of 43 consecutive patients with advanced pelvic recurrence of CRC who had not undergone prior RT, treatment with 5 weeks of 5-FU by infusion concurrent with RT enabled the majority of patients (77%) to undergo re-resection with curative intent.[475] Studies of patients who previously received pelvic radiation show that re-irradiation can be effective, with acceptable rates of toxicity.[476-479] In one such study of 48 patients with recurrent rectal cancer and a history of pelvic radiation, the 3-year rate of grade 3–4 late toxicity was 35%, and 36% of patients treated were able to undergo surgery following radiation.[476] IMRT can be used in this setting of re-irradiation.

Survivorship

The Panel recommends that a prescription for survivorship and transfer of care to the primary care physician be written.[480] The oncologist and primary care provider should have defined roles in the surveillance period, with roles communicated to the patient. The care plan should include an overall summary of treatments received, including surgeries, radiation treatments, and chemotherapy. The possible expected time to resolution of acute toxicities, long-term effects of treatment, and possible late sequelae of treatment should be described. Finally, surveillance and healthy behavior recommendations should be part of the care plan.

Disease preventive measures, such as immunizations; early disease detection through periodic screening for second primary cancers (eg, breast, cervical, or prostate cancers); and routine good medical care and monitoring are recommended (see the NCCN Guidelines for Survivorship [available at www.NCCN.org]). Additional health monitoring should be

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performed as indicated under the care of a primary care physician. Survivors are encouraged to maintain a therapeutic relationship with a primary care physician throughout their lifetime.[481]

Other recommendations include monitoring for late sequelae of rectal cancer or of the treatment of rectal cancer, such as bowel function changes (eg, patients with stoma).[482-487] Urogenital dysfunction following resection and/or pelvic irradiation is common.[482,488-490] Patients should be screened for sexual dysfunction, erectile dysfunction, dyspareunia, vaginal dryness, and urinary incontinence, frequency, and urgency. Referral to a gynecologist or urologist can be considered for persistent symptoms. Other long-term problems common to CRC survivors include oxaliplatin-induced peripheral neuropathy, fatigue, insomnia, cognitive dysfunction, and emotional or social distress.[491-496] Specific management interventions to address side effects of CRC have been described,[497] and a survivorship care plan for patients with CRC has been published.[498]

The NCCN Guidelines for Survivorship (available at www.NCCN.org) provide screening, evaluation, and treatment recommendations for common consequences of cancer and cancer treatment to aid health care professionals who work with survivors of adult-onset cancer in the post-treatment period, including those in specialty cancer survivor clinics and primary care practices. These guidelines include many topics with potential relevance to survivors of CRC, including anxiety, depression, and distress; cognitive dysfunction; fatigue; pain; sexual dysfunction; healthy lifestyles; and immunizations. Concerns related to employment, insurance, and disability are also discussed. The American Cancer Society (ACS) has also established guidelines for the care of survivors of CRC, including surveillance for recurrence, screening for subsequent primary malignancies, the management of physical and psychosocial effects of cancer and its treatment, and promotion of healthy lifestyles.[481] Please refer to *Healthy Lifestyles for Survivors of CRC* and *Secondary Chemoprevention for CRC Survivors* in the NCCN Guidelines for Colon Cancer (available at www.NCCN.org) for more detailed discussion on these topics.

Summary

The NCCN Rectal Cancer Panel believes that a multidisciplinary approach, including representation from gastroenterology, medical oncology, surgical oncology/colorectal surgery, radiation oncology, pathology, and radiology is necessary for treating patients with rectal cancer. Patients with very-early-stage tumors that are node-negative by endorectal ultrasound or endorectal or pelvic MRI and who meet carefully defined criteria can be treated with a transanal local excision or ESD.

Most other rectal cancers are treated by neoadjuvant therapy followed by transabdominal resection. A TNT approach, traditionally consisting of chemoRT/short-course RT and chemotherapy is preferred for the majority of patients with suspected or proven T3–4 disease and/or regional node involvement when RT is being given. However, ongoing clinical trials for rectal cancer are particularly focused on treatment approaches that omit surgery or RT, with the goal of improving outcomes for eligible patients. Careful surveillance is necessary to detect and manage recurrences in a prompt and effective manner. For patients with dMMR/MSI-H rectal cancers, checkpoint inhibitor immunotherapy is increasingly being used as initial treatment, allowing many patients to avoid the toxicities of surgery, RT, and chemotherapy.

The recommended post-treatment surveillance program for patients following treatment for rectal cancer includes serial CEA determinations, as well as periodic chest, abdominal, and pelvic CT scans, and periodic evaluation by colonoscopy. Patients with recurrent localized disease should be considered for resection with chemotherapy and radiation. If

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resection is not possible, then systemic therapy, chemoRT, or RT alone may be given.

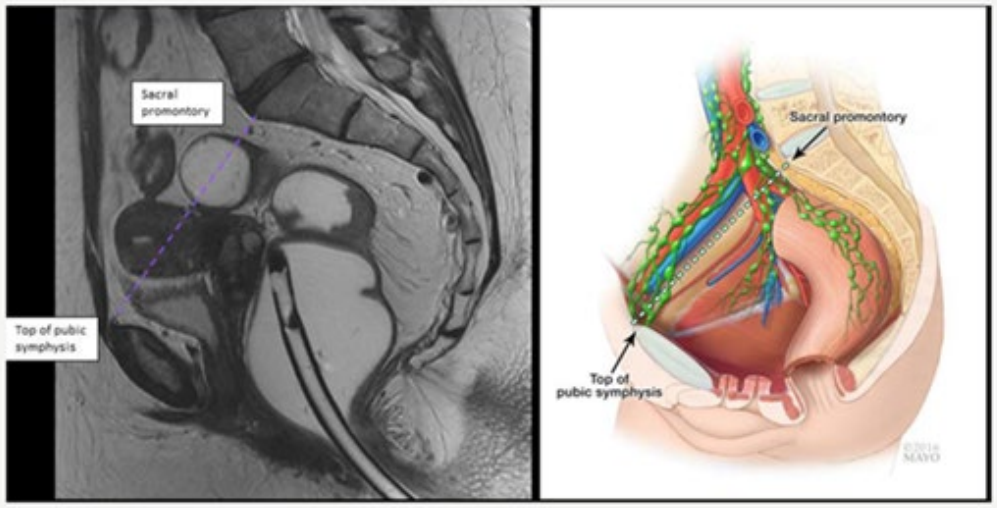
A patient with metastatic disease in the liver or lung should be considered for surgical resection if he or she is a candidate for surgery and if complete resection (R0) can be achieved. Preoperative systemic therapy, and sometimes chemoRT or short-course RT, are used in the synchronous setting, and perioperative chemotherapy is used in the metachronous setting. For patients with dMMR/MSI-H tumors who have not yet received immunotherapy, initial treatment with a checkpoint inhibitor should be considered.

Recommendations for patients with disseminated, unresectable metastatic disease represent a continuum of care in which lines of treatment are blurred rather than discrete. Principles to consider at the start of therapy include pre-planned strategies for altering therapy for patients in both the presence and absence of disease progression and plans for adjusting therapy for patients who experience certain toxicities. Recommended systemic therapy options for advanced or metastatic disease depend on whether the patient is appropriate for intensive therapy; the biomarker status of the tumor; and for patients with progressive disease, the choice of initial therapy.

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Figure 1. Definition of Rectum



"Rectum" is defined as the portion of bowel located below the pelvic inlet (an imaginary line drawn from the sacral promontory to the top of the pubic symphysis) as determined by a dedicated MRI of the pelvis

- Upper rectum: above the anterior peritoneal reflection
- Mid-rectum: at the anterior peritoneal reflection
- Lower-rectum: below the anterior peritoneal reflection

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